



CCNA ALL LABS

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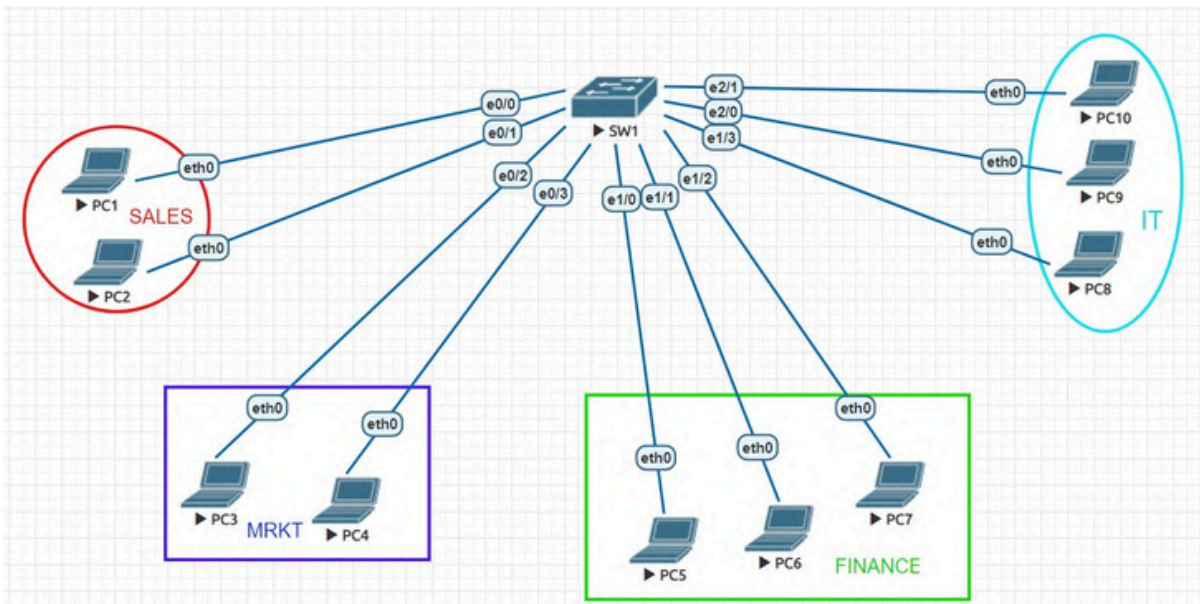
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VLAN

VLAN stands for virtual local area network. VLAN is a technology we use for the purpose of Broadcast isolation or segregation of LAN .

Generally what happen in company there is so many department , for that each and every department require pc and to connect that pc we require Switch .If we buy different switch for each department the cost of IT infra will going to increase ,to save that cost company implement vlan on their switch.so here we go.



Task-1.1 Create VLAN

VLAN 10 sales

VLAN 20 mrkt

VLAN 30 finance

VLAN 40 IT

Task-1.2 IP Range

VLAN 10 - 192.168.10.0/27	For sales department require IP is 30
VLAN 20 - 192.168.20.0/26	For marketing department require IP is 42
VLAN 30 - 192.168.30.0/27	For finance department require IP is 25
VLAN 40 - 192.168.40.0/28	For IT department require IP is 14

Task-1.3 Assign VLAN to the respected port as per diagram & assign IP to the pc

SOLUTION

So, start first with the switch

Switch> enable

Switch#configure terminal

Step 1:- Now we have to create VLAN

Switch(config)#vlan 10

Switch(config)#name sales

Switch(config)#exit

Switch(config)#vlan 20

Switch(config)#name mrkt

Switch(config)#exit

Switch(config)#Vlan 30

Switch(config)#name finance

Switch(config)#exit

Switch(config)#vlan 40

Switch(config)#name IT

Switch(config)#exit

Step 2 :- Assignment of VLAN

Now open interface and assign VLAN

Switch(config)#interface ethernet 0/0

Switch(config)#Switchport mode access

Switch(config)#Switchport access vlan 10

Switch(config)#Exit

Switch(config)#interface ethernet 0/1

Switch(config)#switchport mode access

Switch(config)#switchport access vlan 10

Switch(config)#exit

Switch(config)#interface ethernet 0/2

Switch(config)# switchport mode access

Switch(config)# switchport access vlan 20

Switch(config)# exit

Switch(config)#interface ethernet 0/3

Switch(config)# switchport mode access

Switch(config)# switchport access vlan 20

Switch(config)# exit

```
Switch(config)#interface ethernet 1/0
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 30
Switch(config)# exit
```

```
Switch(config)#interface ethernet 1/1
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 30
Switch(config)# exit
```

```
Switch(config)#interface ethernet 1/2
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 30
Switch(config)# exit
```

```
Switch(config)#interface ethernet 1/3
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 40
Switch(config)# exit
```

```
Switch(config)#interface ethernet 2/0
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 40
Switch(config)# exit
```

```
Switch(config)#interface ethernet 2/1
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 40
Switch(config)# exit
```

Note- Assign IP to the respected VLAN Pc's

VERIFICATION

Switch# show vlan

```
Switch#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Et2/2, Et2/3
10	sales	active	Et0/0, Et0/1
20	mrkt	active	Et0/2, Et0/3
30	finance	active	Et1/0, Et1/1, Et1/2
40	IT	active	Et1/3, Et2/0, Et2/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

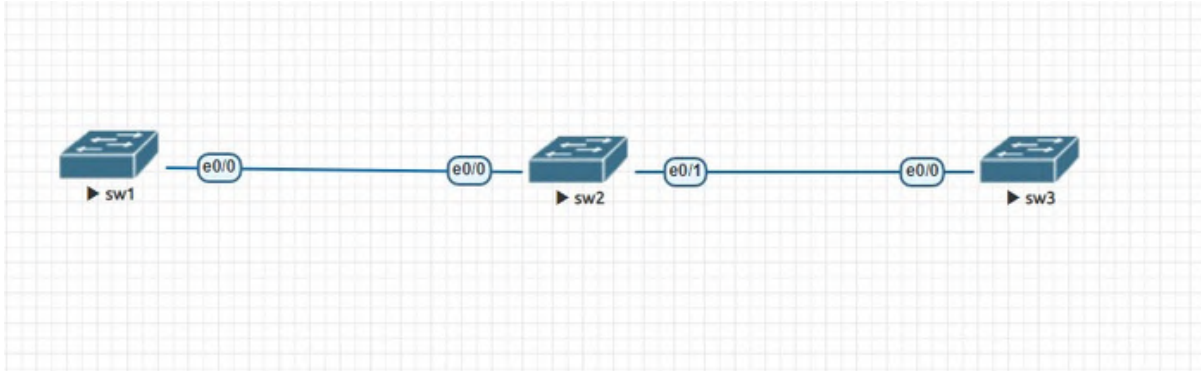
Switch# Show running-config

```
!
interface Ethernet0/0
  switchport access vlan 10
  switchport mode access
!
interface Ethernet0/1
  switchport access vlan 10
!
interface Ethernet0/2
  switchport access vlan 20
  switchport mode access
!
interface Ethernet0/3
  switchport access vlan 20
  switchport mode access
!
interface Ethernet1/0
  switchport access vlan 30
  switchport mode access
!
interface Ethernet1/1
  switchport access vlan 30
  switchport mode access
!
interface Ethernet1/2
  switchport access vlan 30
  switchport mode access
!
interface Ethernet1/3
  switchport access vlan 40
  switchport mode access
!
interface Ethernet2/0
  switchport access vlan 40
  switchport mode access
!
interface Ethernet2/1
  switchport access vlan 40
  switchport mode access
```

VTP

VTP stands for VLAN Trunking Protocol. why we use VTP? This is the important question we generally face. So, in the company there were almost 30-40 switches more or less. As engineer we need to configure each and every switch and if each switch has minimum 10 VLAN on it. It means 40 Switches x 10 VLAN per switch = 400 times you need create vlan.to reduce this task and to make all switches manageable we configure VTP.

So here we go,



Task 2.1- Create VLAN on sw1

Task 2.2- Make interface trunk on sw1 eth0/0 and sw2 eth0/0 & eth0/1 , sw3 eth0/0

Task 2.3- Configure VTP mode, vtp domain, vtp password on all switches.

SOLUTION

Task 2.1

```
sw1(config)#vlan 10
sw1(config-vlan)#name sales
sw1(config-vlan)#exit

sw1(config)#vlan 20
sw1(config-vlan)#name mrkt
sw1(config-vlan)#exit

sw1(config)#vlan 30
sw1(config-vlan)#name IT
sw1(config-vlan)#exit

sw1(config)#vlan 40
sw1(config-vlan)#name DC
sw1(config-vlan)#exit
```

Task 2.2

On switch 1

```
sw1(config)#interface ethernet0/0
sw1(config-if)#switchport trunk encapsulation dot1q
sw1(config-if)#switchport mode trunk
sw1(config-if)#exit
```

On switch 2

```
sw2(config)#interface ethernet0/0
sw2(config-if)#switchport trunk encapsulation dot1q
sw2(config-if)#switchport mode trunk
sw2(config-if)#exit
```

```
sw2(config)#interface ethernet0/1
sw2(config-if)#switchport trunk encapsulation dot1q
sw2(config-if)#switchport mode trunk
sw2(config-if)#exit
```

On switch 3

```
sw3(config)#interface ethernet0/0
sw3(config-if)#switchport trunk encapsulation dot1q
sw3(config-if)#switchport mode trunk
sw3(config-if)#exit
```

Task 2.3

```
sw1(config)#vtp mode server
```

Device mode already VTP Server for VLANs.

```
sw1(config)#vtp domain ccn.com
```

Changing VTP domain name from NULL to ccn.com

```
sw1(config)#vtp password ccn@123
```

Setting device VTP password to ccn@123

```
sw2(config)#vtp mode client
```

Setting device to VTP Client mode for VLANs.

```
sw2(config)#vtp domain ccn.com
```

Domain name already set to ccn.com.

```
sw2(config)#vtp password ccn@123
```

Setting device VTP password to ccn@123

```
sw3(config)#vtp mode client
```

Setting device to VTP Client mode for VLANs.

```
sw3(config)#vtp domain ccn.com
```

Domain name already set to ccn.com.

```
sw3(config)#vtp password ccn@123
```

Setting device VTP password to ccn@123

VERIFICATION

Verification of #show vlan command

Sw1

```
sw1#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Et0/1, Et0/2, Et0/3
10	sales	active	
20	mrkt	active	
30	IT	active	
40	DC	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Sw2

```
sw2#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Et0/2, Et0/3
10	sales	active	
20	mrkt	active	
30	IT	active	
40	DC	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Sw3

```
sw3#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Et0/1, Et0/2, Et0/3
10	sales	active	
20	mrkt	active	
30	IT	active	
40	DC	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Verification of #show vtp status command

Sw1

```
sw1#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : ccn.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : aabb.cc80.1000
Configuration last modified by 0.0.0.0 at 6-1-22 11:38:59
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN:

```
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 9
Configuration Revision    : 4
MD5 digest               : 0x4C 0xBB 0xB6 0x01 0x5F 0x0E 0x8D 0xE8
                          0xC5 0xFC 0xD8 0x72 0x9B 0xB2 0xB5 0x46
```

Sw2

```
sw2#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : ccn.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : aabb.cc80.2000
Configuration last modified by 0.0.0.0 at 6-1-22 11:38:59
```

Feature VLAN:

```
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 1005
Number of existing VLANs : 9
Configuration Revision    : 4
MD5 digest               : 0x4C 0xBB 0xB6 0x01 0x5F 0x0E 0x8D 0xE8
                          0xC5 0xFC 0xD8 0x72 0x9B 0xB2 0xB5 0x46
```

Sw3

```
sw3#show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 1
VTP Domain Name          : ccn.com
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : aabb.cc80.4000
Configuration last modified by 0.0.0.0 at 6-1-22 11:38:59
```

Feature VLAN:

```
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 1005
Number of existing VLANs : 9
Configuration Revision    : 4
MD5 digest               : 0x4C 0xBB 0xB6 0x01 0x5F 0x0E 0x8D 0xE8
                          0xC5 0xFC 0xD8 0x72 0x9B 0xB2 0xB5 0x46
```

Note – After VLAN data forwarding to each and every switch via vtp configuration we can assign VLAN to switchports with the help of [Step 2 :- Assignment of VLAN](#)

EtherChannel

EtherChannel is a technology use to Bundle/Aggregate the Link/Node/Port, with the use of some protocols such as LACP or PAgP.

Why we use EtherChannel?

When we face lack of bandwidth with links then at that time we configure EtherChannel to bundle the link and get combined speed. Like each interface is ethernet means 10MBPS of each then, if we bundle 3 link at a time then the combine speed will be 300 MBPS

EtherChannel uses PAgP or LACP as a Negotiation Protocol

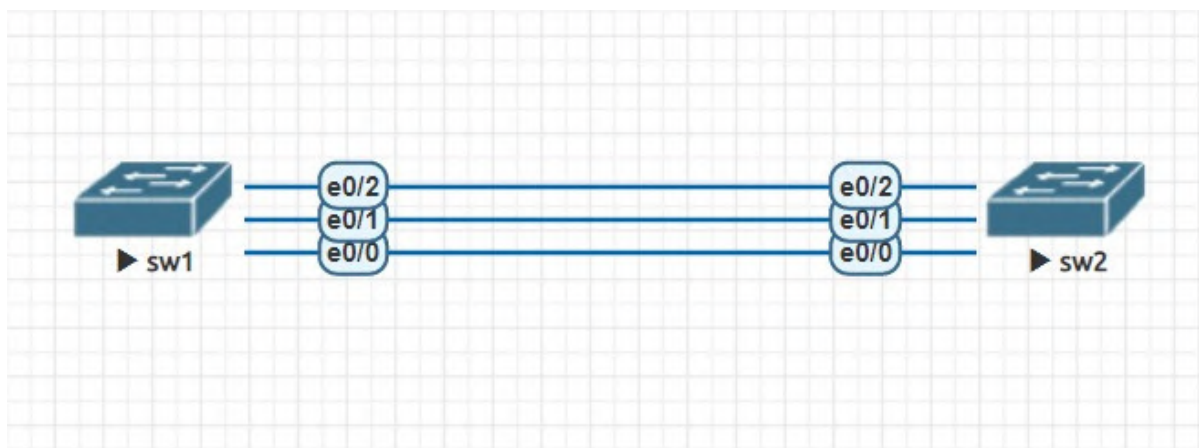
LINK AGGREGATION CONTROL PROTOCOL (LACP)

- ❖ LACP was made by IEEE and define with code of 802.3ad
- ❖ LACP is a Open Source and support all vendors in the market
- ❖ LACP adds up LACPDU to establish a etherchannel
- ❖ LACP has two modes
 1. Active – enable LACP unconditionally
 2. passive – enable LACP when LACP device is detected

PORT AGGREGATION CONTROL PROTOCOL (PAgP)

- ❖ PAgP was made by cisco.
- ❖ PAgP is a Cisco Proprietary Protocol. Hence, it only supports cisco devices
- ❖ PAgP has two modes
 1. Desirable – enable PAgP unconditionally
 2. Auto – enable PAgP when PAgP device is detected

So here we go,



Task 3 Task 3.1 - Configure sw1 interface ethernet 0/0, 0/1, 0/2 for EtherChannel **Task 3.2** - Configure sw2 interface ethernet 0/0, 0/1, 0/2 for EtherChannel **Task 3.3** – Configure sw1 to sw2 link as a trunk

Solution,

Task 3.1

Sw1

```
sw1(config)#interface range ethernet 0/0-2
sw1(config)#channel-protocol lacp
sw1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
sw1(config-if-range)#exit
```

Task 3.2

sw2

```
sw2(config)#interface range ethernet 0/0-2
sw2(config)#channel-protocol lacp
sw2(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
sw2(config-if-range)#exit
```

Task 3.3

Sw1

```
sw1(config)#interface range ethernet 0/0-2
sw1(config-if-range)#switchport trunk encapsulation dot1q
sw1(config-if-range)#switchport mode trunk
sw1(config-if-range)#exit
```

OR

```
sw1(config)#interface port-channel 1
sw1(config-if)#switchport trunk encapsulation dot1q
sw1(config-if)#switchport mode trunk
sw1(config-if)#exit
```

sw2

sw2(config)#interface range ethernet 0/0-2

sw2(config-if-range)#switchport trunk encapsulation dot1q

sw2(config-if-range)#switchport mode trunk

sw2(config-if-range)#exit

VERIFICATION

Sw1

```
sw1#show etherchannel summary
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Et0/0(P)  Et0/1(P)  Et0/2(P)
```

```
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
interface Ethernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
interface Ethernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
```

sw2

```
sw2#show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone  S - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG
```

```
Number of channel-groups in use: 1
Number of aggregators:           1
```

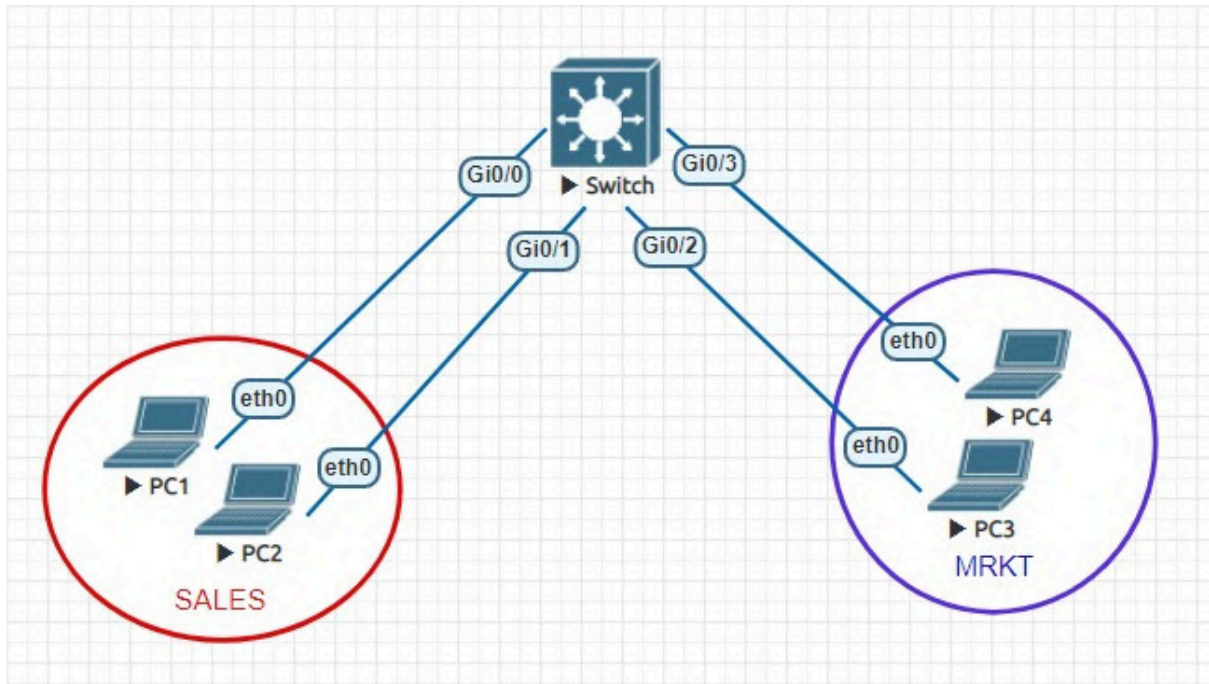
Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Et0/0(P) Et0/1(P) Et0/2(P)

```
!
interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
interface Ethernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
interface Ethernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-protocol lacp
 channel-group 1 mode active
!
```

INTER VLAN ROUTING (IVR)

- ❖ Inter VLAN routing is method we use for the purpose of inter VLAN communication.
- ❖ Inter VLAN Routing is a new method to route between Multiple VLAN's.
- ❖ Inter VLAN Routing require one device which is L3 Switch.

So here we go,



TASK

Task 4.1 Configure switch with VLAN 10 – SALES & VLAN 20 – MRKT

Task 4.2 Implement VLAN on the respected interface as per Diagram

Task 4.3 Create Default Gateway for the VLAN 10 & VLAN 20 PC's

Task 4.4 Assign IP to the PC

SOLUTION

Task 4.1 On switch, first we need to create vlan

```
Switch(config)#vlan 10
```

```
Switch(config-vlan)#name SALES
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name MRKT
```

```
Switch(config-vlan)#exit
```

Task 4.2 We need to apply vlan on interface

```
Switch(config)#interface range gigabitethernet0/0-1
```

```
Switch(config-if-range)#switchport mode access
```

```
Switch(config-if-range)#switchport access vlan 10
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#interface range gigabitethernet0/2-3
```

```
Switch(config-if-range)#switchport mode access
```

```
Switch(config-if-range)#switchport access vlan 20
```

```
Switch(config-if-range)#exit
```

Task 4.3 Now, we need to configure interface vlan as a gateway for the PC's

```
Switch(config)#interface vlan 10
```

```
Switch(config-if)#ip address 192.168.10.254 255.255.255.0
```

```
Switch(config-if)#no shutdown
```

```
Switch(config-if)#exit
```

```
Switch(config)#interface vlan 20
```

```
Switch(config-if)#ip address 192.168.20.254 255.255.255.0
```

```
Switch(config-if)#no shutdown
```

```
Switch(config-if)#exit
```

Note - Now IP given on interface VLAN, will become a gateway for the PC to communicate over the VLAN.

VERIFICATION

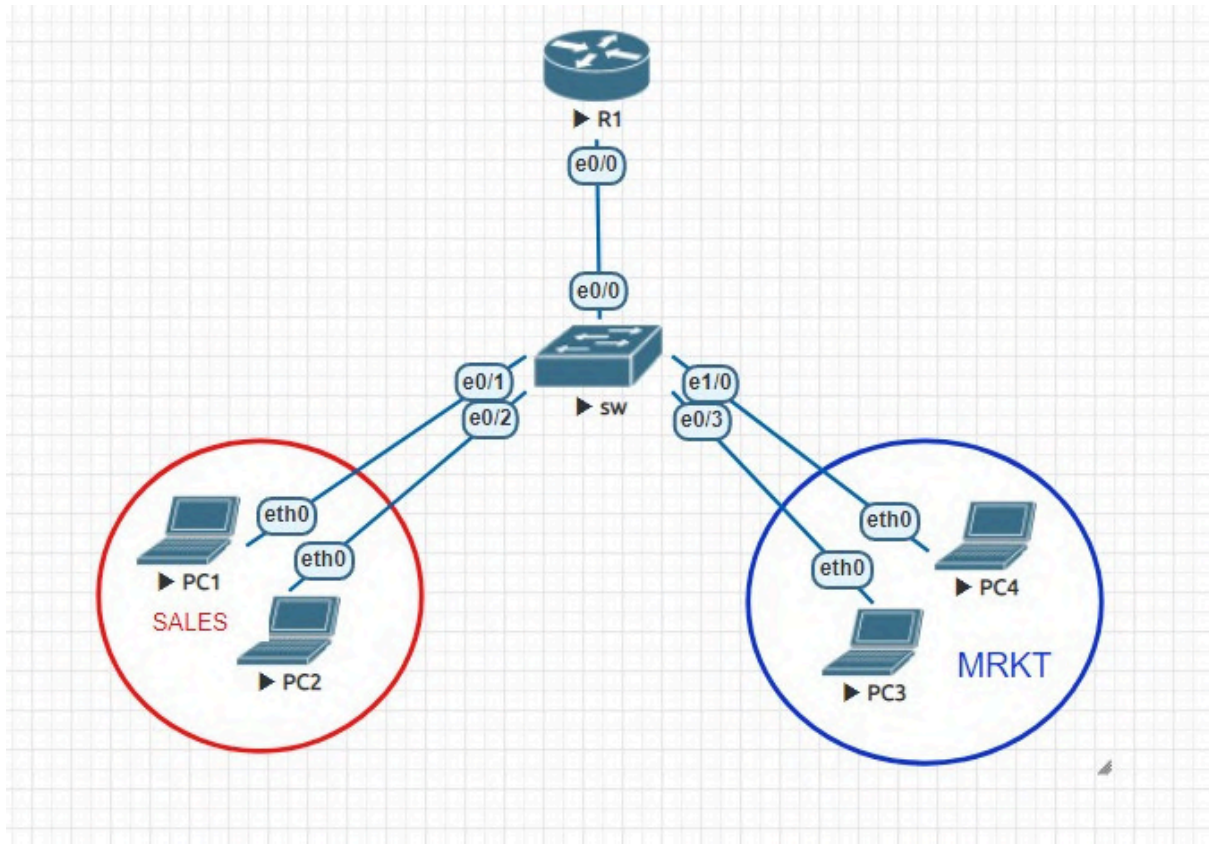
```
Switch#show ip interface brief
Interface          IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0 unassigned      YES unset  up      up
GigabitEthernet0/1 unassigned      YES unset  up      up
GigabitEthernet0/2 unassigned      YES unset  up      up
GigabitEthernet0/3 unassigned      YES unset  up      up
GigabitEthernet1/0 unassigned      YES unset  up      up
GigabitEthernet1/1 unassigned      YES unset  up      up
GigabitEthernet1/2 unassigned      YES unset  up      up
GigabitEthernet1/3 unassigned      YES unset  up      up
Vlan10             192.168.10.254 YES manual up      up
Vlan20             192.168.20.254 YES manual up      up
```

```
Switch#show vlan
VLAN Name                Status    Ports
-----
1    default                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   SALES                  active    Gi0/0, Gi0/1
20   MRKT                   active    Gi0/2, Gi0/3
1002 fddi-default           act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
```

ROUTER ON STICK (ROAS)

- ❖ Router On Stick is a method to establish communication between two or more VLAN.
- ❖ Router On Stick configuration is an old method.
- ❖ It requires 2 devices that is Router and L2 Switch

so here we go,



TASK

Task 5.1 Configure VLAN 10 – SALES & VLAN 20 – MRKT

TASK 5.2 Implement VLAN on respected interface

Task 5.3 Make interface Trunk which is connected to Router

Task 5.4 Create a gateway on Router

Task 5.5 Assign IP's to the PC of respected VLAN

SOLUTION

Task 5.1 First we need to Create VLAN

```
SW(config)#vlan 10
```

```
SW(config-vlan)#name SALES
```

```
SW(config-vlan)#exit
```

```
SW(config)#vlan 20
SW(config-vlan)#name MRKT
SW(config-vlan)#exit
```

Task 5.2 We need to Assign VLAN

```
SW(config)#interface range ethernet 0/1-2
SW(config-if-range)#switchport mode access
SW(config-if-range)#switchport access vlan 10
SW(config-if-range)#exit
```

```
SW(config)#interface range ethernet 0/3 , ethernet 1/0
SW(config-if-range)#switchport mode access
SW(config-if-range)#switchport access vlan 20
SW(config-if-range)#exit
```

Task 5.3 Now we have to create trunk

```
SW(config)#interface ethernet 0/0
SW(config-if)#switchport trunk encapsulation dot1q
SW(config-if)#switchport mode trunk
SW(config-if)#exit
```

Task 5.4 Now, configure router

```
R1(config)#interface ethernet0/0
R1(config-if)#no shutdown
R1(config-if)#exit
```

```
R1(config)#interface ethernet0/0.10
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 192.168.10.254 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
```

```
R1(config)#interface ethernet0/0.20
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 192.168.20.254 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
```

VERIFICATION

On Router

```
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    unassigned      YES unset      up          up
Ethernet0/0.10 192.168.10.254  YES manual    up          up
Ethernet0/0.20 192.168.20.254  YES manual    up          up
Ethernet0/1     unassigned      YES unset      administratively down down
Ethernet0/2     unassigned      YES unset      administratively down down
Ethernet0/3     unassigned      YES unset      administratively down down
```

```
R1#show running-config | section interface Ethernet0/0
interface Ethernet0/0
  no ip address
interface Ethernet0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.254 255.255.255.0
interface Ethernet0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.254 255.255.255.0
```

On Switch

```
SW#show vlan
VLAN Name                Status    Ports
----
1    default             active    Et1/1, Et1/2, Et1/3
10   SALES                active    Et0/1, Et0/2
20   MRKT                 active    Et0/3, Et1/0
1002 fddi-default         act/unsup
1003 token-ring-default  act/unsup
1004 fddinet-default     act/unsup
1005 trnet-default       act/unsup
```

```
SW# show interface trunk

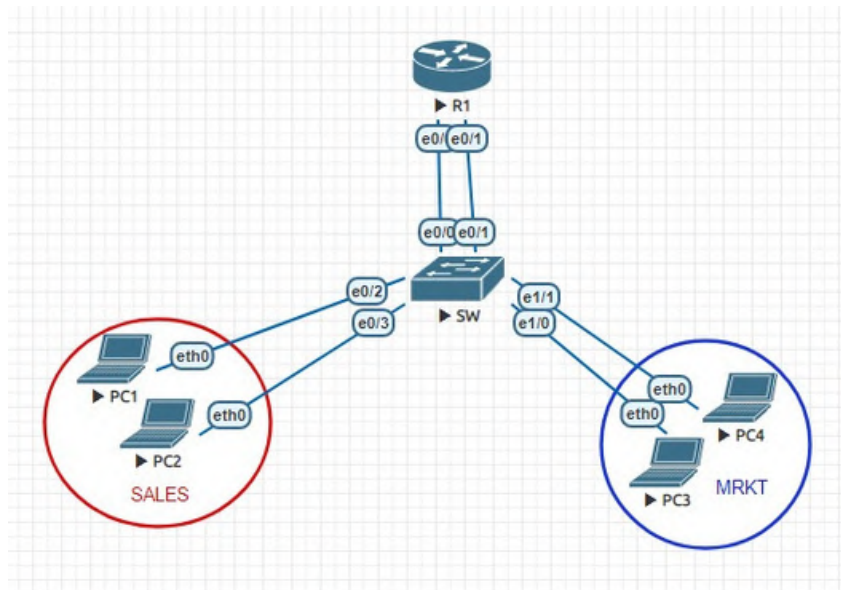
Port      Mode          Encapsulation  Status      Native vlan
Et0/0     on            802.1q         trunking    1

Port      Vlans allowed on trunk
Et0/0     1-4094

Port      Vlans allowed and active in management domain
Et0/0     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Et0/0     1,10,20
```

METHOD 2



Creation of VLAN

```
SW(config)#vlan 10
```

```
SW(config-vlan)#name SALES
```

```
SW(config-vlan)#exit
```

```
SW(config)#vlan 20
```

```
SW(config-vlan)#name MRKT
```

```
SW(config-vlan)#exit
```

VLAN Assignment

```
SW(config)#interface range ethernet 0/2-3
```

```
SW(config-if-range)#switchport mode access
```

```
SW(config-if-range)#switchport access vlan 10
```

```
SW(config-if-range)#exit
```

```
SW(config)#interface range ethernet 1/0-1
```

```
SW(config-if-range)#switchport mode access
```

```
SW(config-if-range)#switchport access vlan 20
```

```
SW(config-if-range)#exit
```

Configure switch interface which is connected to the router

```
SW(config)#interface ethernet 0/0
```

```
SW(config-if)#switchport mode access
```

```
SW(config-if)#switchport access vlan 10
```

```
SW(config-if)#exit
```

```
SW(config)#interface ethernet 0/1
```

```
SW(config-if)#switchport mode access
SW(config-if)#switchport access vlan 20
SW(config-if)#exit
```

Now, configure router

```
R1(config)#interface ethernet 0/0
R1(config-if)#ip address 192.168.10.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface ethernet 0/1
R1(config-if)#ip address 192.168.20.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
```

NOTE:-In this method we are not configuring only one interface for multiple VLAN rather we are configuring multiple interfaces for multiple VLAN .that's why here we are not configuring trunk interface.

VERIFICATION

On Switch

```
SW#show vlan
```

VLAN	Name	Status	Ports
1	default	active	Et1/2, Et1/3
10	sales	active	Et0/0, Et0/2, Et0/3
20	MRKT	active	Et0/1, Et1/0, Et1/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

On Router

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	192.168.10.254	YES	manual	up	up
Ethernet0/1	192.168.20.254	YES	manual	up	up
Ethernet0/2	unassigned	YES	unset	administratively down	down
Ethernet0/3	unassigned	YES	unset	administratively down	down

```
R1#show running-config | section interface Ethernet
interface Ethernet0/0
 ip address 192.168.10.254 255.255.255.0
interface Ethernet0/1
 ip address 192.168.20.254 255.255.255.0
interface Ethernet0/2
 no ip address
 shutdown
interface Ethernet0/3
 no ip address
 shutdown
```

SPANNING TREE PROTOCOL (STP)

Spanning tree is a loop prevention mechanism, generally used to stop broadcast storm in a Switch based network topology. It uses election method to select a **ROOT BRIDGE** in a switch network to have control over a switch network. If switch will be able to choose a root bridge, then all the traffic passes through that root bridge where, it uses

STEP-1 SELECTION OF ROOT BRIDGE

1. lowest priority

- ❖ Default priority of every Cisco switch is 32,768
- ❖ Then it adds up a VLAN ID so default VLAN is 1, hence ultimate value becomes 32,769 (32,768+1)
- ❖ So, the default priority of every switch is same hence it ties up this criteria, then the next criteria is

2. lowest MAC address

The first question is how to calculate lowest MAC address

Here it is, MAC address is Hexadecimal value consist of numbers and alphabets

0 1 2 3 4 5 6 7 8 9 A B C D E F

HERE **0** IS **LOWEST**, THEN MOVE IT FORWARD TO **F** IS **HIGHEST**

NOTE:- IF LOWEST PRIORITY HIT THEN ROOT BRIDGE SELECTED ON THE BASIS OF LOWEST PRIORITY, IF THAT CRITERIA TIES UP THEN MOVE FORWARD TO LOWEST MAC ADDRESS.

STEP-2 SELECTION OF ROOT PORT

Root port gets selected on the basis of port cost which is already assigned on interfaces

Default Port Cost of interface

INTERFACE TYPE	PORT COST
ETHERNET (10MBPS)	100
FAST ETHERNET (100 MBPS)	19
1 GIGABIT ETHERNET (1 GBPS)	4
10 GIGABIT ETHERNET (10 GBPS)	2

Root port defines the nearest path to reach the destination

So, the rule is

Root port = designated port [RP=DP]

Designated port = root port or block port [DP=RP / DP=BP]

Root bridge has each port as designated port, and ports on other switches can be root port or block port

STEP -3 SELECTION OF BLOCK PORT

In the process of selection of block port switch uses a lowest priority and lowest MAC address criteria again, the same criteria applied here as it is. If the lowest priority is equal, then it move toward the lowest mac address.

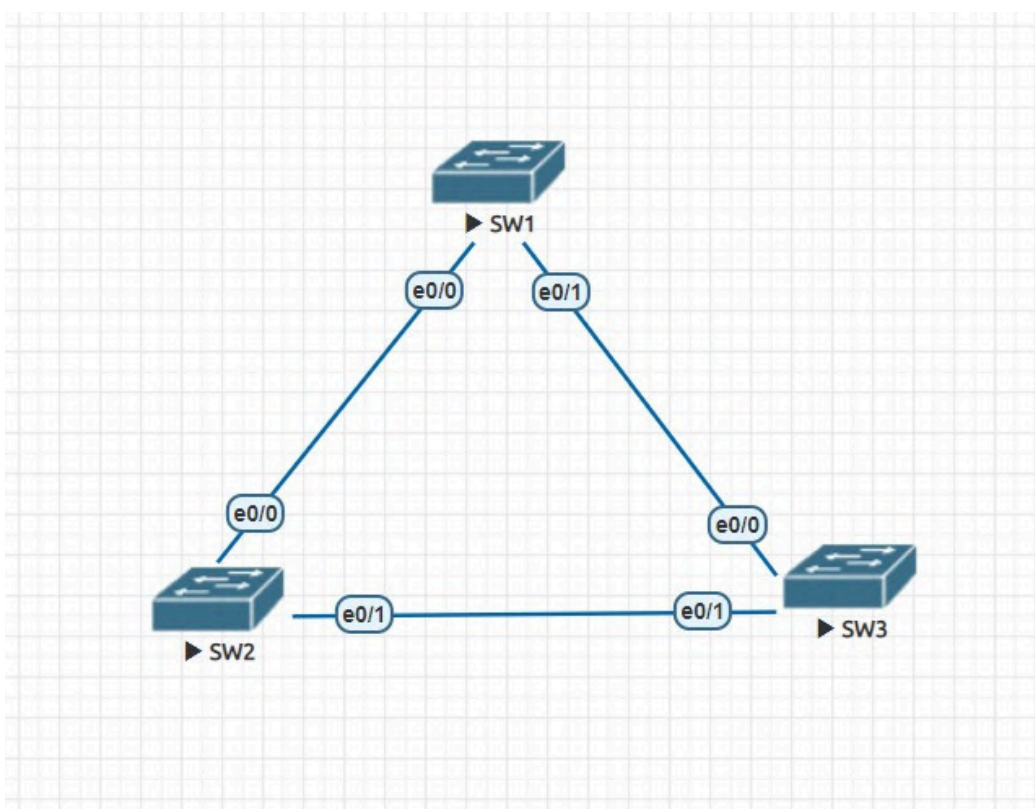
TASK 6

Task 6.1 Verify the root bridge

Task 6.2 Change the root bridge

Task 6.3 Change the port cost

So, let's start



In this picture

Sw-1 Will become a root bridge

```
SW-1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     aabb.cc00.1000
             This bridge is the root
             Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.1000
             Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Desg FWD 100       128.1   Shr
Et0/1                    Desg FWD 100       128.2   Shr
Et0/2                    Desg FWD 100       128.3   Shr
Et0/3                    Desg FWD 100       128.4   Shr
```

Now see switch 1 has all port ad designated port

Now, we are going to change root bridge

We going to plan switch 2 must be root bridge and if switch 2 fails in network switch 3 will become a root bridge

```
SW-2#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     aabb.cc00.1000
             Cost         100
             Port         1 (Ethernet0/0)
             Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.2000
             Hello Time  2 sec    Max Age 20 sec    Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Root FWD 100       128.1   Shr
Et0/1                    Desg FWD 100       128.2   Shr
Et0/2                    Desg FWD 100       128.3   Shr
Et0/3                    Desg FWD 100       128.4   Shr
```

In this switch 2 configuration we can clearly see eth 0/0 selected as root port cause the cost to reach the root bridge is only 100. if switch chooses other route it will take 200 cost, to reach the destination. (from switch 2 > switch 3 > switch 1 = 200 cost)

Same goes with switch 3

```
SW-3#show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     aabb.cc00.1000
             Cost        100
             Port        1 (Ethernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.3000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Root FWD 100       128.1   Shr
Et0/1                    Altn BLK 100       128.2   Shr
Et0/2                    Desg FWD 100       128.3   Shr
Et0/3                    Desg FWD 100       128.4   Shr
```

In this switch eth0/0 will become a root bridge as the cost is 100, whereas if switch chooses a different path to reach the destination it will take more cost than usual (from switch3 > switch 2 > switch 1 = 200 cost)

Now what if we dedicatedly want to change the root bridge role from switch 1 to switch 2

So, we play with switch priority first because we won't change the mac address of the switches.

SW-2(config)#spanning-tree vlan 1 root primary

We make switch 2 as root bridge by giving above command

```
SW-2(config)#do show spanning-tree
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    24577
             Address     aabb.cc00.2000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
             Address     aabb.cc00.2000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Et0/0                    Desg FWD 100       128.1   Shr
Et0/1                    Desg FWD 100       128.2   Shr
Et0/2                    Desg FWD 100       128.3   Shr
Et0/3                    Desg FWD 100       128.4   Shr
```

As a result of that command switch 2 act as a root switch

Now,

SW-3(config)#spanning-tree vlan 1 root secondary

By giving this above command we make switch 3 as a secondary root switch means if switch 2 fails as a root bridge. Then, switch 3 will become a root bridge.

Now how to change path cost

```
SW-3(config)#do show spanning-tree
```

VLAN0001

Spanning tree enabled	protocol rstp
Root ID	Priority 24577
	Address aabb.cc00.2000
	Cost 100
	Port 2 (Ethernet0/1)
	Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID	Priority 28673 (priority 28672 sys-id-ext 1)
	Address aabb.cc00.3000
	Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
	Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	FWD	100	128.1	Shr
Et0/1	Root	FWD	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr
Et0/3	Desg	FWD	100	128.4	Shr

See, switch 3 has root port as eth 0/1 after changing the path cost. we make other port as a root port

SW-3(config)#interface ethernet 0/0

SW-3(config-if)#spanning-tree cost 10

SW-3(config-if)#exit

By applying such command on switch 3 ethernet 0/0, switch 1 – ethernet 0/0 & 0/1, switch 2 ethernet 0/0 we create root port on switch 3 from 0/1 to 0/0

```
SW-3(config)#do show spanning-tree
```

VLAN0001

Spanning tree enabled	protocol rstp
Root ID	Priority 24577
	Address aabb.cc00.2000
	Cost 20
	Port 1 (Ethernet0/0)
	Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID	Priority 28673 (priority 28672 sys-id-ext 1)
	Address aabb.cc00.3000
	Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
	Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Root	FWD	10	128.1	Shr
Et0/1	Altn	BLK	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr
Et0/3	Desg	FWD	100	128.4	Shr

ROUTING

Router is a device which we use for the purpose of routing. Now, there is a question what is routing. routing means to route a data packet between networks .in layman language routing means to show the path to the data packet.

here we go to the routing

routing has generally major two type

1. static routing
2. dynamic routing

STATIC ROUTING – In this method we as a network engineer, have to define each and every route to the destination on every router till the destination.

DYNAMIC ROUTING – In this method we as a network engineer, have to only advertise the network which is directly connected to router. Dynamic Routing has different types of protocol, some of them are RIPv1 , RIPv2 , EIGRP , OSPF , ETC

STATIC ROUTING

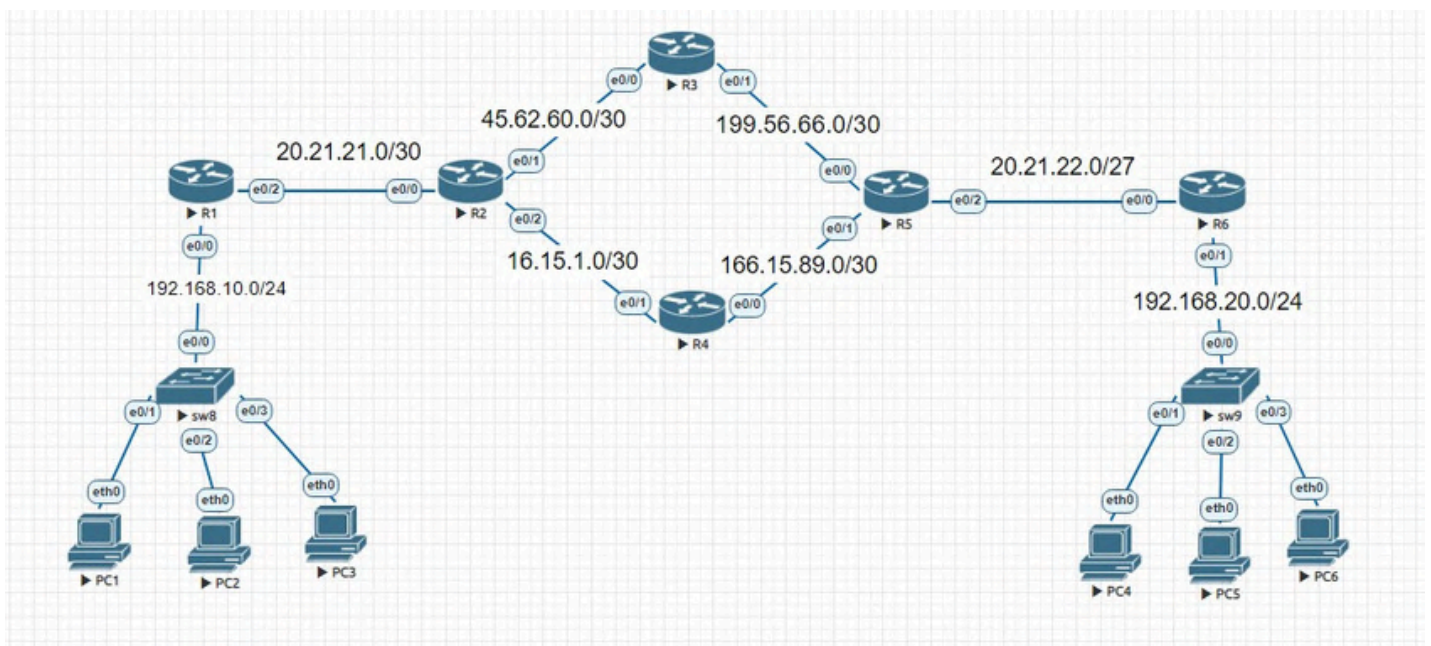
~~LET'S START WITH THE STATIC ROUTING FIRST~~

TASK-7

Task 7.1 provide hostname accordingly

Task 7.2 Assign IP address to interface

Task 7.3 Do static routing, PC-1,2,3 should communicate with PC-4,5,6



IN THE ABOVE LAB CONFIGURE THE GIVEN TASK

R1

```
Router(config)#hostname R1
R1(config)#interface ethernet 0/0
R1(config-if)#ip address 192.168.10.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface ethernet 0/2
R1(config-if)#ip address 20.21.21.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
```

R2

```
Router(config)#hostname R2
R2(config)#interface ethernet 0/0
R2(config-if)#ip address 20.21.21.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#interface ethernet 0/1
R2(config-if)#ip address 45.62.60.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface ethernet 0/2
R2(config-if)#ip address 16.15.1.1 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
```

R3

```
Router(config)#hostname R3
R3(config)#interface ethernet 0/0
R3(config-if)#ip address 45.62.60.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface ethernet 0/1
R3(config-if)#ip address 199.56.66.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
```

R4

```
Router(config)#hostname R4
R4(config)#interface ethernet 0/1
R4(config-if)#ip address 16.15.1.2 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#interface ethernet 0/0
R4(config-if)#ip address 166.15.89.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
```

R5

```
Router(config)#hostname R5
R5(config)#interface ethernet 0/0
R5(config-if)#ip address 199.56.66.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#interface ethernet 0/1
R5(config-if)#ip address 166.15.89.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#interface ethernet 0/2
R5(config-if)#ip address 20.21.22.1 255.255.255.224
R5(config-if)#no shutdown
R5(config-if)#exit
```

R6

```
Router(config)#hostname R6
R6(config)#interface ethernet 0/0
R6(config-if)#ip address 20.21.22.2 255.255.255.224
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#interface ethernet 0/1
R6(config-if)#ip address 192.168.20.254 255.255.255.0
R6(config-if)#no shutdown
R6(config-if)#exit
```

Now, we have done with initial configuration part that is IP'S to the interface and hostname to router.

Well further we have to do is static routing. Let's start with it.

In static routing we have to define path to reach the destination.

Command for static routing

IP ROUTE <....Dest Network Address.....> <.....Dest Subnet Mask....> <Next hop ip address>

```
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.21.21.0/30 is directly connected, Ethernet0/2
L       20.21.21.1/32 is directly connected, Ethernet0/2
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, Ethernet0/0
L       192.168.10.254/32 is directly connected, Ethernet0/0
```

Before starting with routing there is no route to the destination of 192.168.20.0 network. so, we have to provide path for that network.

Start with R1,

R1(config)#ip route 192.168.20.0 255.255.255.0 20.21.21.2

Destination network address- 192.168.20.0

Destination subnet mask – 255.255.255.0

Next hop ip address – 20.21.21.2

After static routing command, there is an additional route start with "S"

```
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.21.21.0/30 is directly connected, Ethernet0/2
L       20.21.21.1/32 is directly connected, Ethernet0/2
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, Ethernet0/0
L       192.168.10.254/32 is directly connected, Ethernet0/0
S       192.168.20.0/24 [1/0] via 20.21.21.2
```

Same thing we have to do with other routers, each router require path to reach the destination.

R2

R2(config)#ip route 192.168.20.0 255.255.255.0 45.62.60.2

R3

```
R3(config)#ip route 192.168.20.0 255.255.255.0 199.56.66.2
```

R5

```
R5(config)#ip route 192.168.20.0 255.255.255.0 20.21.22.2
```

We don't need to configure R6 cause R6 already has route which is directly connected to it. Represented with "C" in routing table.

```
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.21.22.0/27 is directly connected, Ethernet0/0
L    20.21.22.2/32 is directly connected, Ethernet0/0
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, Ethernet0/1
L    192.168.20.254/32 is directly connected, Ethernet0/1
```

When data came in the network of 192.168.20.0/24 then the PC gives backecho-reply to the source device. Because we were trying to ping that network PC.

So, in this case source device is 192.168.10.1 in the network of 192.168.10.0/24, and to give back a reply we don't have return route to the 192.168.10.0/24 network. Now we need to provide path to the destination network of 192.168.10.0/24.

R6

```
R6(config)#ip route 192.168.10.0 255.255.255.0 20.21.22.1
```

R5

```
R5(config)#ip route 192.168.10.0 255.255.255.0 166.15.89.1
```

R4

```
R4(config)#ip route 192.168.10.0 255.255.255.0 16.15.1.1
```

R2

```
R2(config)#ip route 192.168.10.0 255.255.255.0 20.21.21.1
```

Now we don't need to provide path to the R1 for 192.168.10.0 network because that network is directly connected to R1.

Now provide IP to pc and verify it by doing ping

```
VPCS> ip 192.168.10.1/24 192.168.10.254
Checking for duplicate address...
PC1 : 192.168.10.1 255.255.255.0 gateway 192.168.10.254

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=59 time=4.758 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=59 time=3.515 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=59 time=3.374 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=59 time=2.740 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=59 time=3.145 ms
```

Done with the static routing.

DEFAULT ROUTING

we have done static routing here. now there is a concept called default routing which is a part of static routing. Default routing basically forwarding data to some given IP and then that data can be forwarded by the device which is holding a given IP with the help of their routing table. when we do default routing on edge router we only create one default entry rest of the work is done by other router. we use default routing entry when we connect router to internet. And on internet there is a lot of networks which we can't define so there we use default entry to push all the data packets to particular IP which is belongs to ISP and then ISP do the main routing stuff.

Considering the static routing lab, where R1 is companies edge router which is connected with multiple networks.

R1

```
R1(config)#ip route 0.0.0.0 0.0.0.0 20.21.21.2
```

Destination network address- 0.0.0.0 (UNKNOWN)

Destination subnet mask – 0.0.0.0 (UNKNOWN)

Next hop ip address – 20.21.21.2

Here we are telling the router that we don't know the path to reach the destination but we are forwarding that data on IP 20.21.21.2. So, 20.21.21.2 can forward the data with their own routing table. Rest configuration is same.

ROUTING INFORMATION PROTOCOL (RIP)

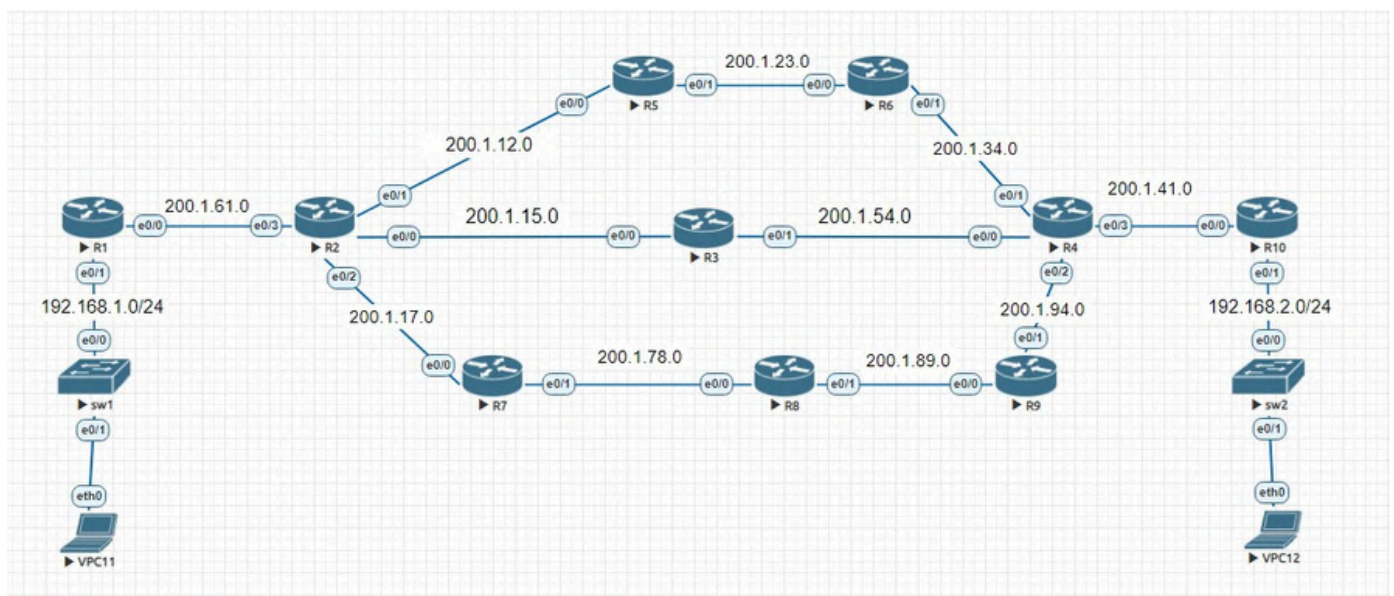
- ❖ Rip is a dynamic routing protocol
- ❖ Rip uses Bellman Ford Algorithm
- ❖ Rip supports max 15 hop count
- ❖ Rip has AD value of 120
- ❖ Rip has a periodic update timer of 30 sec
- ❖ Rip has flush timer of 240 sec out of which 180 sec is a hold down timer & 60 sec is path update timer.
- ❖ Rip has two version- version 1(RIPv1) & version 2(RIPv2)
- ❖ RIPv1 uses broadcast address to send messages, rIPv1 do not support authentication
- ❖ RIPv2 has multicast address – 224.0.0.9, rIPv2 support authentication.

TASK 8

Task 8.1 Assign hostname to the devices

Task 8.2 IP address to interfaces & pc's

Task 8.3 Configure Routing



FROM THE ABOVE GIVEN LAB ASSIGN IP ADDRESS ACCORDINGLY

R1-R2	200.1.61.0/24	R2-R7	200.1.17.0/24
R2-R3	200.1.15.0/24	R7-R8	200.1.78.0/24
R3-R4	200.1.54.0/24	R8-R9	200.1.89.0/24
R4-R10	200.1.41.0/24	R9-R4	200.1.94.0/24
R2-R5	200.1.12.0/24	R1 LAN	192.168.1.0/24
R5-R6	200.1.23.0/24	R10 LAN	192.168.2.0/24
R6-R4	200.1.34.0/24	--	--

After Assigning Ip on The Interfaces, Let's Move Further to The Routing

R1 R1(config)#router rip

R1(config-router)#network 192.168.1.0

R1(config-router)#network 200.1.61.0

R1(config-router)#exit

Here, we only have to advertise the directly connected network on router.

R2

R2(config)#router rip R2(config-router)#network 200.1.12.0 R2(config-

router)#network 200.1.15.0 R2(config-router)#network 200.1.17.0

R2(config-router)#network 200.1.61.0 R2(config-router)#exit Now, we go

through R7 to verify how rip works

R7

R7(config)#router rip

R7(config-router)#network 200.1.17.0

R7(config-router)#network 200.1.78.0

R7(config-router)#exit

R8

R8(config)#router rip

R8(config-router)#network 200.1.78.0

R8(config-router)#network 200.1.89.0

R8(config-router)#exit

R9

R9(config)#router rip

R9(config-router)#network 200.1.89.0

R9(config-router)#network 200.1.94.0 R9(config-router)#exit

R4

R4(config)#router rip R4(config-router)#network 200.1.34.0 R4(config-

router)#network 200.1.54.0

R4(config-router)#network 200.1.94.0

—

```
R4(config-router)#network 200.1.41.0
```

```
R4(config-router)#exit
```

R10

```
R10(config)#router rip
```

```
R10(config-router)#network 200.1.41.0
```

```
R10(config-router)#network 192.168.2.0
```

```
R10(config-router)#exit
```

Lets try to do ping from pc1 – pc2

```
pc1> ping 192.168.2.1
```

```
84 bytes from 192.168.2.1 icmp_seq=1 ttl=58 time=5.210 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=58 time=3.513 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=58 time=3.774 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=58 time=3.974 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=58 time=3.007 ms
```

Now try to do tracing

```
pc1> trace 192.168.2.1
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1  192.168.1.254    1.737 ms  2.019 ms  3.228 ms
 2  200.1.61.2      1.596 ms  1.690 ms  2.507 ms
 3  200.1.17.2      2.792 ms  4.085 ms  2.148 ms
 4  200.1.78.2      1.657 ms  1.949 ms  3.443 ms
 5  200.1.89.2      3.711 ms  2.698 ms  1.550 ms
 6  200.1.94.2      1.866 ms  1.955 ms  4.700 ms
 7  200.1.41.2      4.191 ms  2.329 ms  1.805 ms
 8  *192.168.2.1    3.163 ms (ICMP type:3, code:3, Destination port unreachable)
```

See in the above trace, router chooses a route of R1--R2--R7--R8--R9--R4--R10

Lets go to the R2 to check the route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

R    192.168.1.0/24 [120/1] via 200.1.61.1, 00:00:18, Ethernet0/3
R    192.168.2.0/24 [120/5] via 200.1.17.2, 00:00:07, Ethernet0/2
     200.1.12.0/24 is variably subnetted, 2 subnets, 2 masks
```

Router R2 chooses 200.1.17.2 path to reach the destination of 192.168.2.0 network

NOW CONFIGURE ROUTE OF R1—R2—R5—R6—R4—R10 R1—R2—R4—R10 were

already configured. So, we have to configure the rest R5

```
R5(config)#router rip R5(config-router)#network 200.1.12.0
```

```
R5(config-router)#network 200.1.23.0
```

```
R5(config-router)#exit
```

R6

```
R6(config)#router rip
```

```
R6(config-router)#network 200.1.23.0
```

```
R6(config-router)#network 200.1.34.0 R6(config-router)#exit
```

NOW GO TO R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

R      192.168.1.0/24 [120/1] via 200.1.61.1, 00:00:23, Ethernet0/3
R      192.168.2.0/24 [120/4] via 200.1.12.2, 00:00:14, Ethernet0/1
      200.1.12.0/24 is variably subnetted, 2 subnets, 2 masks
```

Now see R2 chooses 200.1.12.2 path to reach the destination of 192.168.2.0 network

pc1 tracing path to reach 192.168.2.1

```
pc1> trace 192.168.2.1
Trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1  192.168.1.254    0.765 ms  1.730 ms  0.953 ms
 2  200.1.61.2      1.966 ms  1.198 ms  1.947 ms
 3  200.1.12.2      3.509 ms  3.366 ms  2.216 ms
 4  200.1.23.2      2.434 ms  2.088 ms  1.855 ms
 5  200.1.34.2      4.099 ms  4.993 ms  2.753 ms
 6  200.1.41.2      2.610 ms  1.971 ms  3.238 ms
 7  *192.168.2.1    9.625 ms (ICMP type:3, code:3, Destination port unreachable)
```

Now configure the path of R1—R2—R3—R4—R10

R3

R3(config)#router rip R3(config-router)#network

200.1.15.0 R3(config-router)#network 200.1.54.0

R3(config-router)#exit

router R2 for routes update

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

R      192.168.1.0/24 [120/1] via 200.1.61.1, 00:00:23, Ethernet0/3
R      192.168.2.0/24 [120/3] via 200.1.15.2, 00:00:17, Ethernet0/0
      200.1.12.0/24 is variably subnetted, 2 subnets, 2 masks
```

Now let's look at pc1 tracing

```
pc1> trace 192.168.2.1
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1  192.168.1.254    0.731 ms  0.963 ms  0.646 ms
 2  200.1.61.2      2.170 ms  2.511 ms  1.038 ms
 3  200.1.15.2      3.672 ms  1.400 ms  2.135 ms
 4  200.1.54.2      1.572 ms  2.741 ms  3.123 ms
 5  200.1.41.2      6.363 ms  1.894 ms  1.711 ms
 6  *192.168.2.1    7.881 ms (ICMP type:3, code:3, Destination port unreachable)
```

So, here we are with the conclusion of RIP chooses lowest hop count path to reach the destination.

ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL(EIGRP)

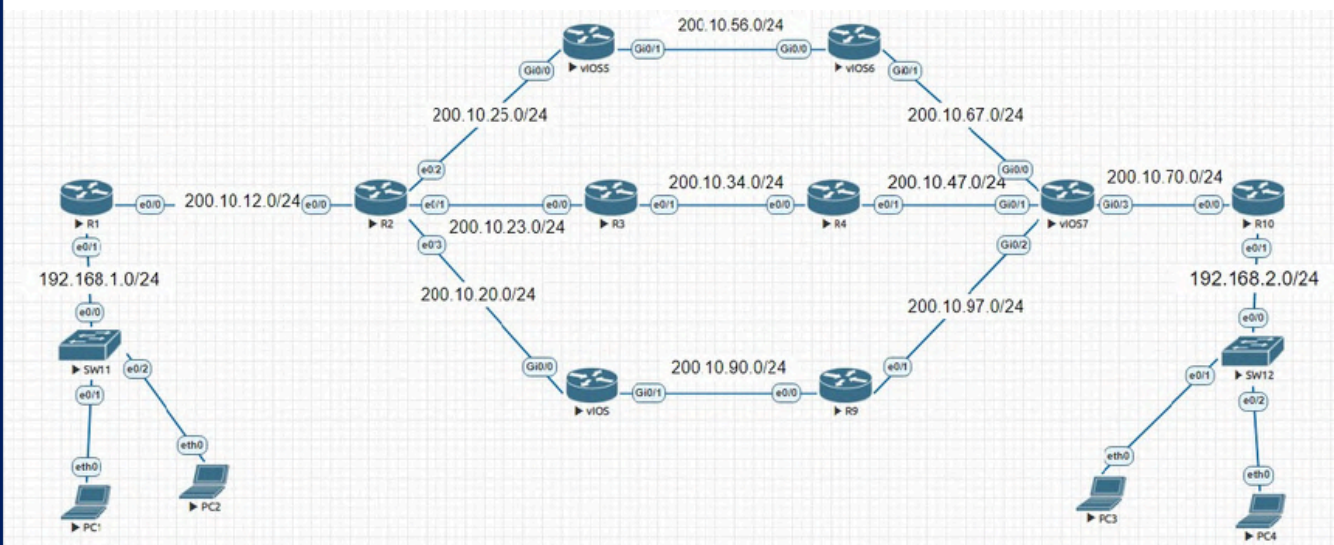
- ❖ Eigrp isa Dynamic routing protocol. Under thisit is a Hybrid Protocol.
- ❖ Eigrp uses DUALALGORITHEM (Diffusing Update Algorithm)
- ❖ Eigrp support classless IP addressing, also support Authentication.
- ❖ Eigrp has AD value of 90
- ❖ Eigrp has hello timer of 5 sec and hold down timer is 15 sec. Eigrp do incremental update.
- ❖ Eigrp has Multicast address 224.0.0.10
- ❖ Eigrp has default hop count is 100. We can increase the limit till 255 hops.
- ❖ Eigrp uses Metric : Bandwidth + Delay (+MTU+Reliability+load)
- ❖ Eigrp make three table :- 1.Neighbor table 2.Topology table 3.Routing table

TASK9

Task9.1 Assign ip to the interfaces

Task9.2 Assign hostname on devices

Task9.3 Configure EIGRP routing



Assign ip and hostname to the devices Now lets move to the Routing part

Firstly we choose middle part to route data (R1—R2—R3—R4—vIOS7—R10)

R1

```
R1(config)#router eigrp 100 R1(config-router)#network 192.168.1.0
0.0.0.255 R1(config-router)#network 200.10.12.0 0.0.0.255 R1(config-
router)#exit
```

R2 R2(config)#router eigrp 100

R2(config-router)#network 200.10.12.0 0.0.0.255

R2(config-router)#network 200.10.25.0 0.0.0.255

R2(config-router)#network 200.10.23.0 0.0.0.255

R2(config-router)#network 200.10.20.0 0.0.0.255

R2(config-router)#exit

R3

R3(config)#router eigrp 100

R3(config-router)#network 200.10.23.0 0.0.0.255

R3(config-router)#network 200.10.34.0 0.0.0.255

R3(config-router)#exit

R4 R4(config)#router eigrp 100

R4(config-router)#network 200.10.34.0 0.0.0.255

R4(config-router)#network 200.10.47.0 0.0.0.255

R4(config-router)#exit

vIOS7

vIOS7(config)#router eigrp 100

vIOS7(config-router)#network 200.10.47.0 0.0.0.255

~~vIOS7~~(config-router)#network 200.10.67.0 0.0.0.255

vIOS7(config-router)#network 200.10.97.0 0.0.0.255

vIOS7(config-router)#network 200.10.70.0 0.0.0.255

vIOS7(config-router)#exit **R10** R10(config)#router

eigrp 100

R10(config-router)#network 200.10.70.2 0.0.0.255

R10(config-router)#network 192.168.2.0 0.0.0.255

R10(config-router)#network 192.168.2.0 0.0.0.255

R10(config-router)#exit

Now lets ping from PC1 – PC3

```
PC1> ping 192.168.2.1
```

```
84 bytes from 192.168.2.1 icmp_seq=1 ttl=58 time=33.137 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=58 time=6.274 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=58 time=16.597 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=58 time=14.116 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=58 time=12.713 ms
```

Now trace for path

```
PC1> trace 192.168.2.1
```

```
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1  192.168.1.254    2.678 ms  5.613 ms  0.790 ms
 2  200.10.12.2     4.754 ms  2.201 ms  1.600 ms
 3  200.10.23.2     2.589 ms  1.996 ms  2.432 ms
 4  200.10.34.2     3.744 ms  5.336 ms  2.788 ms
 5  200.10.47.2    10.887 ms  8.903 ms  8.951 ms
 6  200.10.70.2     5.661 ms  3.093 ms  8.780 ms
 7  *192.168.2.1    8.572 ms  (ICMP type:3, code:3, Destination
port unreachable)
```

Now we have to configure new path R1—R2—vIOS—R9—vIOS7—R10

vIOS

```
vIOS(config)#router eigrp 100
```

```
vIOS(config-router)#network 200.10.90.0 0.0.0.255
```

```
vIOS(config-router)#network 200.10.20.0 0.0.0.255
```

```
vIOS(config-router)#exit
```

R9

```
R9(config)#router eigrp 100 R9(config-
```

```
router)#network 200.10.90.0 0.0.0.255 R9(config-
```

```
router)#network 200.10.97.0 0.0.0.255 R9(config-
router)#exit
```

Now eigrp chooses route of R1—R2—vIOS—R9—vIOS7—R10

trace from PC1 to PC3

```
PC1> trace 192.168.2.1
```

```
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop
 1  192.168.1.254    1.053 ms  0.569 ms  0.585 ms
 2  200.10.12.2     0.993 ms  0.797 ms  0.862 ms
 3  200.10.20.2    15.397 ms  6.707 ms  4.480 ms
 4  200.10.90.2     5.058 ms  8.661 ms  5.224 ms
 5  200.10.97.2    32.548 ms 12.786 ms 12.535 ms
 6  200.10.70.2    14.830 ms 15.371 ms 12.013 ms
 7  *192.168.2.1   13.717 ms (ICMP type:3, code:3, Destination port unreachable)
```

Now configure upper route

vIOS5

```
vIOS5(config)#router eigrp 100 vIOS5(config-  
router)#network 200.10.25.0 0.0.0.255 vIOS5(config-  
router)#network 200.10.56.0 0.0.0.255 vIOS5(config-  
router)#exit
```

vIOS6

```
vIOS6(config)#router eigrp 100  
vIOS6(config-router)#network 200.10.56.0 0.0.0.255  
vIOS6(config-router)#network 200.10.67.0 0.0.0.255  
vIOS6(config-router)#exit
```

Now router R2 chooses a upper path

```
R2(config)#do show ip route  
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override  
  
Gateway of last resort is not set  
  
D 192.168.1.0/24 [90/307200] via 200.10.12.1, 00:00:09, Ethernet0/0  
D 192.168.2.0/24 [90/307968] via 200.10.25.2, 00:00:14, Ethernet0/2  
200.10.12.0/24 is variably subnetted, 2 subnets, 2 masks
```

Now let's trace from PC1 to PC3

```
PC1> trace 192.168.2.1  
trace to 192.168.2.1, 8 hops max, press Ctrl+C to stop  
 1 192.168.1.254 0.823 ms 0.648 ms 0.686 ms  
 2 200.10.12.2 0.976 ms 1.823 ms 1.363 ms  
 3 200.10.25.2 19.500 ms 4.837 ms 10.388 ms  
 4 200.10.56.2 15.936 ms 8.064 ms 13.643 ms  
 5 200.10.67.2 20.538 ms 18.308 ms 22.251 ms  
 6 200.10.70.2 20.940 ms 17.046 ms 13.785 ms  
 7 *192.168.2.1 20.780 ms (ICMP type:3, code:3, Destination port unreachable)
```

OPEN SHORTEST PATH FIRST (OSPF)

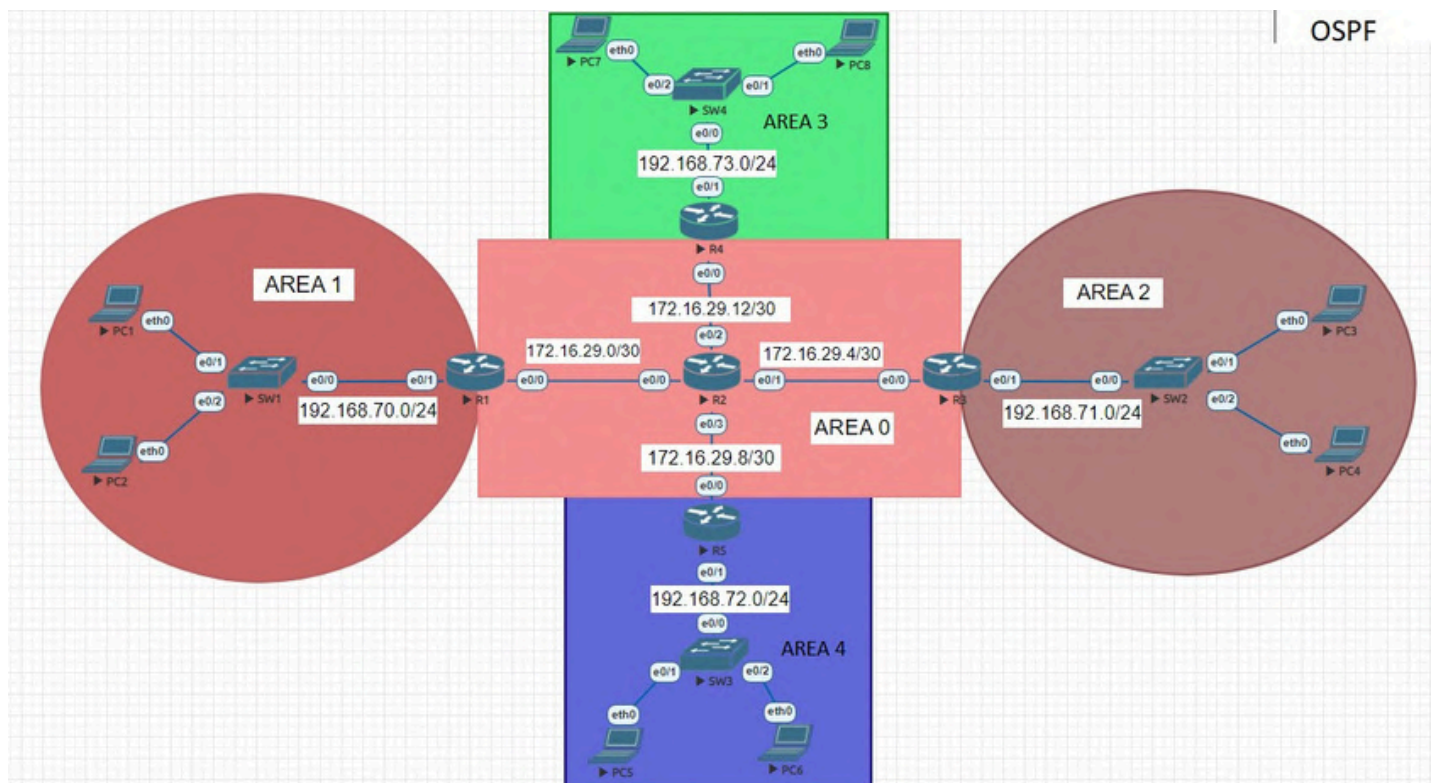
- ❖ OSPF is dynamic routing protocol, under this is a link state protocol.
- ❖ OSPF uses DIJKSTRA ALGORITHM or SPF (SHORTEST PATH FIRST) ALGORITHM
- ❖ OSPF has AD value of 110
- ❖ OSPF uses multicast address 224.0.0.5 and 224.0.0.6
- ❖ OSPF supports classless routing and also supports authentication
- ❖ OSPF creates 3 Routing Table
 1. Neighbor table
 2. Database table
 3. Routing table
- ❖ OSPF do trigger update (incremental update + every 30 min whole routing table update)
- ❖ OSPF uses lowest bandwidth as a cost to reach the destination

TASK 10

Task 10.1 Assign IP to the interfaces

Task 10.2 Assign hostname on devices

Task 10.3 Configure OSPF routing



WE HAVE ALREADY GIVEN AN IP ADDRESS AND HOSTNAME TO THE DEVICES

R1

```
R1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        172.16.29.1     YES manual up          up
Ethernet0/1        192.168.70.254  YES manual up          up
Ethernet0/2        unassigned      YES unset  administratively down down
Ethernet0/3        unassigned      YES unset  administratively down down
```

R2

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        172.16.29.2     YES manual up          up
Ethernet0/1        172.16.29.6     YES manual up          up
Ethernet0/2        172.16.29.14    YES manual up          up
Ethernet0/3        172.16.29.10    YES manual up          up
```

R3

```
R3#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        172.16.29.5     YES manual up          up
Ethernet0/1        192.168.71.254  YES manual up          up
Ethernet0/2        unassigned      YES unset  administratively down down
Ethernet0/3        unassigned      YES unset  administratively down down
```

R4

```
R4#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        172.16.29.13    YES manual up          up
Ethernet0/1        192.168.73.254  YES manual up          up
Ethernet0/2        unassigned      YES unset  administratively down down
Ethernet0/3        unassigned      YES unset  administratively down down
```

R5

```
R5#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        172.16.29.9     YES manual up          up
Ethernet0/1        192.168.72.254  YES manual up          up
Ethernet0/2        unassigned      YES unset  administratively down down
Ethernet0/3        unassigned      YES unset  administratively down down
```

We done with the verification of ip and hostname on the devices

Let's move to the OSPF configuration

R1

R1(config)#router ospf 100

R1(config-router)#network 192.168.70.0 0.0.0.255 area 1

R1(config-router)#network 172.16.29.0 0.0.0.3 area 0

R1(config-router)#exit

R2

```
R2(config)#router ospf 100 R2(config-router)#network
172.16.29.0 0.0.0.3 area 0 R2(config-router)#network
172.16.29.4 0.0.0.3 area 0 R2(config-router)#network
172.16.29.8 0.0.0.3 area 0 R2(config-router)#network
172.16.29.12 0.0.0.3 area 0 R2(config-router)#exit
```

R3

```
R3(config)#router ospf 100
R3(config-router)#network 172.16.29.4 0.0.0.3 area 0
R3(config-router)#network 192.168.71.0 0.0.0.255 area 2
R3(config-router)#exit R4
```

```
R4(config)#router ospf 100 R4(config-router)#network
172.16.29.12 0.0.0.3 area 0 R4(config-router)# network
192.168.73.0 0.0.0.255 area 3 R4(config-router)#exit
```

R5

```
R5(config)#router ospf 100
R5(config-router)#network 172.16.29.8 0.0.0.3 area 0
R5(config-router)#network 192.168.72.0 0.0.0.255 area 4
R5(config-router)#exit For VERIFICATION,
```

#show ip route

R1

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C    172.16.29.0/30 is directly connected, Ethernet0/0
L    172.16.29.1/32 is directly connected, Ethernet0/0
O    172.16.29.4/30 [110/20] via 172.16.29.2, 00:11:15, Ethernet0/0
O    172.16.29.8/30 [110/20] via 172.16.29.2, 00:11:15, Ethernet0/0
O    172.16.29.12/30 [110/20] via 172.16.29.2, 00:11:03, Ethernet0/0
192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.70.0/24 is directly connected, Ethernet0/1
L    192.168.70.254/32 is directly connected, Ethernet0/1
O IA 192.168.71.0/24 [110/30] via 172.16.29.2, 00:09:58, Ethernet0/0
O IA 192.168.72.0/24 [110/30] via 172.16.29.2, 00:06:45, Ethernet0/0
```

R2

```
172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C    172.16.29.0/30 is directly connected, Ethernet0/0
L    172.16.29.2/32 is directly connected, Ethernet0/0
C    172.16.29.4/30 is directly connected, Ethernet0/1
L    172.16.29.6/32 is directly connected, Ethernet0/1
C    172.16.29.8/30 is directly connected, Ethernet0/3
L    172.16.29.10/32 is directly connected, Ethernet0/3
C    172.16.29.12/30 is directly connected, Ethernet0/2
L    172.16.29.14/32 is directly connected, Ethernet0/2
O IA 192.168.70.0/24 [110/20] via 172.16.29.1, 00:16:53, Ethernet0/0
O IA 192.168.71.0/24 [110/20] via 172.16.29.5, 00:15:26, Ethernet0/1
O IA 192.168.72.0/24 [110/20] via 172.16.29.9, 00:12:13, Ethernet0/3
```

R3

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O    172.16.29.0/30 [110/20] via 172.16.29.6, 00:17:55, Ethernet0/0
C    172.16.29.4/30 is directly connected, Ethernet0/0
L    172.16.29.5/32 is directly connected, Ethernet0/0
O    172.16.29.8/30 [110/20] via 172.16.29.6, 00:17:55, Ethernet0/0
O    172.16.29.12/30 [110/20] via 172.16.29.6, 00:17:55, Ethernet0/0
O IA 192.168.70.0/24 [110/30] via 172.16.29.6, 00:17:55, Ethernet0/0
192.168.71.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.71.0/24 is directly connected, Ethernet0/1
L    192.168.71.254/32 is directly connected, Ethernet0/1
O IA 192.168.72.0/24 [110/30] via 172.16.29.6, 00:14:31, Ethernet0/0
```

R4

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O    172.16.29.0/30 [110/20] via 172.16.29.14, 00:16:57, Ethernet0/0
O    172.16.29.4/30 [110/20] via 172.16.29.14, 00:16:57, Ethernet0/0
O    172.16.29.8/30 [110/20] via 172.16.29.14, 00:16:57, Ethernet0/0
C    172.16.29.12/30 is directly connected, Ethernet0/0
L    172.16.29.13/32 is directly connected, Ethernet0/0
O IA 192.168.70.0/24 [110/30] via 172.16.29.14, 00:16:57, Ethernet0/0
O IA 192.168.71.0/24 [110/30] via 172.16.29.14, 00:16:57, Ethernet0/0
O IA 192.168.72.0/24 [110/30] via 172.16.29.14, 00:15:39, Ethernet0/0
192.168.73.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.73.0/24 is directly connected, Ethernet0/1
L    192.168.73.254/32 is directly connected, Ethernet0/1
```

R5

```
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O    172.16.29.0/30 [110/20] via 172.16.29.10, 00:16:37, Ethernet0/0
O    172.16.29.4/30 [110/20] via 172.16.29.10, 00:16:37, Ethernet0/0
C    172.16.29.8/30 is directly connected, Ethernet0/0
L    172.16.29.9/32 is directly connected, Ethernet0/0
O    172.16.29.12/30 [110/20] via 172.16.29.10, 00:16:37, Ethernet0/0
O IA 192.168.70.0/24 [110/30] via 172.16.29.10, 00:16:37, Ethernet0/0
O IA 192.168.71.0/24 [110/30] via 172.16.29.10, 00:16:37, Ethernet0/0
192.168.72.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.72.0/24 is directly connected, Ethernet0/1
L    192.168.72.254/32 is directly connected, Ethernet0/1
```

We have verify the routing, now let's try to ping form PC1 to PC4

```
PC1> ping 192.168.71.2

84 bytes from 192.168.71.2 icmp_seq=1 ttl=61 time=5.260 ms
84 bytes from 192.168.71.2 icmp_seq=2 ttl=61 time=1.640 ms
84 bytes from 192.168.71.2 icmp_seq=3 ttl=61 time=2.058 ms
84 bytes from 192.168.71.2 icmp_seq=4 ttl=61 time=2.167 ms
84 bytes from 192.168.71.2 icmp_seq=5 ttl=61 time=2.199 ms
```

From PC1 to PC7

```
PC1> ping 192.168.73.1

84 bytes from 192.168.73.1 icmp_seq=1 ttl=61 time=4.014 ms
84 bytes from 192.168.73.1 icmp_seq=2 ttl=61 time=2.274 ms
84 bytes from 192.168.73.1 icmp_seq=3 ttl=61 time=3.176 ms
84 bytes from 192.168.73.1 icmp_seq=4 ttl=61 time=2.759 ms
84 bytes from 192.168.73.1 icmp_seq=5 ttl=61 time=2.184 ms
```

From PC1 to PC6

```
PC1> ping 192.168.72.2

84 bytes from 192.168.72.2 icmp_seq=1 ttl=61 time=4.491 ms
84 bytes from 192.168.72.2 icmp_seq=2 ttl=61 time=2.397 ms
84 bytes from 192.168.72.2 icmp_seq=3 ttl=61 time=2.732 ms
84 bytes from 192.168.72.2 icmp_seq=4 ttl=61 time=2.613 ms
84 bytes from 192.168.72.2 icmp_seq=5 ttl=61 time=3.566 ms
```

So, here we done with OSPF.

Till the time we have done routing with each single protocol like only lab with RIP/ EIGRP / OSPF individually. What if we want to do routing between multiple protocol like routing between RIP, EIGRP, OSPF at a same time, that's what we called "**REDISTRIBUTION**".

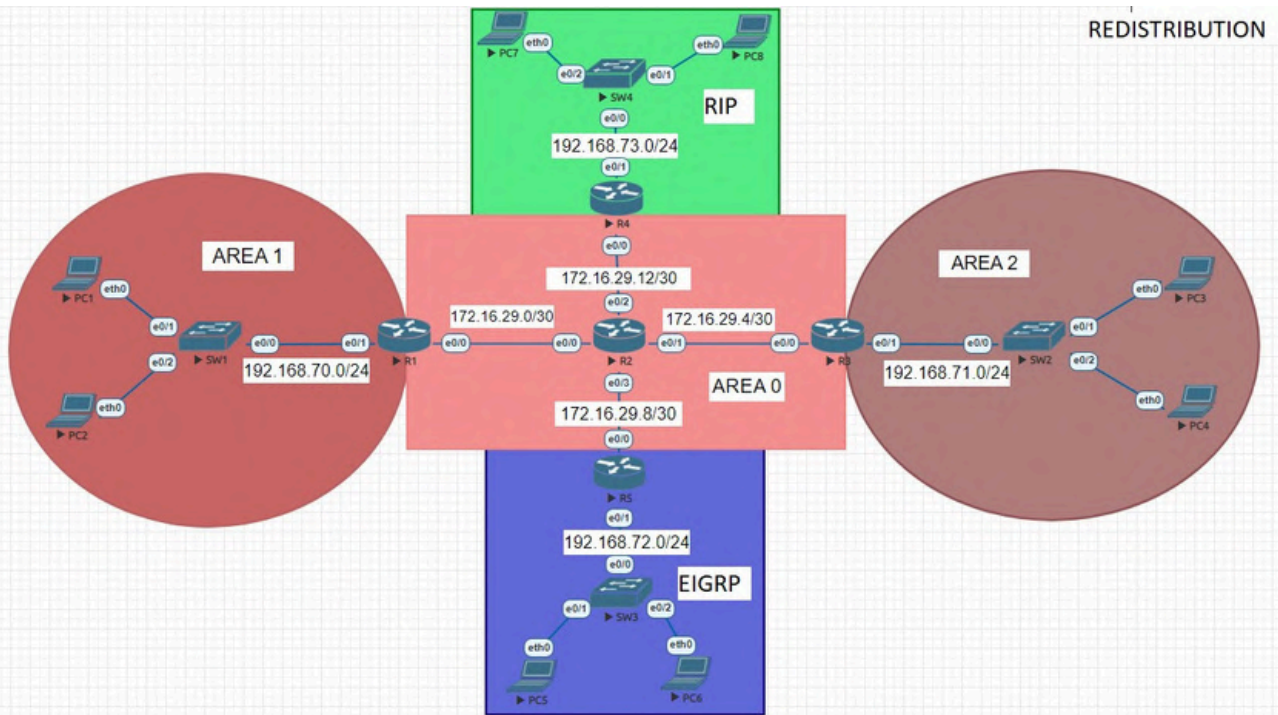
CONSIDER THIS LAB FOR REDISTRIBUTION

WE HAVE MADE SLIGHT CHANGES IN THE OSPF LAB FOR MAKE IT FIT FOR THE PURPOSE OF REDISTRIBUTION

TASK 10

Task 10.4 R4 ROUTER NOW PROCESS TWO PROTOCOL OF OSPF & RIPv2

Task 10.5 R5 ROUTER NOW PROCESS TWO PROTOCOL OF OSPF 100 & EIGRP 90



REST CONFIGURATION IS SAME.EXCEPT R4 and R5

R4

Interface ethernet 0/0 consist network of 172.16.29.12/30 will be advertise in OSPF and Interface ethernet 0/1 consist network of 192.168.73.0/24 will be advertise in RIP .

R4(config)#router ospf 100

R4(config-router)#network 172.16.29.13 0.0.0.3 area 0

R4(config-router)#exit

R4(config)#router rip

R4(config-router)#version 2

R4(config-router)#network 192.168.73.0

R4(config-router)#exit

See there is no route on R2 router for 192.168.73.0/24 network cause R2 is in ospf and 192.168.73.0/24 is in RIP

```

172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C    172.16.29.0/30 is directly connected, Ethernet0/0
L    172.16.29.2/32 is directly connected, Ethernet0/0
C    172.16.29.4/30 is directly connected, Ethernet0/1
L    172.16.29.6/32 is directly connected, Ethernet0/1
C    172.16.29.8/30 is directly connected, Ethernet0/3
L    172.16.29.10/32 is directly connected, Ethernet0/3
C    172.16.29.12/30 is directly connected, Ethernet0/2
L    172.16.29.14/32 is directly connected, Ethernet0/2
O IA 192.168.70.0/24 [110/20] via 172.16.29.1, 01:08:29, Ethernet0/0
O IA 192.168.71.0/24 [110/20] via 172.16.29.5, 01:07:02, Ethernet0/1

```

Let's do it

```
R4(config)#router ospf 100
```

```
R4(config-router)#redistribute rip subnets
```

```
R4(config-router)#exit
```

```
R4(config)#router rip
```

```
R4(config-router)#redistribute ospf 100 metric 1
```

```
R4(config-router)#exit
```

Now see R2 will have 192.168.73.0/24 network in his routing table

```
172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C    172.16.29.0/30 is directly connected, Ethernet0/0
L    172.16.29.2/32 is directly connected, Ethernet0/0
C    172.16.29.4/30 is directly connected, Ethernet0/1
L    172.16.29.6/32 is directly connected, Ethernet0/1
C    172.16.29.8/30 is directly connected, Ethernet0/3
L    172.16.29.10/32 is directly connected, Ethernet0/3
C    172.16.29.12/30 is directly connected, Ethernet0/2
L    172.16.29.14/32 is directly connected, Ethernet0/2
O IA 192.168.70.0/24 [110/20] via 172.16.29.1, 01:17:35, Ethernet0/0
O IA 192.168.71.0/24 [110/20] via 172.16.29.5, 01:16:08, Ethernet0/1
O E2 192.168.73.0/24 [110/20] via 172.16.29.13, 00:04:25, Ethernet0/2
```

R5

Interface ethernet 0/0 consist network of 172.16.29.8/30 will be advertise in OSPF and

Interface ethernet 0/1 consist network of 192.168.72.0/24 will be advertise in EIGRP .

```
R5(config)#router ospf 100
```

```
R5(config-router)#network 172.16.29.8 0.0.0.3 area 0
```

```
R5(config-router)#exit
```

```
R5(config)#router eigrp 90
```

```
R5(config-router)#network 192.168.72.0 0.0.0.255
```

```
R5(config-router)#exit
```

See there is no route on R2 router for 192.168.72.0/24 network cause R2 is in ospf and 192.168.72.0/24 is in EIGRP

```
172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C    172.16.29.0/30 is directly connected, Ethernet0/0
L    172.16.29.2/32 is directly connected, Ethernet0/0
C    172.16.29.4/30 is directly connected, Ethernet0/1
L    172.16.29.6/32 is directly connected, Ethernet0/1
C    172.16.29.8/30 is directly connected, Ethernet0/3
L    172.16.29.10/32 is directly connected, Ethernet0/3
C    172.16.29.12/30 is directly connected, Ethernet0/2
L    172.16.29.14/32 is directly connected, Ethernet0/2
O IA 192.168.70.0/24 [110/20] via 172.16.29.1, 01:29:20, Ethernet0/0
O IA 192.168.71.0/24 [110/20] via 172.16.29.5, 01:27:53, Ethernet0/1
O E2 192.168.73.0/24 [110/20] via 172.16.29.13, 00:16:10, Ethernet0/2
```

Now we have to advertise EIGRP network in OSPF and vice versa

```
R5(config)#router eigrp 90
```

```
R5(config-router)#redistribute ospf 100 metric 10000 1000 255 255 1500
```

```
R5(config-router)#exit
```

Here 10000 is a bandwidth of interface, 1000 is a delay of the ethernet interface, 255 is a reliability, 255 is a load, 1500 is MTU

```
R5(config)#router ospf 100
```

```
R5(config-router)#redistribute eigrp 90 subnets
```

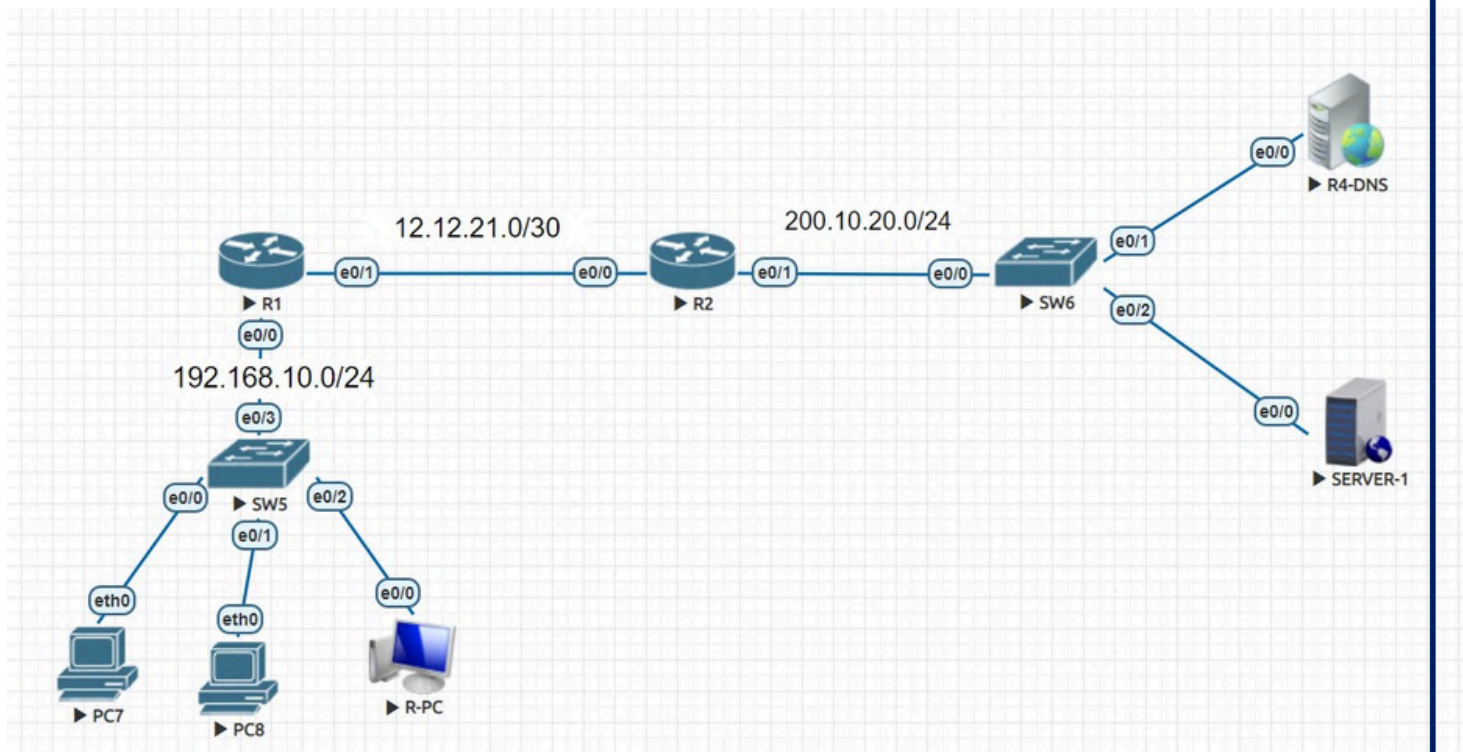
```
R5(config-router)#exit
```

Now we can see 192.168.72.0/24 network in R2's routing table

```
172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C       172.16.29.0/30 is directly connected, Ethernet0/0
L       172.16.29.2/32 is directly connected, Ethernet0/0
C       172.16.29.4/30 is directly connected, Ethernet0/1
L       172.16.29.6/32 is directly connected, Ethernet0/1
C       172.16.29.8/30 is directly connected, Ethernet0/3
L       172.16.29.10/32 is directly connected, Ethernet0/3
C       172.16.29.12/30 is directly connected, Ethernet0/2
L       172.16.29.14/32 is directly connected, Ethernet0/2
O IA    192.168.70.0/24 [110/20] via 172.16.29.1, 01:35:56, Ethernet0/0
O IA    192.168.71.0/24 [110/20] via 172.16.29.5, 01:34:29, Ethernet0/1
O E2    192.168.72.0/24 [110/20] via 172.16.29.9, 00:00:12, Ethernet0/3
O E2    192.168.73.0/24 [110/20] via 172.16.29.13, 00:22:46, Ethernet0/2
```

We have done with the REDISTRIBUTION task.

DHCP and DNS



IN THIS ABOVE LAB we are going to configure DHCP, DNS first

So what is DHCP (dynamic host configuration protocol), it is the protocol we generally used to assign IP address automatically.

And what is DNS (domain name system/server), it is the service we uses to resolve domain name to IP address .like if we put www.google.com in our web browser ,it redirect us to that website but in actual scenario we are heading towards the ip address of that perticular web site instead .cause devices wont understand domain name rather they recognize ip address which is associated with that domain name just like

Example :- www.google.com = 142.250.183.4

TASK 11

Task 11.1 Configure R1 LAN network with the IP range of 192.168.10.0/24

Task 11.2 Assign ip to the pc through DHCP

Task 11.3 Then, do routing on all devices

Task 11.4 Configure R4-DNS as a DNS server

Task 11.5 Try to ping www.ccn.com which is SERVER-1

IP's of the devices

R1	Eth0/0 – 192.168.10.254/24	Eth0/1 – 12.12.21.1/30
R2	Eth0/0 – 12.12.21.2/30	Eth0/1 – 200.10.20.30/24
R4-DNS	ETH 0/0 – 200.10.20.10/24	
SERVER-1	ETH 0/0 – 200.10.20.20/24	

Let's do it

R1(config)#ip dhcp pool CCN-LAN

R1(dhcp-config)#network 192.168.10.0 255.255.255.0

R1(dhcp-config)#default-router 192.168.10.254

R1(dhcp-config)#dns-server 200.10.20.10

R1(dhcp-config)#lease 0 12

R1(dhcp-config)#exit

- To configure DHCP we need 4 basic things that is IP, subnet mask, default gateway and DNS
- From network command we provide IP and subnet mask
- From default-router command we have provide a default gateway
- From DNS-SERVER command we provided a DNS server IP
- And by giving lease command we provided a lease period for that assigning IP

R1(config)#ip dhcp excluded-address 192.168.10.254

(If we want to remove any IP from that DHCP POOL, just to avoid duplicate IP's)

VERIFICATION

```
R1#show ip dhcp pool
Pool CCN-LAN :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                    : 254
  Leased addresses                   : 1
  Pending event                     : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  192.168.10.2       192.168.10.1 - 192.168.10.254      1
```

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.1     0100.5079.6668.07  Feb 15 2023 02:48 PM  Automatic
```

Then, how to ip to the VPC and R-PC

VPC

```
VPCS> ip dhcp  
DDORA IP 192.168.10.1/24 GW 192.168.10.254
```

R-PC, this is actually a router considering and acting as a PC

R-PC(config)#interface ethernet 0/0

R-PC(config-if)#ip address dhcp

R-PC(config-if)#no shutdown

R-PC(config-if)#exit

To verify that give command

```
R-PC#show ip interface brief  
Interface      IP-Address      OK? Method Status      Protocol  
Ethernet0/0    192.168.10.2    YES DHCP      up          up  
Ethernet0/1    unassigned      YES unset     administratively down down  
Ethernet0/2    unassigned      YES unset     administratively down down  
Ethernet0/3    unassigned      YES unset     administratively down down
```

3 so now do routing on all devices here we running eigrp on router you can do any routing

R1

R1(config)#router eigrp 100

R1(config-router)#network 192.168.10.0 0.0.0.255

R1(config-router)#network 12.12.21.0 0.0.0.3

R1(config-router)#exit

R2

R2(config)#router eigrp 100

R2(config-router)#network 12.12.21.0 0.0.0.3

R2(config-router)#network 200.10.20.0 0.0.0.255

R2(config-router)#exit

R-PC

R-PC(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.254

R-DNS

R4-DNS(config)#ip route 0.0.0.0 0.0.0.0 200.10.20.30

SERVER-1 (www.ccn.com)

www.ccn.com(config)#ip route 0.0.0.0 0.0.0.0 200.10.20.3

4 now start the dns server on R4-DNS

R4-DNS(config)#ip dns server

R4-DNS(config)#ip host www.ccn.com 200.10.20.20

First command is to start DNS server port number udp-53

And second command is to bind domain name with ip address

5 now ping the www.ccn.com consist ip 200.10.20.20

```
VPCS> ping www.ccn.com
www.ccn.com resolved to 200.10.20.20

84 bytes from 200.10.20.20 icmp_seq=1 ttl=253 time=7.019 ms
84 bytes from 200.10.20.20 icmp_seq=2 ttl=253 time=3.204 ms
84 bytes from 200.10.20.20 icmp_seq=3 ttl=253 time=4.116 ms
84 bytes from 200.10.20.20 icmp_seq=4 ttl=253 time=1.287 ms
84 bytes from 200.10.20.20 icmp_seq=5 ttl=253 time=4.267 ms
```

NETWORK TIME PROTOCOL (NTP)

- ❖ NTP is a network time uses UDP port -123
- ❖ Basically, NTP uses on all LAN as well as WAN devices just to sync time of their devices with NTP server, if time of those won't match then internet won't be able to work on that device.
- ❖ With the help of NTP we generate logs of the devices to solve the issues which generally occur in Enterprise environment. if time is mismatching then the troubleshoot will be more difficult.

We are going to consider lab which we used for the DHCP & DNS configuration

R2 (NTP Master)

```
R2(config)#interface loopback 1
```

```
R2(config-if)#ip address 10.10.10.1 255.255.255.255
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#exit
```

```
R2(config)#ntp master 1
```

```
R2(config)#ntp source loopback 1
```

Here we specify the NTP master stratum number 1 which mean it is most trusted NTP server in a network and we specify the source interface from where NTP broadcast will happen

R1(NTP Client)

```
R1(config)#ntp server 10.10.10.1
```

Here we specify the NTP servers ip address which is R2's loopback 1 interface ip address

Now the NTP authentication is totally optional

R1

```
R1(config)#ntp authentication-key 1 md5 ccn
```

```
R1(config)#ntp authenticate
```

R2

```
R2(config)#ntp authentication-key 1 md5 ccn
```

```
R2(config)#ntp authenticate
```

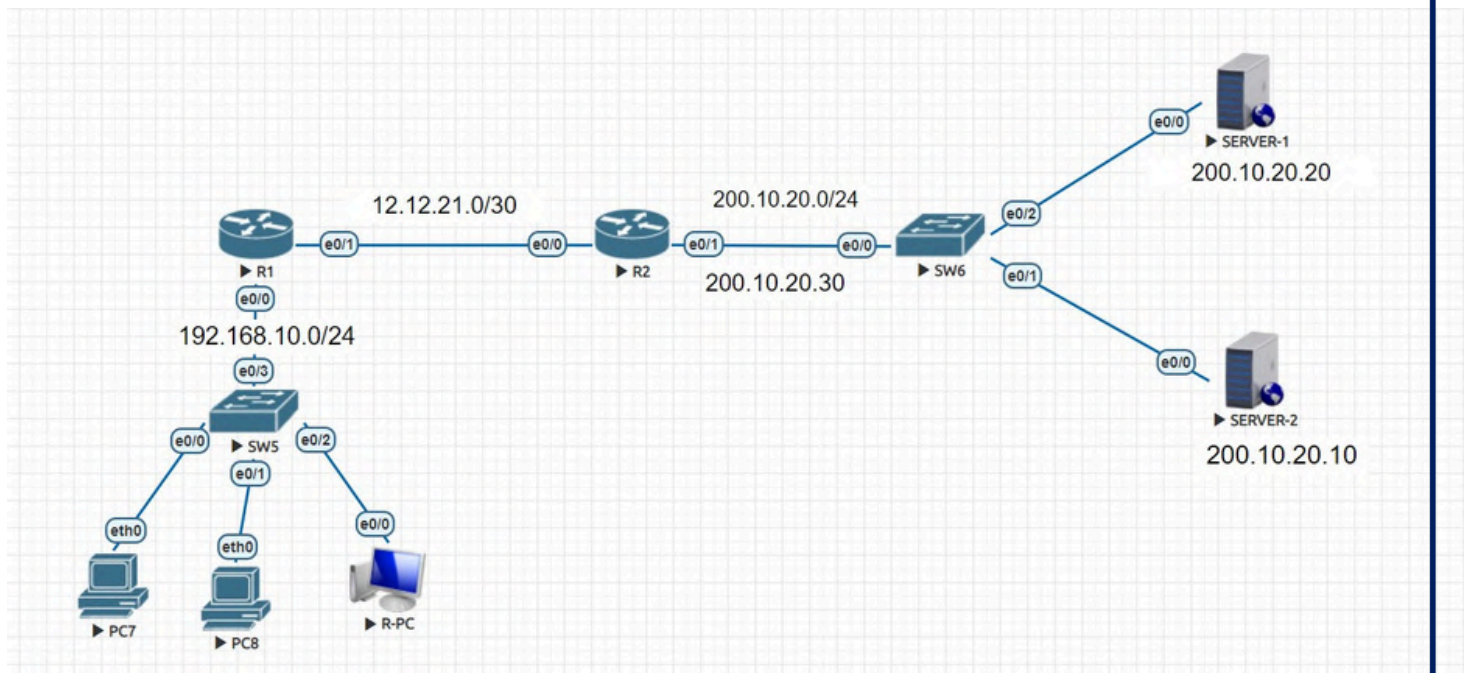
VERIFICATION

```
R1#show ntp status
```

OR

```
R1#show ntp association
```

TELNET, SSH, HTTP, HTTPS



TELNET

Telnet is a remote access service works under TCP and uses port number 23 .generally telnet sends data in clear text format.

SSH

SSH is also a remote access service works under tcp and uses port number 22 . Genrally , ssh send data in encrypted format over the networks .it encrypts data through rsa algorithm , and consist key size of 2048 bit which is market standard.

Now configure the TELNET, SSH, HTTP, HTTPS on devices

Task 12

Task 12.1 Assign IP according to networks represents in diagram

Task 12.2 Configure SERVER 1 for TELENT service

Task 12.3 Configure SERVER 2 for SSH service

Task 12.4 Configure SERVER 1 for HTTP service

Task 12.5 Configure SERVER 2 for HTTPS service

R1	ETH 0/0 192.168.10.254/24		ETH 0/1 12.12.21.1/30
R1- INTERNAL NETWORK PC	PC7 192.168.10.1/24	PC8 192.168.10.2/24	R-PC 192.168.10.3/24
R2	ETH 0/0 12.12.21.2/24		ETH 0/1 200.10.20.30/24
SERVER 1	ETH 0/0 200.10.20.20/24		
SERVER 2	ETH 0/0 200.10.20.10		

NOW WE ARE CONFIGURING THE SERVER 1 FOR TELNET

ON SERVER 1

SERVER-1(config)#line vty 0 4

SERVER-1(config-line)#transport input telnet

SERVER-1(config-line)#login local

SERVER-1(config-line)#exit

Local username & password creation

SERVER-1(config)#username ccn privilege 15 password ccn

Here, line vty means virtual teletype which means a connection over a networks or connection which is virtual. console access is direct access but telnet access is virtual access on device. transport input telnet means the data travels over a network should only connect to a telnet service activated on a device .and, login local means at the time of login device should ask username & password which is verified from local user database which is created on device.

How to take access?

For that we need a router(R-PC) in eve-ng setup

Router# telnet <destination_ip_address> <desired_service_port>

```
R-PC#telnet 200.10.20.20 23
Trying 200.10.20.20 ... Open
```

User Access Verification

```
Username: ccn
Password:
SERVER-1#
```

SSH

SSH on server 2

```
SERVER-2(config)#line vty 0 4
```

```
SERVER-2(config-line)#transport input ssh
```

```
SERVER-2(config-line)#login local
```

```
SERVER-2(config-line)#exit
```

```
SERVER-2(config)#crypto key generate rsa modulus 2048 label mypubkey
```

Username & password

```
SERVER-2(config)#username ccn privilege 15 password ccn
```

Here line vty means virtual teletype which means a connection over a networks or connection which is virtual .then , 0 4 means at a time 5 line connection can be activate .transport input ssh means the data travels over a network should only connect to a ssh service activated on a device .and , login local means at the time of login device should ask username & password which is verified from local user database which is created on device. Crypto key command is to start the encryption process of the data through rsa algorithm with standard key Of 2048 bit.

How to take access?

```
Router# ssh -l <username> <destination_ip_address>
```

```
R-PC#ssh -l ccn 200.10.20.10
Password:

SERVER-2#
SERVER-2#
SERVER-2#
```

HTTP

HTTP on server 1

SERVER-1(config)#ip http server

By applying above command, we are enabling TCP port 80 which we generally use to take GUI access.

VERIFICATION

Router# telnet <destination_ip_address> <desired_port>

With the command we get a output of [.....open] means http is enable on device

```
R-PC#telnet 200.10.20.20 80
Trying 200.10.20.20, 80 ... Open
```

HTTPS

https on server 2

SERVER-2(config)#ip http secure-server

By applying this command we are enabling tcp port 443 on device which we genrally use to take secure GUI access

To additional security we enable authentication

SERVER-2(config)#ip http authentication local

By applying this command we are enabling authentication on devices through local user database.

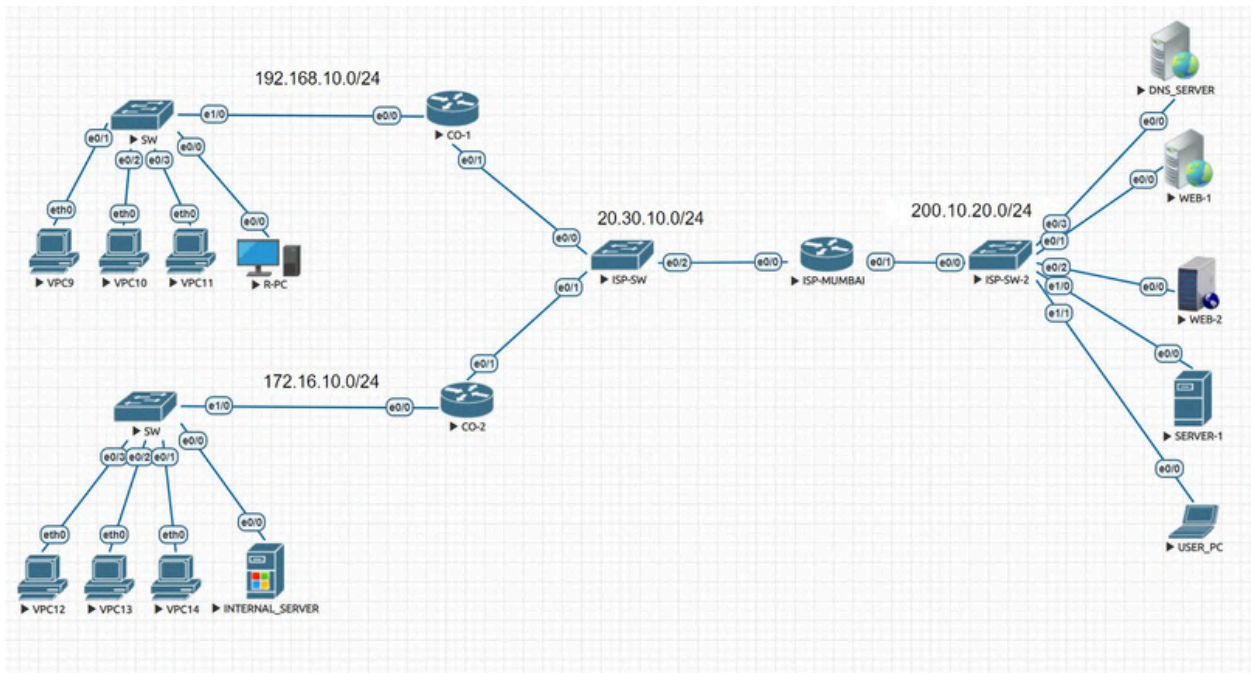
VERIFICATION

Router# telnet <destination_ip_address> <desired_port>

With the command we get a output of [.....open] means https is enable on device

```
R-PC#telnet 200.10.20.10 443
Trying 200.10.20.10, 443 ... Open
```

ACCESS-CONTROL LIST (ACL)



TASK -

13

Task 13.1 Provide IP to the devices.

SECTION	DEVICE NAME	IP ON INTERFACE ETH0/0=
COMPANY - 1	CO-1	ETH0/0= 192.168.10.254/24 ETH0/1= 20.30.10.10/24
	VPC9	192.168.10.1/24
	VPC10	192.168.10.2/24
	VPC11	192.168.10.3/24
	R-PC	192.168.10.100/24
COMPANY - 2	CO-2	ETH0/0= 172.16.10.254/24 ETH 0/1 = 20.30.10.20/24
	VPC12	172.16.10.1/24
	VPC13	172.16.10.2/24
	VPC14	172.16.10.3/24
	INTERNAL_SERVER	172.16.10.100/24
ISP	ISP-MUMBAI	ETH0/0 = 20.30.10.30/24 ETH0/1= 200.10.20.10/24
OTHERS	DNS_SERVER	200.10.20.20/24
	WEB-1	200.10.20.30/24
	WEB-2	200.10.20.40/24
	SERVER-1	200.10.20.50/24
	USER_PC	200.10.20.60/24

Task 13.2 Do Routing on devices (Any kind of Routing)

Task 13.3 Create a ACL for company - 1

1. R-PC should TELNET WEB-1 and WEB-2
2. R-PC should SSH WEB-1 and WEB-2
3. R-PC should HTTP WEB-1 and WEB-2
4. R-PC should HTTPS WEB-1 and WEB-2
5. Network of 192.168.10.0/24 should PING DNS_SERVER & SERVER-1
6. R-PC should PING/TELNET/SSH/HTTP/HTTPS to SERVER-1 (www.ccn.com)

Task 13.4 Create A ACL For Company – 2

1. USER_PC should TELNET INTERNAL_SERVER
2. USER_PC should SSH INTERNAL_SERVER
3. USER_PC should HTTP INTERNAL_SERVER
4. USER_PC should HTTPS INTERNAL_SERVER
5. USER_PC should PING a Network of 172.16.10.0/24

SOLUTION

COMPANY 1 ACL

1) R-PC SHOULD TELNET WEB-1 AND WEB-2

ip access-list extended in-out

- 1 permit tcp host 192.168.10.100 host 200.10.20.30 eq telnet
- 2 permit tcp host 192.168.10.100 host 200.10.20.40 eq telnet

ip access-list extended out-in

- 1 permit tcp host 200.10.20.30 host 192.168.10.100 ack
- 2 permit tcp host 200.10.20.40 host 192.168.10.100 ack

2) R-PC SHOULD SSH WEB-1 AND WEB-2

ip access-list extended in-out

- 3 permit tcp host 192.168.10.100 host 200.10.20.30 eq 22
- 4 permit tcp host 192.168.10.100 host 200.10.20.40 eq 22

In first entry we are provided permission of ack to the TCP given IP, so again we don't need to do it

3) R-PC SHOULD HTTP WEB-1 AND WEB-2

ip access-list extended in-out

- 5 permit tcp host 192.168.10.100 host 200.10.20.30 eq 80
- 6 permit tcp host 192.168.10.100 host 200.10.20.40 eq 80

4) R-PC SHOULD HTTPS WEB-1 AND WEB-2

ip access-list extended in-out 7 permit tcp host 192.168.10.100 host 200.10.20.30 eq 443 8 permit tcp host 192.168.10.100 host 200.10.20.40 eq 443

5) NETWORK OF 192.168.10.0/24 SHOULD PING DNS_SERVER & SERVER-1

ip access-list extended in-out

9 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.20 echo

10 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.50 echo

ip access-list extended out-in

3 permit icmp host 200.10.20.20 192.168.10.0 0.0.0.255 echo-reply

4 permit icmp host 200.10.20.50 192.168.10.0 0.0.0.255 echo-reply

6) R-PC SHOULD PING/TELNET/SSH/HTTP/HTTPS TO SERVER-1 (www.ccn.com)

ip access-list extended in-out

11 permit tcp host 192.168.10.100 host 200.10.20.50 eq www

12 permit tcp host 192.168.10.100 host 200.10.20.50 eq 443

13 permit tcp host 192.168.10.100 host 200.10.20.50 eq 22

14 permit tcp host 192.168.10.100 host 200.10.20.50 eq telnet

15 permit icmp host 192.168.10.100 host 200.10.20.50 echo

ip access-list extended out-in

5 permit tcp host 200.10.20.50 host 192.168.10.100 ack

6 permit icmp host 200.10.20.50 host 192.168.10.100 echo-reply

COMPANY 2 ACL

1) USER_PC SHOULD TELNET INTERNAL_SERVER

ip access-list extended out-in

permit tcp host 200.10.20.60 host 172.16.10.100 eq 23

2) USER_PC SHOULD SSH INTERNAL_SERVER

ip access-list extended out-in

permit tcp host 200.10.20.60 host 172.16.10.100 eq 22

3) USER_PC SHOULD HTTP INTERNAL_SERVER

ip access-list extended out-in

permit tcp host 200.10.20.60 host 172.16.10.100 eq 80

4) USER_PC SHOULD HTTPS INTERNAL_SERVER

ip access-list extended out-in

permit tcp host 200.10.20.60 host 172.16.10.100 eq 443

5) USER_PC SHOULD PING NETWORK OF 172.16.10.0/24

ip access-list extended out-in

permit icmp host 200.10.20.60 172.16.10.0 0.0.0.255 echo

WE HAVE TO PERMIT ROUTING PROTOCOL BECAUSE THAT PROTOCOL SENDING HELLO MESSAGES AND BLOCKED BY ACL

ip access-list extended out-in

permit eigrp any any

VERIFICATION

COMPANY-1

```
CO-1#show ip access-lists
Extended IP access list in-out
 10 permit tcp host 192.168.10.100 host 200.10.20.30 eq telnet
 20 permit tcp host 192.168.10.100 host 200.10.20.40 eq telnet
 30 permit tcp host 192.168.10.100 host 200.10.20.30 eq 22
 40 permit tcp host 192.168.10.100 host 200.10.20.40 eq 22
 50 permit tcp host 192.168.10.100 host 200.10.20.30 eq www
 60 permit tcp host 192.168.10.100 host 200.10.20.40 eq www
 70 permit tcp host 192.168.10.100 host 200.10.20.30 eq 443
 80 permit tcp host 192.168.10.100 host 200.10.20.40 eq 443
 90 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.20 echo
100 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.50 echo
Extended IP access list out-in
 10 permit tcp host 200.10.20.30 host 192.168.10.100 ack
 20 permit tcp host 200.10.20.40 host 192.168.10.100 ack
 30 permit icmp host 200.10.20.20 192.168.10.0 0.0.0.255 echo-reply
 40 permit icmp host 200.10.20.50 192.168.10.0 0.0.0.255 echo-reply
 50 permit eigrp any any (964 matches)
```

COMPANY-2

```
CO-2#show ip access-lists
Extended IP access list in-out
 10 permit tcp host 172.16.10.100 host 200.10.20.60 ack
 20 permit icmp host 172.16.10.100 host 200.10.20.60 echo-reply (10 matches)
Extended IP access list out-in
 10 permit tcp host 200.10.20.60 host 172.16.10.100 eq telnet
 20 permit tcp host 200.10.20.60 host 172.16.10.100 eq 22
 30 permit tcp host 200.10.20.60 host 172.16.10.100 eq www
 40 permit tcp host 200.10.20.60 host 172.16.10.100 eq 443
 50 permit icmp host 200.10.20.60 172.16.10.0 0.0.0.255 echo (10 matches)
 60 permit eigrp any any (396 matches)
```

STANDARD ACL

- ❖ Basically, We Do Make Standard Access-Control List For Permit/Deny Source Traffic Only Unlike Extended ACL Where We Permit/Deny Source To Destination Traffic With The Control Of Protocol & Services As Well
- ❖ Standard ACL permit source traffic and allow all kinds of traffic on that give ip or a network/subnet

Let's make Standard ACL

CREATION

Router(config)#ip access-list standard lan-wan

Router(config)#permit host 192.168.10.1(permit single ip with all source traffic)

OR

Router(config)#permit 192.168.10.0 0.0.0.255...permit whole network with all source traffic)

Router(config)#deny any

IMPLEMENTATION

Router(config)#interface ethernet 0/0

Router(config)#ip access-group lan-wan in

Router(config)#exit

VERIFICATION

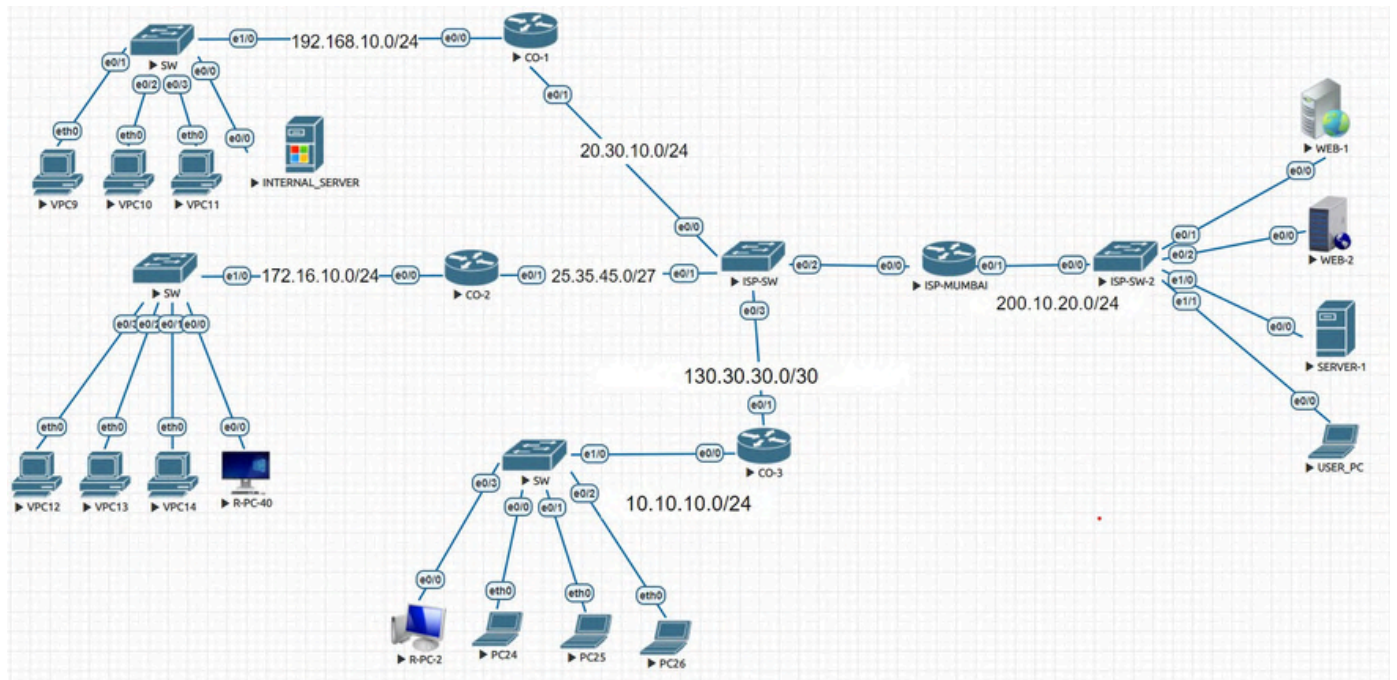
Router# show ip access-list(show all acl)

OR

Router#show ip access-list lan-wan(show only lan-wan acl)

Allowing all kinds of source traffic will make it more vulnerable that's why we use extended ACL, otherwise we use Standard ACL for NAT/PAT configuration or VPN configuration.

NETWORK ADDRESS TRANSLATION (NAT) & PORT ADDRESS TRANSLATION (PAT)



- ❖ We use a NAT technology to translate private IP to public IP. Then, there are many things in a NAT that is Static NAT, Dynamic NAT, PAT (Port Address Translation)
- ❖ **Static NAT** – in Static NAT translate private IP to public IP, it is one to one translate and we do manual entry for that. with Static NAT bidirectional connection can be made.
- ❖ **Dynamic NAT**- dynamic NAT also translate private IP to public IP but in different way. Here we create pool of public IP then random private IP can be translate to public IP out of this pool. It creates unidirectional connection.
- ❖ **PAT** – PAT also translate private IP to public IP but adds up port number. Many private IP translate into one public IP with the effect of port number. pat provide unidirectional connection.

TASKS – 14

IP

CO-1	ETH 0/0=192.168.10.254/24	ETH 0/1=20.30.10.1/24
CO-1 PC	VPC9	192.168.10.1
	VPC10	192.168.10.2
	VPC11	192.168.10.3

	INTERNAL_SERVER	192.168.10.100
CO-2	ETH0/0 = 172.16.10.254/24	ETH0/1 = 25.35.45.1/27
CO-2 PC	VPC12	172.16.10.1
	VPC13	172.16.10.2
	VPC14	172.16.10.3
	R-PC-40	172.16.10.100
CO-3	ETH0/0=10.10.10.254/24	ETH0/1=130.30.30.1/30
CO-3 PC	R-PC-2	10.10.10.100
	PC24	10.10.10.1
	PC25	10.10.10.2
	PC26	10.10.10.3
ISP-MUMBAI		
	ETH0/0.111	20.30.10.2/24
	ETH0/0.219	25.35.45.2/27
	ETH0/0.389	130.30.30.2/30
	ETH0/1	200.10.20.1/24
ISP-SW		
	ETH0/0	VLAN 111 = CO-1_ISP
	ETH0/1	VLAN 219 = CO-2_ISP
	ETH0/3	VLAN 389 = CO-3_ISP
	ETH0/2	TRUNK
OTHERS		
	WEB-1	200.10.20.10/24
	WEB-2	200.10.20.20/24
	SERVER-1	200.10.20.30/24
	USER_PC	200.10.20.40/24

ROUTING EIGRP 199

Task 14.1 CO-1

VPC-9, VPC-10 , VPC-11 , INTERNAL SERVER Should Assign Static Mapped With The Ip Of 20.30.10.10 ,20.30.10.20 , 20.30.10.30 , 20.30.10.40

Do telnet on IP 20.30.10.40 it redirects to the INTERNAL SERVER

Task 14.2 – CO-2

VPC-12 , VPC-13 , VPC-14 , R-PC-40 Should Dynamically Mapped With The Given Pool Of IP 25.35.45.3 – 25.35.45.7

Task 14.3 -CO-3

A Whole Network of Company 3 Should Mapped With The Ip Of ETH 0/1 (CO-3)

LET'S DO IT **TASK – 14.1** 1.Do Static NAT entry on router

```
CO-1(config)#ip nat inside source static 192.168.10.1 20.30.10.10 CO-
```

```
1(config)#ip nat inside source static 192.168.10.2 20.30.10.20
```

```
CO-1(config)#ip nat inside source static 192.168.10.3 20.30.10.30
```

```
CO-1(config)#ip nat inside source static 192.168.10.100 20.30.10.40
```

2.then activate NAT on interface

```
CO-1(config)#interface ethernet 0/0
```

```
CO-1(config-if)#ip nat inside
```

```
CO-1(config-if)#exit CO-1(config)#interface ethernet 0/1 CO-1(config-if)#ip nat  
outside
```

```
CO-1(config-if)#exit
```

VERIFICATION ON ROUTER

```
CO-1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 20.30.10.10         192.168.10.1      ---               ---
--- 20.30.10.20         192.168.10.2      ---               ---
--- 20.30.10.30         192.168.10.3      ---               ---
--- 20.30.10.40         192.168.10.100    ---               ---
```

Debug on web-1 A ping via vpc-9

WEB-1#debug ip packet

```
*Mar 3 10:44:53.526: IP: s=20.30.10.10 (Ethernet0/0), d=200.10.20.10 (Ethernet0/0), len 84, rcvd 3
*Mar 3 10:44:53.526: IP: s=20.30.10.10 (Ethernet0/0), d=200.10.20.10, len 84, stop process pak for for
s packet
*Mar 3 10:44:53.526: IP: s=200.10.20.10 (local), d=20.30.10.10 (Ethernet0/0), len 84, sending
*Mar 3 10:44:53.526: IP: s=200.10.20.10 (local), d=20.30.10.10 (Ethernet0/0)
WEB-1#, len 84, sending full packet
```

To stop Debugging

WEB-1#undebug all

Telnet

```
USER_PC#telnet 20.30.10.40 23
Trying 20.30.10.40 ...
% Connection refused by remote host

USER_PC#telnet 20.30.10.40 23
Trying 20.30.10.40 ... Open
```

User Access Verification

```
Username: ccn
Password:
INTERNAL_SERVER#
INTERNAL_SERVER#
```

TASK – 14.2

To Combine all Private IP we create Standard ACL

```
CO-2(config)#ip access-list standard DYN_NAT-LIST
CO-2(config-std-nacl)#permit 172.16.10.0 0.0.0.255
CO-2(config-std-nacl)#exit
```

To create NAT POOL

```
CO-2(config)# ip nat pool CCN-DYN_NAT 25.35.45.3 25.35.45.7 netmask 255.255.255.224
```

To match up Private IP 's list (ACL) with NAT POOL (pool)

```
CO-2(config)#ip nat inside source list DYN_NAT-LIST pool CCN-DYN_NAT
```

To Implement NAT

```
CO-2(config)#interface ethernet 0/0
CO-2(config-if)#ip nat inside
CO-2(config-if)#exit
CO-2(config)#interface ethernet 0/1
CO-2(config-if)#ip nat outside
CO-2(config-if)#exit
```

VERIFICATION

Did ping from vpc-12 to web-1

```
CO-2#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 25.35.45.3:14043  172.16.10.1:14043 200.10.20.10:14043 200.10.20.10:14043
icmp 25.35.45.3:14299  172.16.10.1:14299 200.10.20.10:14299 200.10.20.10:14299
icmp 25.35.45.3:14555  172.16.10.1:14555 200.10.20.10:14555 200.10.20.10:14555
icmp 25.35.45.3:14811  172.16.10.1:14811 200.10.20.10:14811 200.10.20.10:14811
```

TASK – 14.3

To combine all Private IP we create Standard ACL

```
CO-3(config)#ip access-list standard PAT-LIST
```

```
CO-3(config-std-nacl)#permit 10.10.10.0 0.0.0.255
```

```
CO-3(config-std-nacl)#exit
```

To combine ACL with Interface for PAT

```
CO-3(config)# ip nat inside source list PAT-LIST interface ethernet 0/1 overload
```

To Implement NAT/PAT

```
CO-3(config)#interface ethernet 0/0
```

```
CO-3(config-if)#ip nat inside
```

```
CO-3(config-if)#exit
```

```
CO-3(config)#interface ethernet 0/1
```

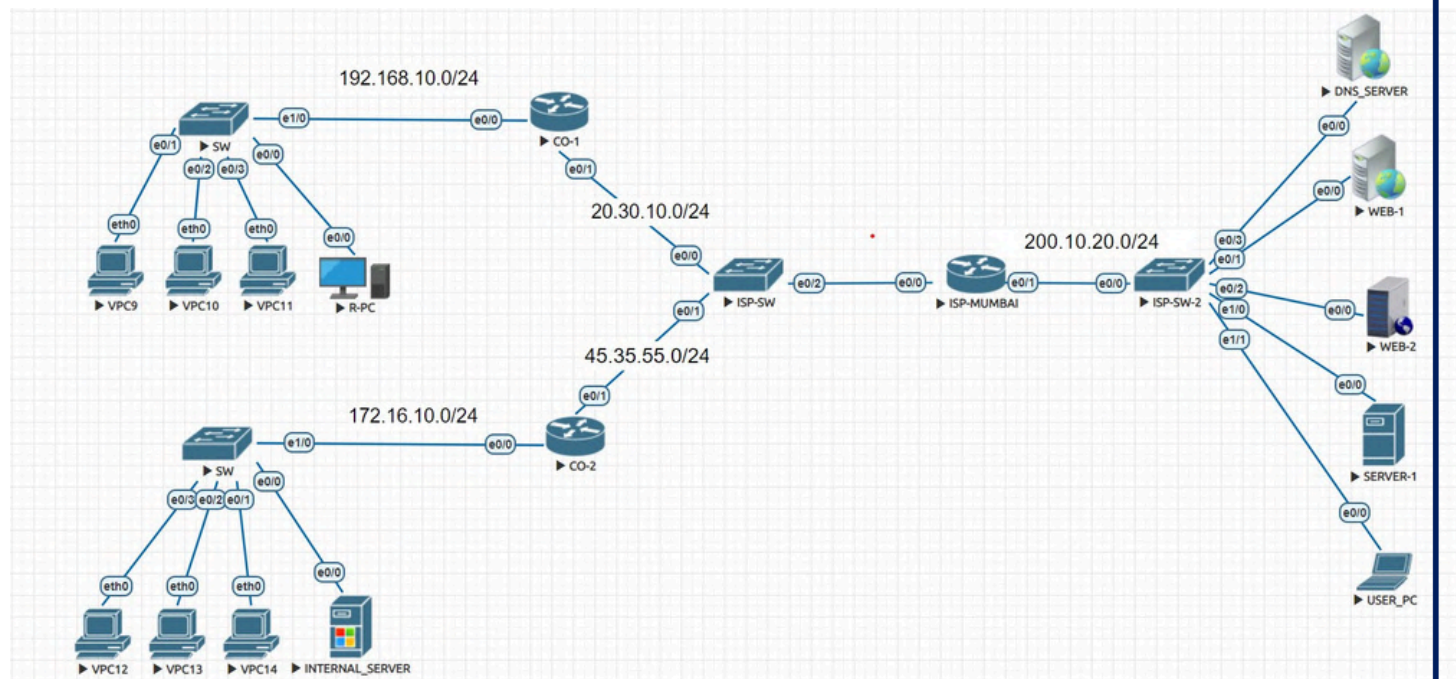
```
CO-3(config-if)#ip nat outside
```

```
CO-3(config-if)#exit
```

VERIFICATION

```
CO-3#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 130.30.30.1:2013  10.10.10.1:2013   200.10.20.1:2013   200.10.20.1:2013
icmp 130.30.30.1:2269  10.10.10.1:2269   200.10.20.1:2269   200.10.20.1:2269
icmp 130.30.30.1:2525  10.10.10.1:2525   200.10.20.1:2525   200.10.20.1:2525
icmp 130.30.30.1:2781  10.10.10.1:2781   200.10.20.1:2781   200.10.20.1:2781
icmp 130.30.30.1:3037  10.10.10.1:3037   200.10.20.1:3037   200.10.20.1:3037
icmp 130.30.30.1:3805  10.10.10.1:3805   200.10.20.10:3805   200.10.20.10:3805
icmp 130.30.30.1:4061  10.10.10.1:4061   200.10.20.10:4061   200.10.20.10:4061
icmp 130.30.30.1:4317  10.10.10.1:4317   200.10.20.10:4317   200.10.20.10:4317
icmp 130.30.30.1:4573  10.10.10.1:4573   200.10.20.10:4573   200.10.20.10:4573
icmp 130.30.30.1:4829  10.10.10.1:4829   200.10.20.10:4829   200.10.20.10:4829
```

ACL AND NAT



CO-1	ETH0/0 = 192.168.10.254/24		ETH0/1 = 20.30.10.10/24	
	VPC-9	192.168.10.1/24		
	VPC-10	192.168.10.2/24		
	VPC-11	192.168.10.3/24		
	R-PC	192.168.10.100/24		
CO-2	ETH0/0= 172.16.10.254/24		ETH0/1 = 45.35.55.65/24	
	VPC-12	172.16.10.1/24		
	VPC-13 =	172.16.10.2/24		
	VPC-14	172.16.10.3/24		
	INTERNAL_SERVER	172.16.10.100/24		
ISP-SW				
ETH0/0		VLAN 99 => ISP_CO-1		

	ETH0/1	VLAN 199 => ISP_CO-2
	ETH 0/2	TRUNK
ISP-MUMBAI	ETH0/0	NO SHUT / NO IP ADDRESS
	ETH0/0.99 =	20.30.10.20/24
	ETH0/0.199	45.35.55.75/24
	ETH0/1	200.10.20.1/24
OTHERS		
	DNS_SERVER	200.10.20.10/24
	WEB-1	200.10.20.20
	WEB-2	200.10.20.30
	SERVER-1	200.10.20.40
	USER_PC	200.10.20.50

ROUTING

- ❖ CO-1 > EIGRP 199 on both side
- ❖ CO-2 > OSPF 100 AREA 1 both side
- ❖ ISP_MUMBAI > EIGRP 199 / OSPF 100 AREA 1 for CO-1 & CO-2's other side RIP
- ❖ DNS_SERVER / WEB-1 / WEB-2 / SERVER-1 / USER_PC / R-PC / INTERNAL_SERVER =
DEFAULT ROUTING

TASK – 15

CO-1

Task – 15.1 ACL TASK

1. A whole network can PING WEB-1 AND WEB-2
2. R-PC can TELNET & SSH WEB-1 & WEB-2
3. R-PC can do HTTP & HTTPS TO SERVER-1

Task – 15.2 STATIC NAT TASK

VPC-9, VPC-10, VPC-11, R-PC should assign static mapped IP OF 20.30.10.1, 20.30.10.2, 20.30.10.3, 20.30.10.100 respectively

CO-2

Task – 15.3 ACL TASK

- 1.USER_PC can PING a whole network
- 2.USER_PC can TELNET, SSH, HTTP , HTTPS to INTERNAL SERVER

Task – 15.4 STATIC NAT TASK

VPC-12, VPC-13, VPC-14 , INTERNAL SERVER should assign static ip of 45.35.55.10, 45.35.55.20 , 45.35.55.30 , 45.35.55.100

LET's DO THE TASK

CO-1

First we are going to do NAT Task

Task - 15.2

```
CO-1(config)#ip nat inside source static 192.168.10.1 20.30.10.1
CO-1(config)#ip nat inside source static 192.168.10.2 20.30.10.2
CO-1(config)#ip nat inside source static 192.168.10.3 20.30.10.3
CO-1(config)#ip nat inside source static 192.168.10.100 20.30.10.100
```

Implementation of NAT

```
CO-1(config)#interface ethernet 0/0
CO-1(config-if)#ip nat inside
CO-1(config-if)#exit
CO-1(config)#interface ethernet 0/1
CO-1(config-if)#ip nat outside
CO-1(config-if)#exit
```

ACL

Task 15.1 = 1)

```
CO-1(config)#ip access-list extended in-out
CO-1(config-ext-nacl)#1 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.20 echo
```

```
CO-1(config-ext-nacl)#ip access-list extended out-in
CO-1(config-ext-nacl)#1 permit icmp host 200.10.20.20 host 20.30.10.1 echo-reply
CO-1(config-ext-nacl)#2 permit icmp host 200.10.20.20 host 20.30.10.2 echo-reply
CO-1(config-ext-nacl)#3 permit icmp host 200.10.20.20 host 20.30.10.3 echo-reply
CO-1(config-ext-nacl)#4 permit icmp host 200.10.20.20 host 20.30.10.100 echo-reply
```

Task 15.1 = 2) CO-1(config-ext-nacl)#ip access-list extended in-out CO-1(config-ext-nacl)#2 permit tcp host 192.168.10.100 host 200.10.20.20 eq 23 CO-1(config-ext-nacl)#3 permit tcp host 192.168.10.100 host 200.10.20.20 eq 22 CO-1(config-ext-nacl)#2 permit tcp host 192.168.10.100 host 200.10.20.30 eq 23 CO-1(config-ext-nacl)#3 permit tcp host 192.168.10.100 host 200.10.20.30 eq 22 CO-1(config-ext-nacl)#ip access-list extended out-in CO-1(config-ext-nacl)#5 permit tcp host 200.10.20.20 host 20.30.10.100 ack CO-1(config-ext-nacl)#6 permit tcp host 200.10.20.30 host 20.30.10.100 ack **Task 15.1 = 3)** CO-1(config-ext-nacl)#ip access-list extended in-out CO-1(config-ext-nacl)#4 permit tcp host 192.168.10.100 host 200.10.20.40 eq 80 CO-1(config-ext-nacl)#5 permit tcp host 192.168.10.100 host 200.10.20.40 eq 443

CO-1(config-ext-nacl)#ip access-list extended out-in

CO-1(config-ext-nacl)#7 permit tcp host 200.10.20.40 host 20.30.10.100 ack

Permit routing Eigrp on out-in

CO-1(config-ext-nacl)#ip access-list extended out-in

CO-1(config-ext-nacl)#8 permit eigrp any any

Implementation of ACL

CO-1(config-ext-nacl)#interface ethernet 0/0

CO-1(config-if)#ip access-group in-out in

CO-1(config-if)#exit

CO-1(config)#interface ethernet 0/1

CO-1(config-if)#ip access-group out-in in

CO-1(config-if)#exit

VERIFICATION

CO-1(config)#Show ip access-list

```

CO-1(config)#do show ip access-lists
Extended IP access list in-out
 1 permit icmp 192.168.10.0 0.0.0.255 host 200.10.20.20 echo
 2 permit tcp host 192.168.10.100 host 200.10.20.20 eq telnet (37 matches)
 3 permit tcp host 192.168.10.100 host 200.10.20.20 eq 22
 4 permit tcp host 192.168.10.100 host 200.10.20.40 eq www (7 matches)
 5 permit tcp host 192.168.10.100 host 200.10.20.40 eq 443 (8 matches)
Extended IP access list out-in
 1 permit icmp host 200.10.20.20 host 20.30.10.1 echo-reply
 2 permit icmp host 200.10.20.20 host 20.30.10.2 echo-reply
 3 permit icmp host 200.10.20.20 host 20.30.10.3 echo-reply
 4 permit icmp host 200.10.20.20 host 20.30.10.100 echo-reply
 5 permit tcp host 200.10.20.20 host 20.30.10.100 ack (28 matches)
 6 permit tcp host 200.10.20.30 host 20.30.10.100 ack
 7 permit tcp host 200.10.20.40 host 20.30.10.100 ack (12 matches)
 8 permit eigrp any any (66 matches)

```

CO – 2 First we are going to do NAT **Task -15.4** NAT CO-2(config)#ip nat inside source static 172.16.10.1 45.35.55.10

CO-2(config)#ip nat inside source static 172.16.10.2 45.35.55.20

CO-2(config)#ip nat inside source static 172.16.10.3 45.35.55.30

CO-2(config)#ip nat inside source static 172.16.10.100 45.35.55.100

Implementation of NAT

CO-2(config)#interface ethernet 0/0

CO-2(config-if)#ip nat inside

CO-2(config-if)#exit

CO-2(config)#interface ethernet 0/1

CO-2(config-if)#ip nat outside

CO-2(config-if)#exit

ACL

Task 15.3 = 1) CO-2(config)#ip access-list extended wan-lan

CO-2(config-ext-nacl)#1 permit icmp host 200.10.20.50 45.35.55.0 0.0.0.255 echo

CO-2(config)#ip access-list extended lan-wan

CO-2(config-ext-nacl)#1 permit icmp 172.16.10.0 0.0.0.255 host 200.10.20.50 echo-reply

Task 15.3 = 2)

CO-2(config)#ip access-list extended wan-lan

CO-2(config-ext-nacl)#2 permit tcp host 200.10.20.50 host 45.35.55.100 eq 22 23 80 44

CO-2(config)#ip access-list extended lan-wan

CO-2(config-ext-nacl)#2 permit tcp host 172.16.10.100 host 200.10.20.50 ack
Implementation of ACL CO-2(config)#interface ethernet 0/0 CO-2(config-if)#ip
 access-group lan-wan in CO-2(config-if)#exit CO-2(config)#interface ethernet
 0/1 CO-2(config-if)#ip access-group wan-lan in CO-2(config-if)#exit Permit
 routing ospf CO-2(config)#ip access-list extended wan-lan CO-2(config-ext-
 nacl)#3 permit ospf any any **VERIFICATION** CO-2#Show ip access-list

```
CO-2#show ip access-lists
Extended IP access list lan-wan
  1 permit icmp 172.16.10.0 0.0.0.255 host 200.10.20.50 echo-reply
  2 permit tcp host 172.16.10.100 host 200.10.20.50 ack
Extended IP access list wan-lan
  1 permit icmp host 200.10.20.50 45.35.55.0 0.0.0.255 echo
  2 permit tcp host 200.10.20.50 host 45.35.55.100 eq 22 telnet www 443
  3 permit ospf any any (21 matches)
```

ISP-MUMBAI

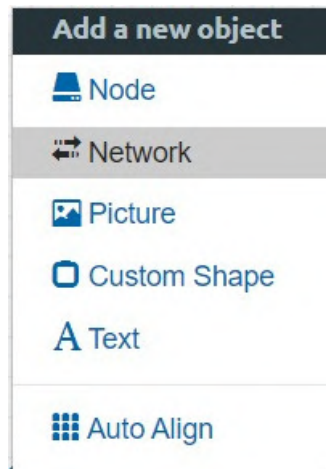
```
ISP-MUMBAI#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	unassigned	YES	NVRAM	up	up
Ethernet0/0.99	20.30.10.20	YES	NVRAM	up	up
Ethernet0/0.199	45.35.55.75	YES	NVRAM	up	up
Ethernet0/1	200.10.20.1	YES	NVRAM	up	up
Ethernet0/2	unassigned	YES	NVRAM	administratively down	down
Ethernet0/3	unassigned	YES	NVRAM	administratively down	down

```
router eigrp 199
  network 20.30.10.0 0.0.0.255
  redistribute rip metric 10000 1000 255 1 1500
  redistribute ospf 100 metric 10000 1000 255 1 1500
!
router ospf 100
  redistribute eigrp 199 subnets
  redistribute rip subnets
  network 45.35.55.0 0.0.0.255 area 1
!
router rip
  redistribute eigrp 199 metric 1
  redistribute ospf 100 metric 1
  network 200.10.20.0
.
```

HOW TO ADD YOUR LAPTOP/PC TO EVE-NG LAB SETUP

1.DO RIGHT CLICK SELECT ADD A NEW OBJECT >“NETWORK”.

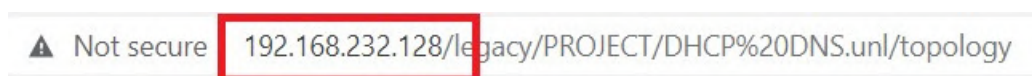


Then, NAME/PREFIX - MY_PC >TYPE – Management(cloud0)

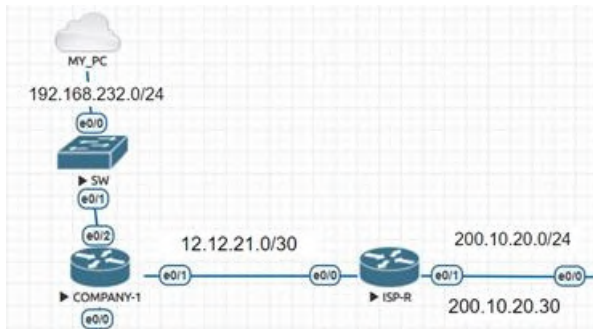
ADD A NEW NETWORK

Number of networks to add	1
Name/Prefix	MY_PC
Type	Management(Cloud0)
Left	750
Top	74
Save Cancel	

2.Here IP of the network is 192.168.232.0/24, here we put IP of that eve-ng network .so your IP might be change according to your eve-ng network IP



3. Assign one IP of that network to router interface ethernet 0/2=192.168.232.100
255.255.255.0



COMPANY-1(config)#interface ethernet 0/2

COMPANY-1(config-if)#ip address 192.168.232.100 255.255.255.0

COMPANY-1(config-if)#no shutdown

COMPANY-1(config-if)#exit

4. Advertise that route via routing of any means here we using Router EIGRP

```
router eigrp 100
 network 12.12.21.0 0.0.0.3
 network 192.168.10.0
 network 192.168.232.0
```

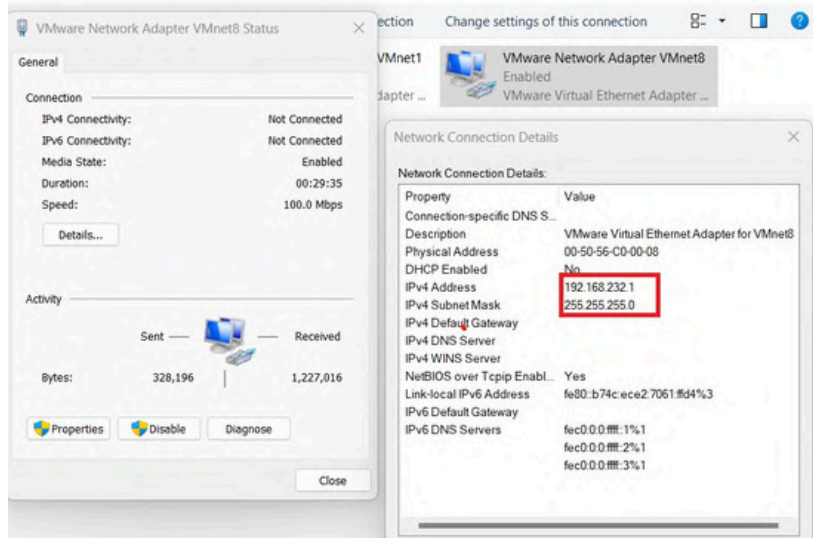
5. Then, go to your real PC/LAPTOP open **command prompt – run as administrator** and add a manual route to the eve-ng network

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>route add 0.0.0.0 mask 0.0.0.0 192.168.232.1
OK!

C:\Windows\System32>
```

The IP of 192.168.232.1 is the IP of your VMnet8 adapters



6. Then try to ping IP 192.168.232.100(router's IP)

```
C:\Windows\System32>ping 192.168.232.100

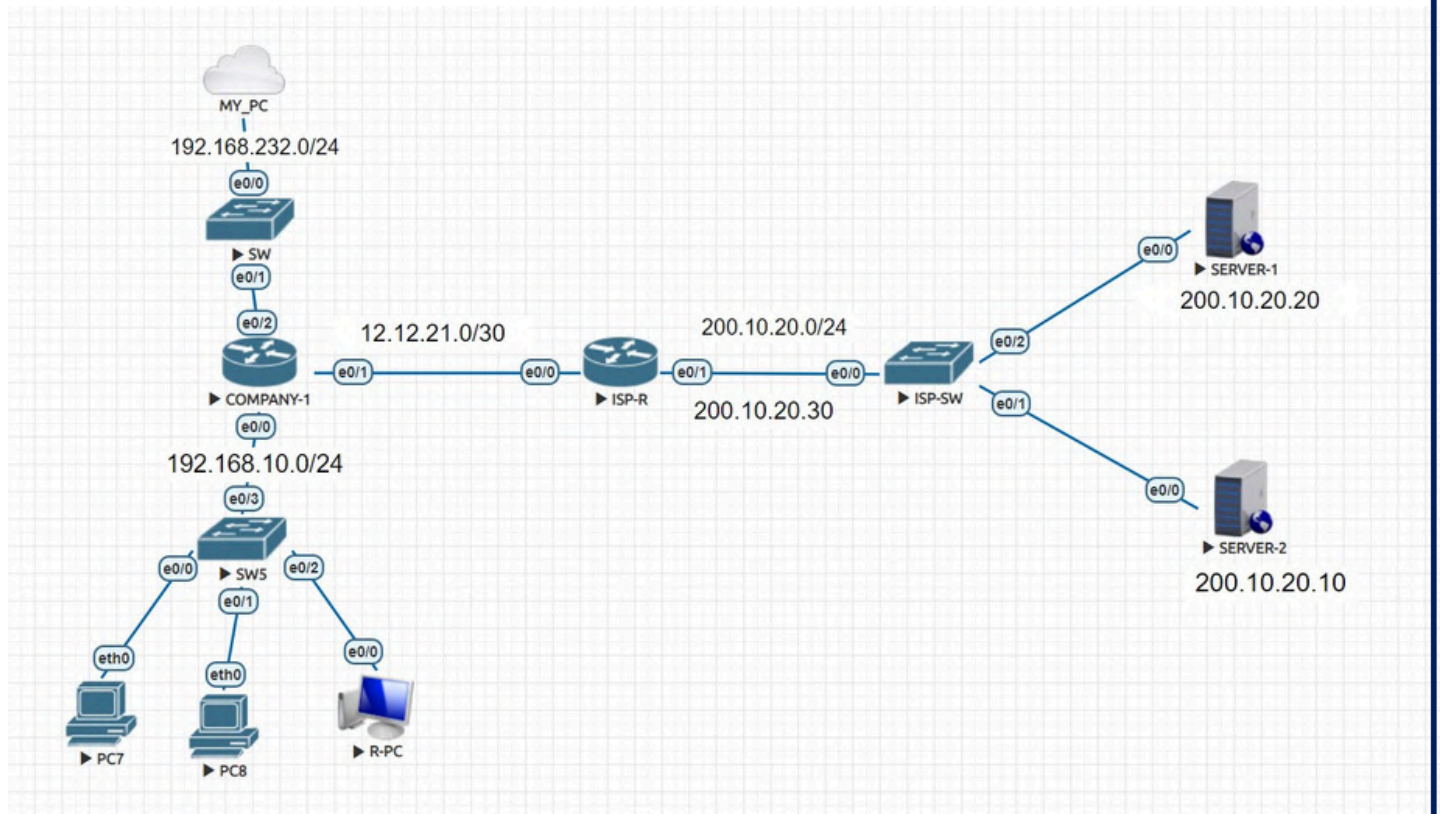
Pinging 192.168.232.100 with 32 bytes of data:
Reply from 192.168.232.100: bytes=32 time=2ms TTL=255
Reply from 192.168.232.100: bytes=32 time=1ms TTL=255
Reply from 192.168.232.100: bytes=32 time=1ms TTL=255
Reply from 192.168.232.100: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.232.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

If it is pinging it means your real PC/LAPTOP got connected with the LAB setup

SYSLOG

- ❖ Syslog means system logs. we generally use this log to identify the problem or trouble shoot the issue, to monitor the network, to identify the problematic area
- ❖ Syslog has basically **THREE METHODS**
 1. CONSOLE- all the logs generated by the device will show it on screen/terminal like putty, Secure-CRT
 2. BUFFER- All the logs generated by the device will stored it in buffer/memory of the device
 3. TRAP- All the logs generated by the device will trap and send it given specific IP



CONSOLE

COMPANY-1(config)#logging on(to start the logging on devices)

COMPANY-1(config)#logging console informational

BUFFER

COMPANY-1(config)#logging on

COMPANY-1(config)#logging buffered 4096 (allocate memory size)

COMPANY-1(config)#logging buffered informational (logs message till informational level)

Verification

COMPANY-1# show logging

TRAP

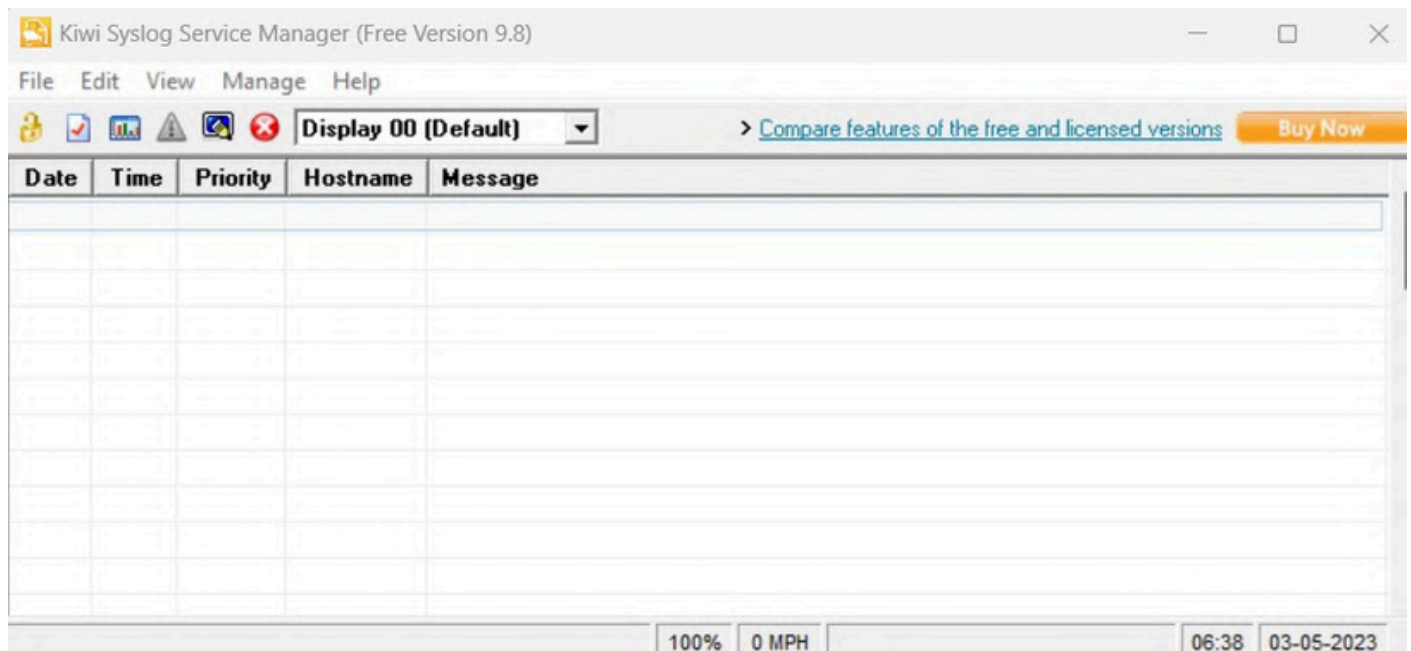
COMPANY-1(config)#logging on

COMPANY-1(config)#logging trap informational

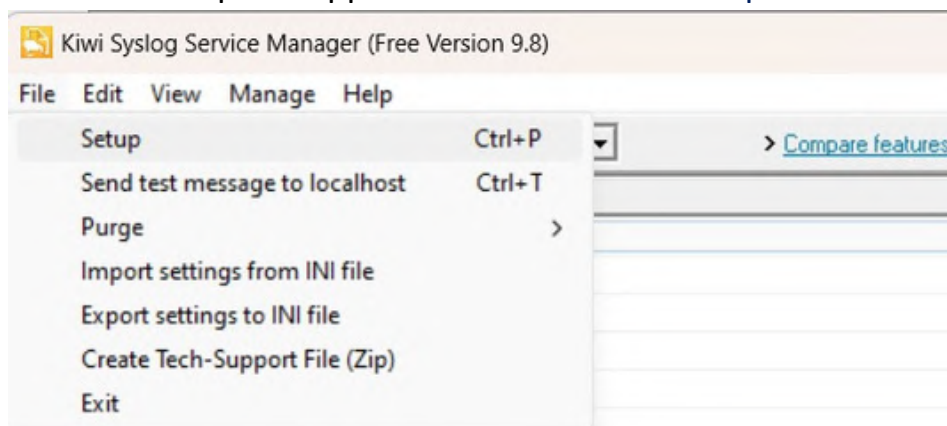
COMPANY-1(config)#logging host 192.168.232.1(logs are trapped and send it to this ip)

For trapped messages we need some tools to view it so here we use the 'kiwi syslog server'

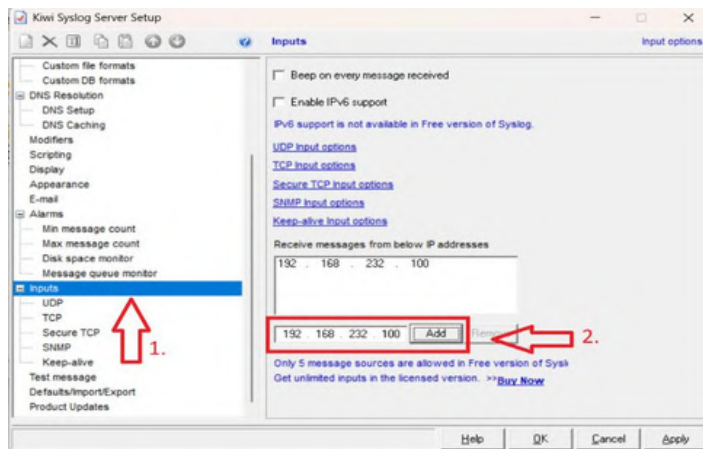
1.This is the kiwi syslog server look



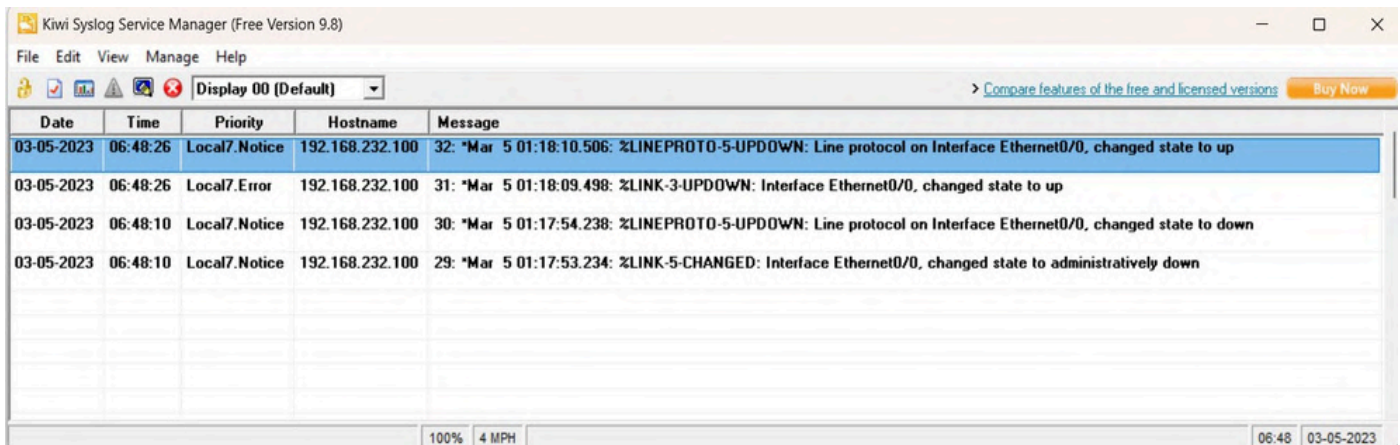
2.how to setup this application. Go to File > Setup



3.then , go to 1.Input > 2.Add IP address of the log generating device > Add > Apply > OK



4.finally you get message over here like this if event occur. here we done manual interface shutdown just to check messages are trapped or not. so, we get the messages.



CDP

- ❖ CDP stands for Cisco Discovery Protocol uses to find neighboring cisco devices only.
- ❖ Default, CDP timer is 60 sec, whereas CDP hold down timer is 180 sec.
- ❖ CDP is default running on cisco devices.

Commands

To run CDP on Global mode

```
COMPANY-1(config)#cdp run
```

To set CDP timer

```
COMPANY-1(config)#cdp timer .... <5-254>.....
```

To set CDP Holdtime

```
COMPANY-1(config)#cdp holdtime .....<10-255>.....
```

To enable CDP on interface

```
COMPANY-1(config)#interface ethernet 0/1
```

```
COMPANY-1(config-if)#cdp enable
```

```
COMPANY-1(config-if)#exit
```

```
COMPANY-1(config)#interface ethernet 0/0
```

```
COMPANY-1(config-if)#cdp enable
```

```
COMPANY-1(config-if)#exit
```

VERIFICATION

```
COMPANY-1#show cdp neighbors
```

```
COMPANY-1#show cdp neighbors detail
```

```
COMPANY-1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
Switch            Eth 0/2        158        R S I       Linux Uni  Eth 0/1
Switch            Eth 0/0        140        R S I       Linux Uni  Eth 0/3
R2                Eth 0/1        174        R B         Linux Uni  Eth 0/0

Total cdp entries displayed : 3
```

LLDP

LLDP is a Link Layer Discovery Protocol we use this to find out neighbor devices in a network. it is a Layer 2 protocol of OSI model. LLDP timer is 30 sec. whereas, LLDP holdtime is 120 sec.

Commands

To start LLDP Globally

```
COMPANY-1(config)#lldp run
```

To LLDP on interface (first start it Globally)

```
COMPANY-1(config-if)#interface ethernet 0/0
```

```
COMPANY-1(config-if)#lldp transmit
```

```
COMPANY-1(config-if)#lldp receive
```

```
COMPANY-1(config-if)#exit
```

```
COMPANY-1(config)#interface ethernet 0/1
```

```
COMPANY-1(config-if)#lldp transmit
```

```
COMPANY-1(config-if)#lldp receive
```

```
COMPANY-1(config-if)#exit
```

VERIFICATION

```
COMPANY-1#show lldp neighbors
```

```
COMPANY-1#show lldp neighbors detail
```

```
COMPANY-1#show lldp neighbors
Capability codes:
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID           Local Intf      Hold-time  Capability      Port ID
R2                   Et0/1          120        R               Et0/0

Total entries displayed: 1
```

PASSWORD ASSIGNMENT

How to assign password on router and switches?
the commands are same no changes

ROUTER/SWITCH

To Give Password to CONSOLE PORT

Router(config)#line con 0

Router(config-line)#password ccn

Router(config-line)#login

Router(config-line)#exit

User Access Verification

Password:

Password:

Router>

Router>

Router>

Router>

NOTE:-By Applying This Command Device Will Only Ask For The Password

If Company Wants To Ask For **Username And Password** To There Employee Then We Can Set Accordingly

Router(config)#line con 0

Router(config-line)#login local(by this it will ask for username & password at the time of console login)

Router(config-line)#exit

For that we have to create username and password

Router(config)#username ccn privilege 1 password ccn(make entry in clear text format)

OR

Router(config)#username ccn privilege 1 secret ccn123(make entry in clear encrypted format)

User Access Verification

```
Username: ccn
Password:
Router>
Router>
```

Applying password at PRIVILEGE MODE

Router(config)#enable password ccn1234

This will enable password at privilege mode with clear text format

```
Router(config)#do show running-config | section enable
enable password ccn1234
```

VERIFICATION

User Access Verification

```
Username: ccn
Password:
Router>enable
Password:
Password:
Password:
Router#
```

For that we have to encrypt this password

Router(config)#service password-encryption

```
Router(config)#do show running-config | section enable
enable password 7 03075805575D7218
```

We can crack this password

<https://www.firewall.cx/cisco-technical-knowledgebase/cisco-routers/358-cisco-type7-password-crack.html>

to decrypt this password go to above link and put this value on Encrypted Password section “ 03075805575D7218 ” and press submit button.

Enter Your Encrypted Password Below:

Encrypted Password:

Decrypted Password:

Now, because of this weakness we most probably assign password in encrypted format

Router(config)#enable secret ccn123

```
Router(config)#do show running-config | section enable
enable secret 5 $1$36UY$1D4YFKJc.LsdkzzrZSKzy/
```

This password is now encrypted in md5 hash value. Now, this value is not reversible.

PASSWORD RECOVERY

if we fail to remember the password then the only condition is to recover the password this situation is not that often in company. we need to know what to do in such condition.

ROUTER PASSWORD RECOVERY

1. First Check Router Register Value

Router#show version

In this command we can see the register value

Configuration register is 0x2102

- 0x2102 is a default value by reading this value router read the startup-config
- 0x2142 is another value by reading this value router skip the startup-config

So we have to change this value but we can't get inside the router CLI .so what else we can do is

Go to routers rommon mode (read only memory monitor mode).

1. Restart the router

2. When router is decompressing the iso image press button

CTRL + C

OR

CTRL + PAUSE/BREAK

3. And enter in rommon mode

Now change the register value

rommon 1 > confreg 0x2142

rommon 2 > boot

after booting the router go to privilege type **#show version**

Configuration register is 0x2142

It will show you 0x2142

Now, bring startup-config to running-config

Router#copy startup-config running-config

4. It is the time to change the password or remove the password

5. After changing the password change the register value as well

Router(config)#config-register 0x2102

Now, to verify

Router#show version

It will show you output like this

Configuration register is 0x2142 (will be 0x2102 at next reload)

After all this save the configuration

Router# write memory

SWITCH PASSWORD RECOVERY

Because of password we can't enter in switch so we have to recover the password by entering in switch mode. by clicking mode button on cisco switch while decompressing the image. We will enter in switch mode.



switch: flash_init(to initiate flash)

switch:dir flash:(to check the flash content)

switch store all the configuration in config.text file so if we change the file extension then the file will not be in readable format

switch: rename flash:config.text config:config.aaa

switch:reset(to start the switch)

after booting the switch,
check out the flash of switch

switch#dir flash:

now you can see the **config.aaa** file > change it back to **config.text**

switch#rename flash:config.aaa flash:config.text

now,get back your startup-config

switch#copy startup-config running-config

change all the password and then save the configuration

IOS UPGRADATION

IOS stands for **INTERNETWORK OPERATING SYSTEM** which is a Basic OS for all cisco Router and Switches.

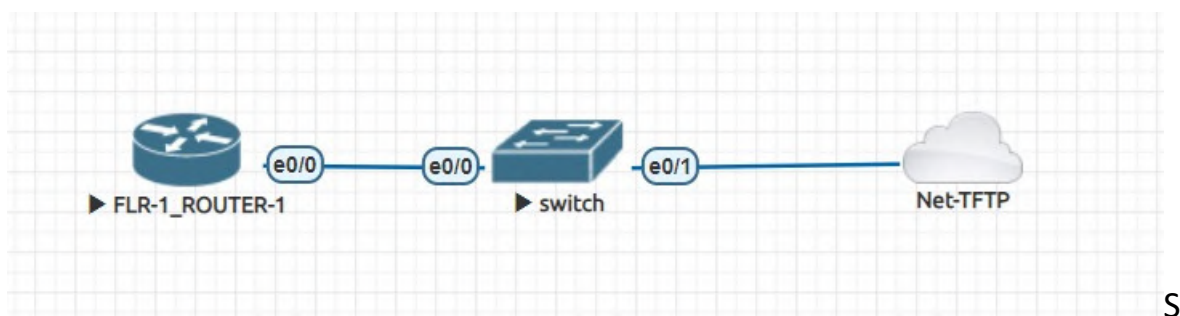
Certainly, Cisco launches a New updates in their operating system like current version of cisco IOS is 15.1.1 after some fixes cisco launches a new update which will be like 15.1.2. so here we have to upgrade the old OS with new one.

How to do it?

Let's see,

On ROUTER,

Here is the reference diagram,



In the above reference lab “FLR-1_ROUTER-1” is the main router. then, that Net-TFTP is our real pc connected in a network

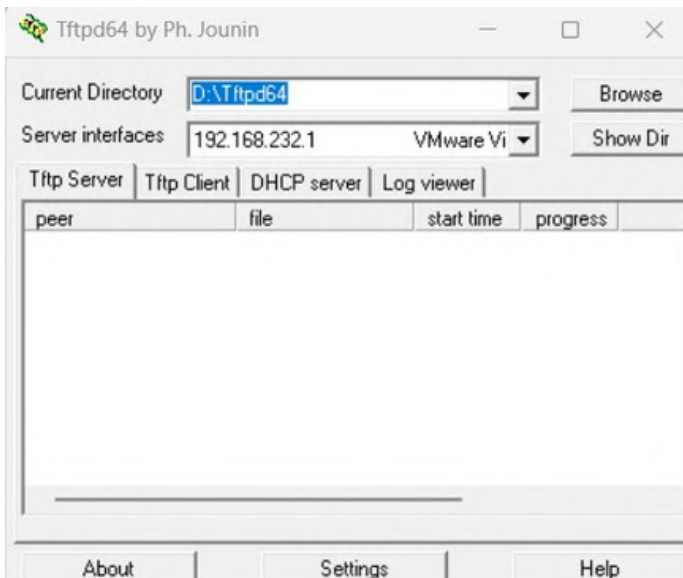
Before going for upgradation we took backup of the configuration file and current IOS version just to take care from future uncertainty.

1. Take Configuration backup of Device

```
FLR-1_ROUTER-1#copy running-config tftp:
Address or name of remote host []? 192.168.232.1
Destination filename [flr-1_router-1-config]?
!!
1121 bytes copied in 0.055 secs (20382 bytes/sec)
```

we are using tftpd64 tool to create our pc as a TFTP se

NOTE:- All the upcoming task like IOS upgradation , configuration file and IOS file backup .all are done on real devices or we can do it cisco packet tracer if you don't have devices to co



Here is the first look of the tftpd64 tool

2. Take IOS back up on device

FLR-1_ROUTER-1#copy flash: tftp:

Then, provide remote address of the host means address of TFTP server => now provide source file which will be going copy from routers flash to

```
FLR-1_ROUTER-1#copy flash: tftp
Source filename []? c2800nm-advispervicesk9-mz.151-4.M1.bin
Address or name of remote host []? 192.168.10.1
Destination filename [c2800nm-advispervicesk9-mz.151-4.M1.bin]?
```

Here we can take backup of IOS file

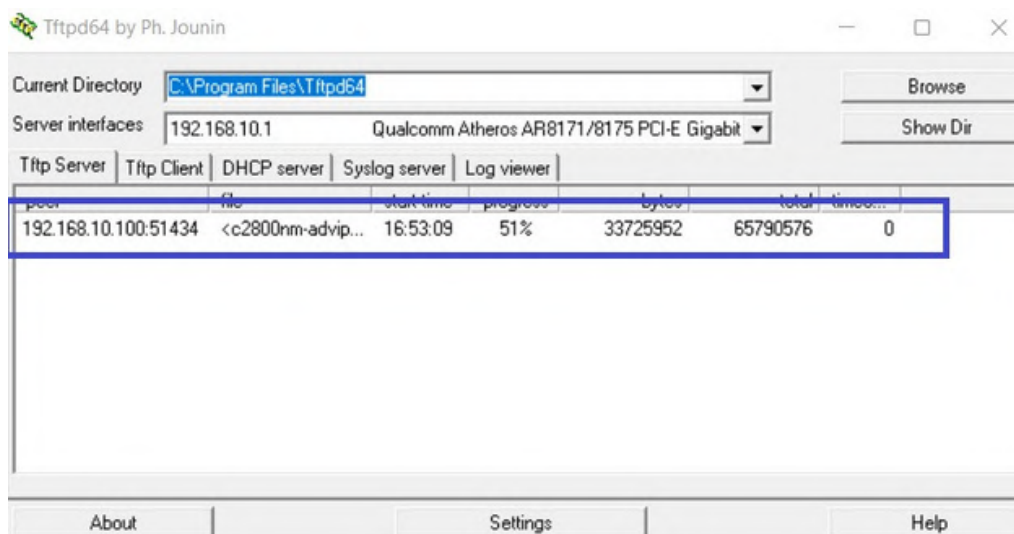
Now, how to load IOS in device (it only support actual device)

FLR-1_ROUTER-1#copy tftp: flash:

```
FLR-1#copy tftp: flash:
Address or name of remote host []? 192.168.10.1
Source filename []? c2800nm-adviservicesk9-mz.151-4.M1.bin
Destination filename [c2800nm-adviservicesk9-mz.151-4.M1.bin]?
%Warning:There is a file already existing with this name
Do you want to over write? [confirm]
Accessing tftp://192.168.10.1/c2800nm-adviservicesk9-mz.151-4.M1.bin...
Erase flash: before copying? [confirm]
Erasing the flash filesystem will remove all files! Continue? [confirm]
Erasing device... 
.....
...erased
Erase of flash: complete
Loading c2800nm-adviservicesk9-mz.151-4.M1.bin from 192.168.10.1 (via FastEthernet0/0): !!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 65790576 bytes]

Verifying checksum... cccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc
cccccccccccc OK (0xAA5)
65790576 bytes copied in 1854.112 secs (35484 bytes/sec)
```

While copying of this data, tftpd64 looks like



Provide them remote address of tftp server and file name of the IOS which we want to load. Then it will start copying the file. It will take some time to copy data.

Similarly, we can do reversal of configuration file.

Provide them remote address of the tftp server. And source file name.

WHAT IF WE DO HAVE TWO OPERATING SYSTEM ON DEVICE. then,

```
FLR-1_ROUTER-1(config)#boot system flash c2800nm-advipservicesk9-mz.151-4.M1.b$
```

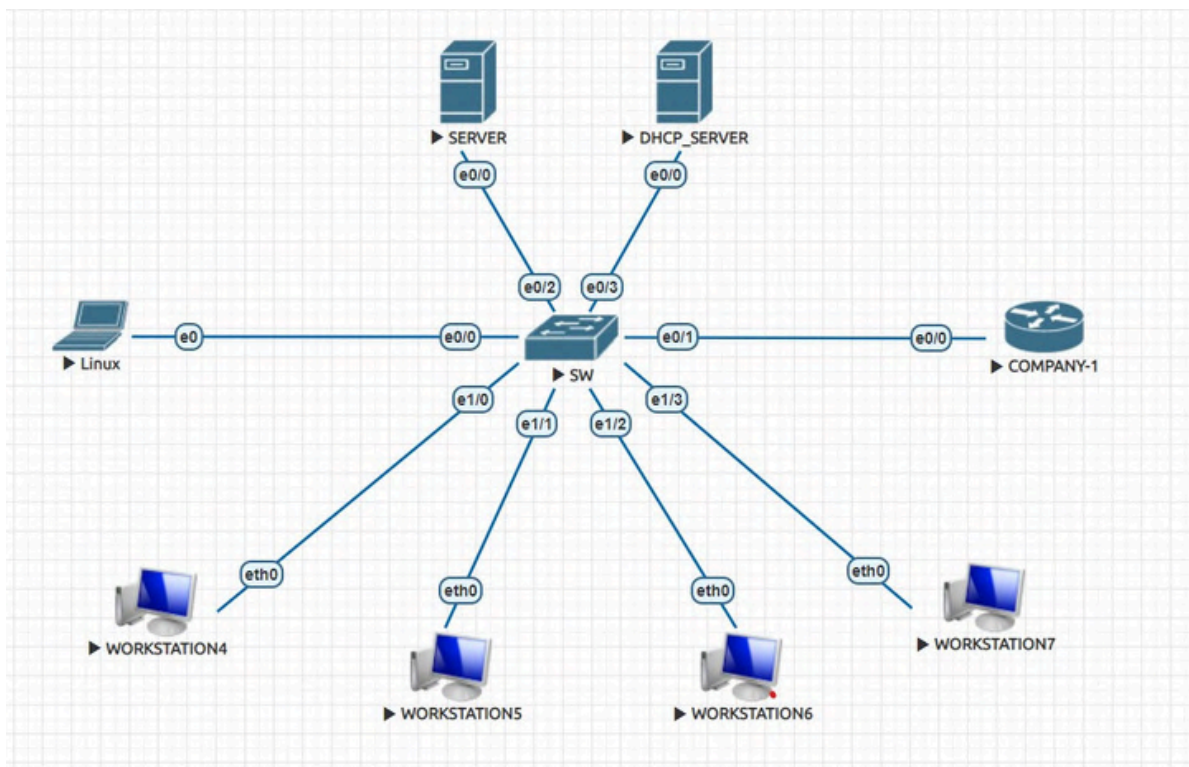
By applying the above command, we can specify the OS which will be loading at the time of booting.

L2 SECURITY

- ❖ L2 security means overall switch security. when it comes to LAN network we have to secure the network switches and when it comes to internet we have to secure routers by doing proper configuration.
- ❖ So, in LAN network there is possibility that a rouge employee can attack a company's IT Infra with some malicious intention. Just like to steal company's data, to create jitter in a network, to redirect company's employees to malicious site just to gather information.
- ❖ in such situation how can we secure our organization, so here is the solution – L2 security

In This Scenario We Have to Consider Potential Threats And Make Our Lan Network More Secure Against The Attacks

<u>ATTACKS</u>	<u>SECURITY</u>
DHCP starvation attack	SWITCHPORT SECUIRTY
MAC spoofing	
MAC flooding	
CDP flooding	
Rouge DHCP server / DHCP spoofing	DHCP SNOOPING
IP spoofing	IP SOURCE GUARD
ARP poisoning	DYNAMIC ARP INSPECTION (DAI)
STP attacks	BPDU GUARD / BPDU FILTER / ROOT GUARD

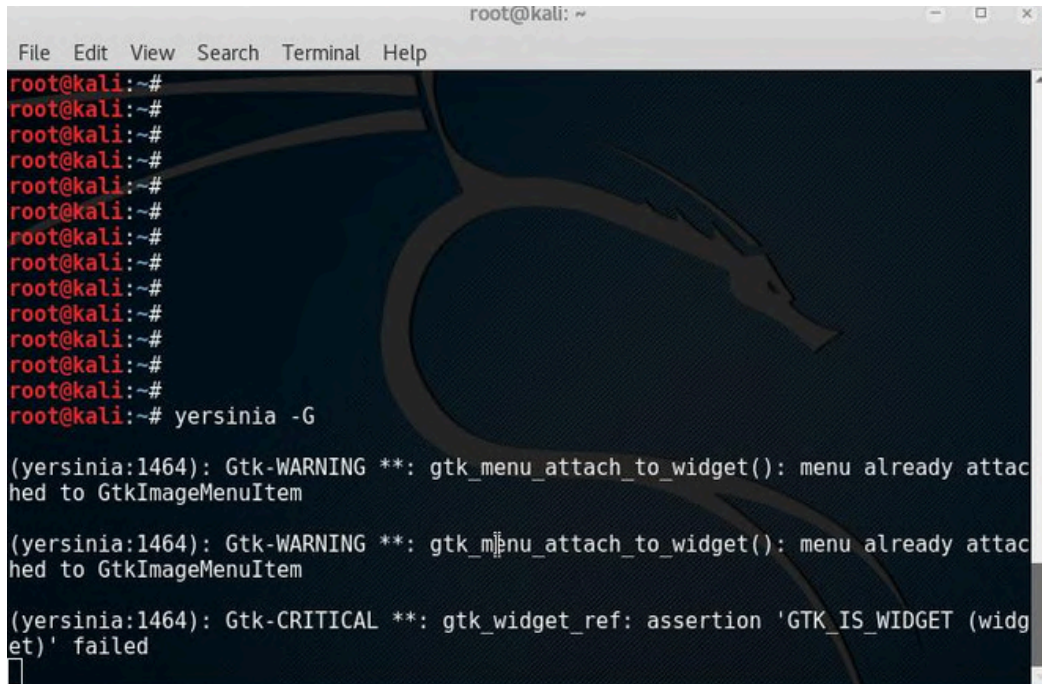


1.DHCP STRAVATION ATTACK

In this attack linux PC fetch all the IP of DHCP pool and left it with 0 IP's to assign to genuine user.

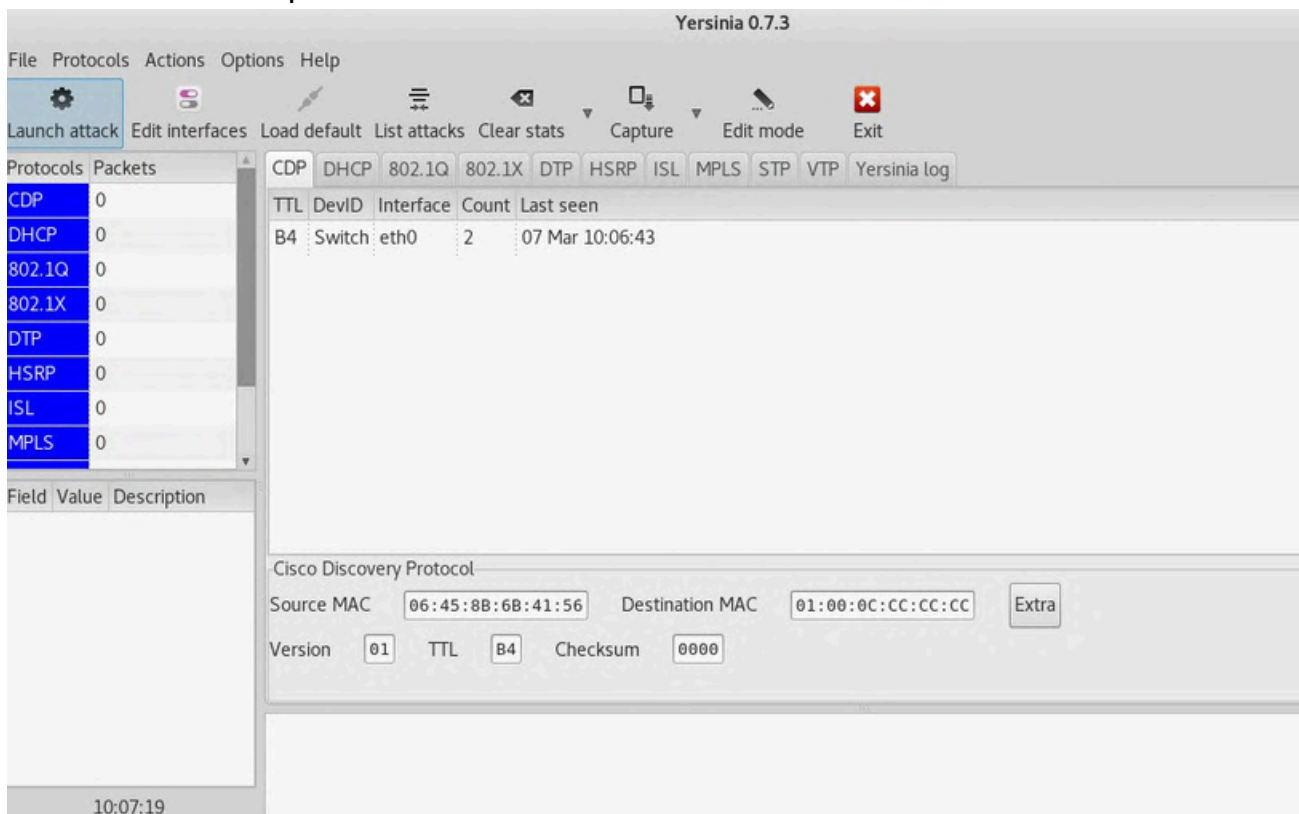
How to perform this attack

Go to linux =>Terminal => yersinia -G



```
root@kali: ~  
File Edit View Search Terminal Help  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~#  
root@kali:~# yersinia -G  
  
(yersinia:1464): Gtk-WARNING **: gtk_menu_attach_to_widget(): menu already attached to GtkImageMenuItem  
  
(yersinia:1464): Gtk-WARNING **: gtk_menu_attach_to_widget(): menu already attached to GtkImageMenuItem  
  
(yersinia:1464): Gtk-CRITICAL **: gtk_widget_ref: assertion 'GTK_IS_WIDGET (widget)' failed  
█
```

Tool like this will open



This tool can perform different types of attacks

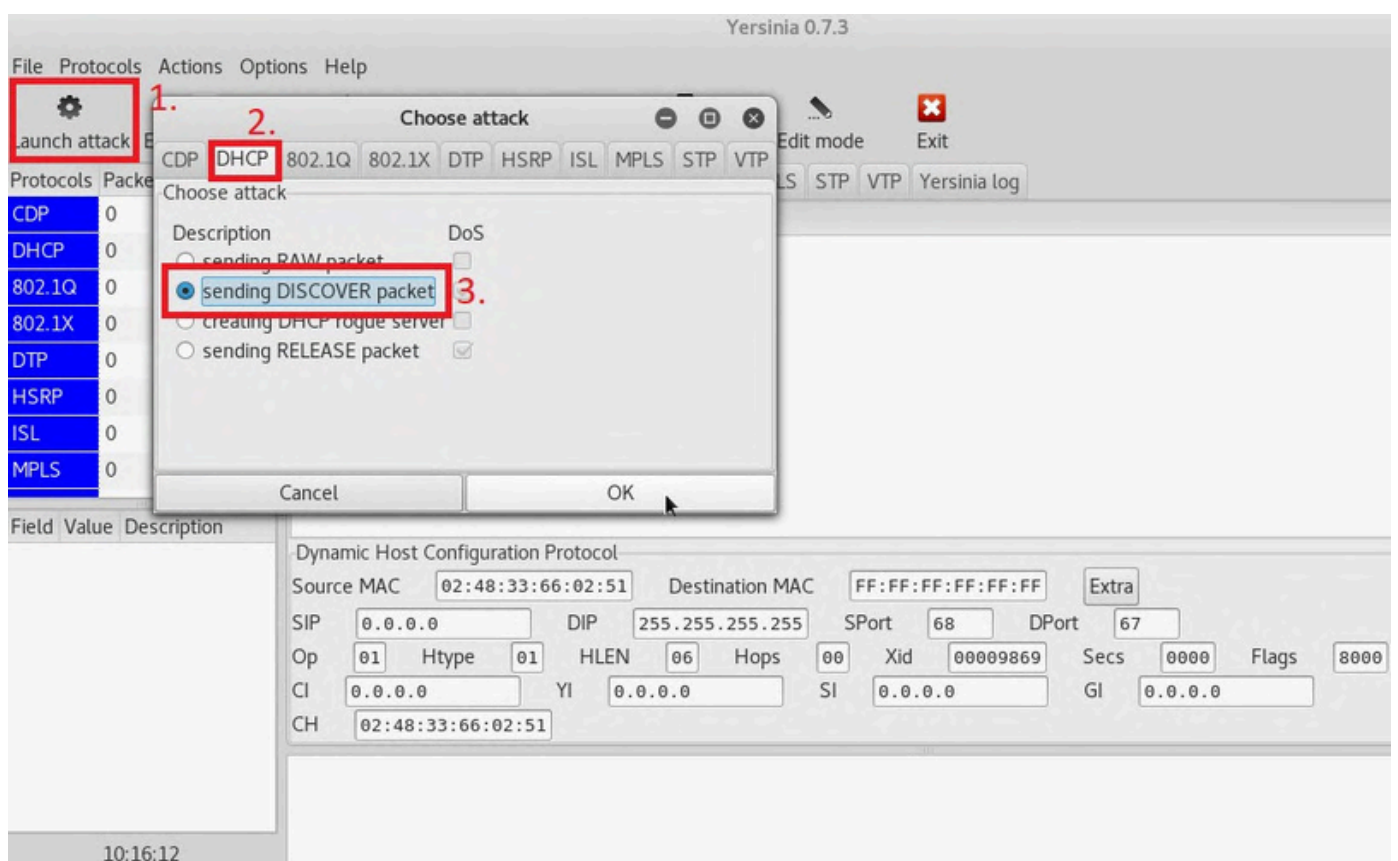
First, we are going to do DHCP starvation attack

Before this attack, the DHCP server allotted only two IP's

```
DHCP_SERVER#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/
                    Hardware address/
                    User name
192.168.10.1         0050.0000.0900      Mar 08 2023 04:59 PM  Automatic
192.168.10.2         0100.5079.6668.04   Mar 08 2023 05:11 PM  Automatic
```

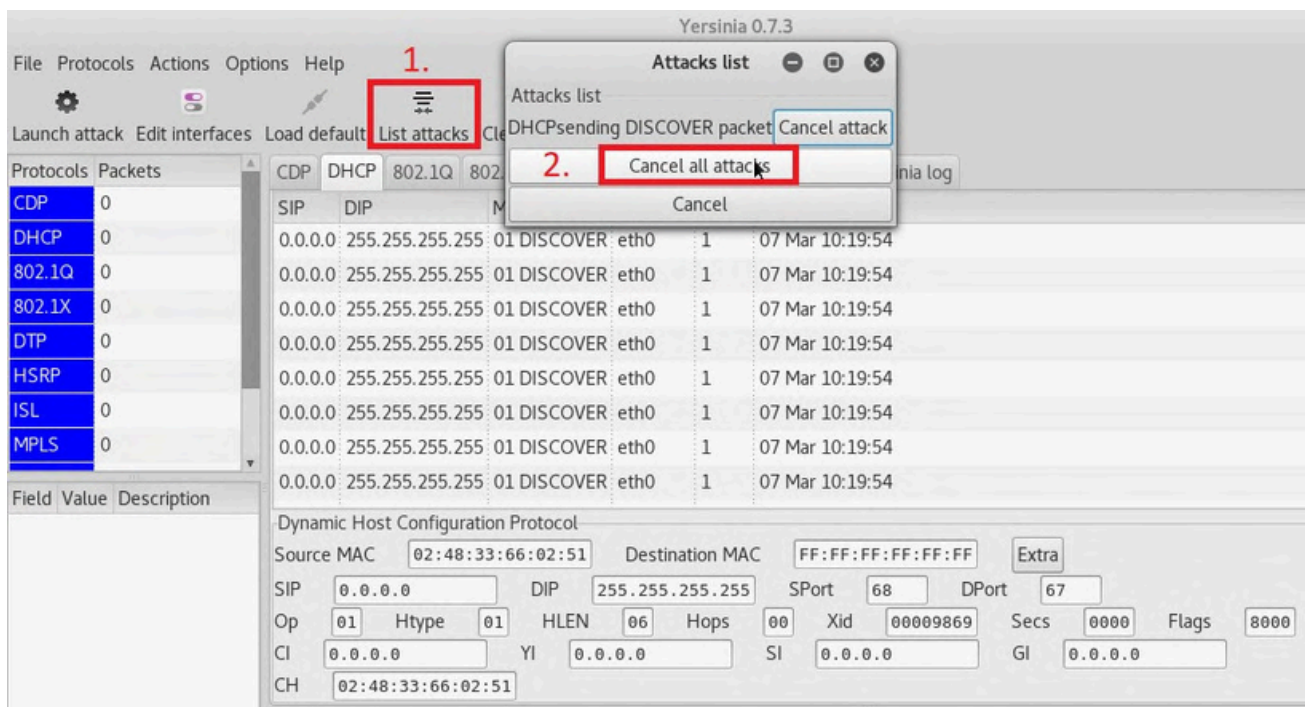
Let's perform attack

1. launch attack => 2. DHCP tab => 3. Sending DISCOVER packet => OK



How to stop attack

1. list attack => 2. Cancel all attacks



Effect of this attack is pool of DHCP server will get empty

```
DHCP_SERVER# show ip dhcp pool

Pool CCN-LAN :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 251
  Pending event                    : none
  1 subnet is currently in the pool :
  Current index      IP address range  Leased addresses
  0.0.0.0           192.168.10.1 - 192.168.10.254 251
```

See leased addresses is 251 means all IP's of this pool is allotted.

Effect on switch

```
Switch#show mac address-table count

Mac Entries for Vlan 1:
-----
Dynamic Address Count : 13635
Static Address Count  : 0
Total Mac Addresses   : 13635

Total Mac Address Space Available: 212588024
```

See dynamic address count is 13,635 it means switch reads multiple MAC addresses. we also can see it with command

#show mac address-table

2.MAC SPOOFING

Mac spoofing means to mask someone's mac address and put it on our interface.

How to perform this attack

Select a target = workstation4 is our target => command - show ip => observe MAC – 00:50:79:66:68:04

```
WRKST4> show ip

NAME           : WRKST4[1]
IP/MASK        : 192.168.10.2/24
GATEWAY        : 192.168.10.254
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.253
DHCP LEASE     : 82504, 86400/43200/75600
DOMAIN NAME    : ccn.com
MAC            : 00:50:79:66:68:04
LPORT          : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500
```

Observe this mac of workstation because we are spoofing this mac and putting it on linux pc

```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.1 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::250:ff:fe00:900 prefixlen 64 scopeid 0x20<link>
    ether 00:50:00:00:09:00 txqueuelen 1000 (Ethernet)
    RX packets 3472 bytes 306366 (299.1 KiB)
    RX errors 0 dropped 23 overruns 0 frame 0
    TX packets 576290 bytes 164810004 (157.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Default mac address of this linux pc 00:50:00:00:09:00

Now we have to change linux mac address with workstation4 mac address

With the use of **macchanger** tool command => **macchanger -m 00:50:79:66:68:04 eth0**

```
root@kali:~# macchanger -m 00:50:79:66:68:04 eth0
Current MAC: 00:50:00:00:09:00 (NEXO COMMUNICATIONS, INC.)
Permanent MAC: 00:50:00:00:09:00 (NEXO COMMUNICATIONS, INC.)
New MAC: 00:50:79:66:68:04 (PRIVATE)
```

New spoofed mac address of linux pc

```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.1 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::250:ff:fe00:900 prefixlen 64 scopeid 0x20<link>
    ether 00:50:79:66:68:04 txqueuelen 1000 (Ethernet)
    RX packets 3472 bytes 306366 (299.1 KiB)
    RX errors 0 dropped 23 overruns 0 frame 0
    TX packets 585498 bytes 165307468 (157.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

This is how we can spoof the mac address

How to change back it to normal

Same macchanger tool with command of > **macchanger -p eth0**

```
root@kali:~# macchanger -p eth0
Current MAC: 00:50:79:66:68:04 (PRIVATE)
Permanent MAC: 00:50:00:00:09:00 (NEXO COMMUNICATIONS, INC.)
New MAC: 00:50:00:00:09:00 (NEXO COMMUNICATIONS, INC.)
```

Check it on interface properties

```
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.1 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::250:ff:fe00:900 prefixlen 64 scopeid 0x20<link>
    ether 00:50:00:00:09:00 txqueuelen 1000 (Ethernet)
    RX packets 3472 bytes 306366 (299.1 KiB)
    RX errors 0 dropped 23 overruns 0 frame 0
    TX packets 576290 bytes 164810004 (157.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3.MAC FLOODING

In mac flooding attack we flood the mac address table of the switch.

Before this attack switch mac table

```
Switch#show mac address-table count

Mac Entries for Vlan 1:
-----
Dynamic Address Count      : 3
Static Address Count       : 0
Total Mac Addresses        : 3

Total Mac Address Space Available: 212588024
```

Now perform an attack with the command of ~#**macof -i eth0**

```
root@kali:~# macof -i eth0
```

To stop this attack press **CTRL+C**

After attack switch MAC Address table get flooded

```
Switch#show mac address-table count

Mac Entries for Vlan 1:
-----
Dynamic Address Count      : 13030
Static Address Count       : 0
Total Mac Addresses        : 13030

Total Mac Address Space Available: 212588024
```

4.CDP FLOODING

In the CDP Flooding Attack, we flood the CDP table of the switch with fake entries

Before starting the attack

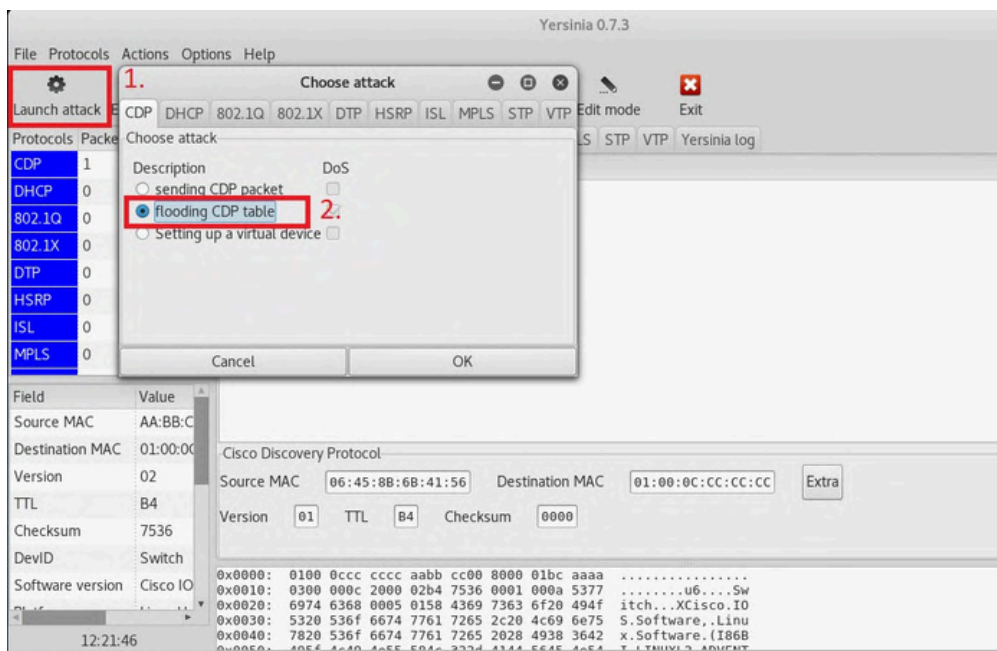
```
Switch#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
SERVER            Eth 0/2         156        R B          Linux Uni  Eth 0/0
DHCP_SERVER       Eth 0/3         156        R B          Linux Uni  Eth 0/0
COMPANY-1         Eth 0/1         166        R B          Linux Uni  Eth 0/0

Total cdp entries displayed : 3
```

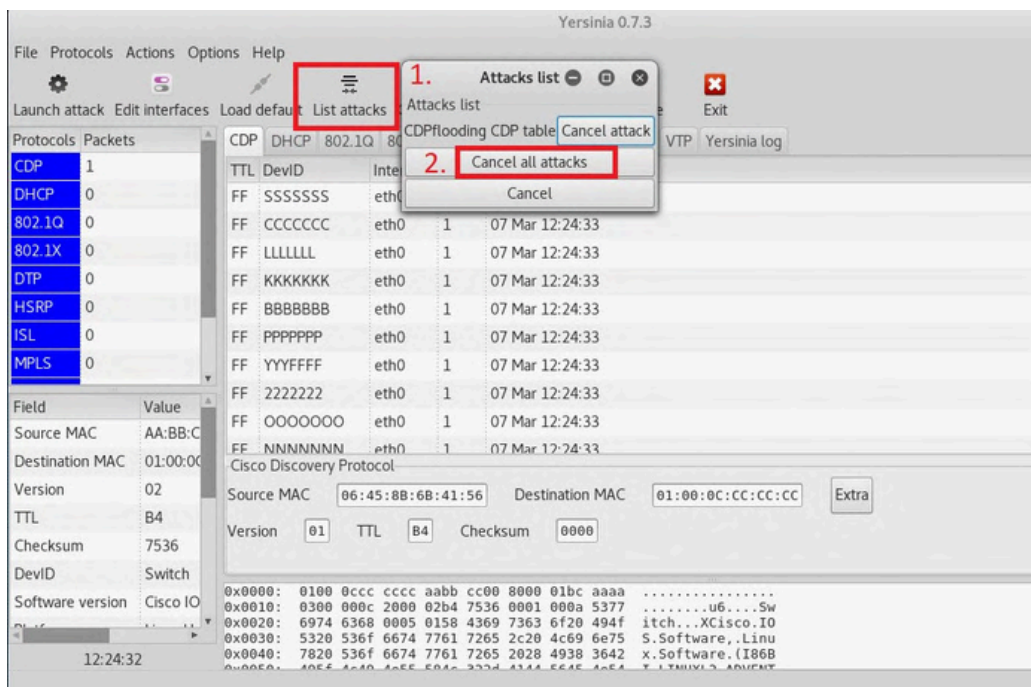
Check it out, at last CDP entries is 3 Now, perform CDP flooding attacks [Go to](#)

terminal => yersinia -G > Move to CDP tab => flooding CDP table > OK



To stop attack

List attacks => cancel all attacks



After this attack check CDP table, you will see it flooded.

Now, how can we secure switches from such type of attacks cause allof this attacks are mac based attacks.

Here we use **SWITCHPORT SECURITY**

Switch(config)#interface eth 0/0

Switch(config-if)#switchport mode access

Switch(config-if)#switchport port-security mac-address 00:50:00:00:09:00(manual mac)

OR

Switch(config-if)#switchport port-security mac-address sticky(sticky means any mac address)

Switch(config-if)#switchport port-security maximum 1(max mac accepted is 1)

Switch(config-if)#switchport port-security violation protect ..(take action if max limit violate)

Switch(config-if)#switchport port-security(to activate this above parameter)

For violation we do have 3 options. we can use anyone of them as per policy.

- **Protect**—This mode permits traffic from known MAC addresses to continue to be forwarded while dropping traffic from unknown MAC addresses when over the allowed MAC address limit. When configured with this mode, no notification action is taken when traffic is dropped.
- **Restrict**—This mode permits traffic from known MAC addresses to continue to be forwarded while dropping traffic from unknown MAC addresses and make interface status as error-disable. when over the allowed MAC address limit. When configured with this mode, a syslog message is logged, a Simple Network Management Protocol (SNMP) trap is sent, and a violation counter is incremented when traffic is dropped.
- **Shutdown**—This mode is the default violation mode; when in this mode, the switch will automatically force the switchport into an error disabled (*err-disable*) state when a violation occurs. While in this state, the switchport forwards no traffic. The switchport can be brought out of this state by issuing the *errdisable recovery cause* CLI command or by disabling and reenabling the switchport.

Basically, we activate switchport security on all the Access Ports interfaces except the trunk interface.

5. ROUGE DHCP SERVER/DHCP SPOOFING

In this attack we create a fake/rouge DHCP server to redirect users from its original network. This attack is a part of MITM (man in the middle) attack.

Let's perform this attack

First, we have to do DHCP starvation attack and fetch all the IP's. so that genuine user won't get IP from original DHCP server. now the real task is to generate Rouge DHCP in a network and PC's in a network get Rouge DHCP network's IP.

Go to Linux - terminal => **ettercap -G**

```
root@kali:~# ettercap -G
```

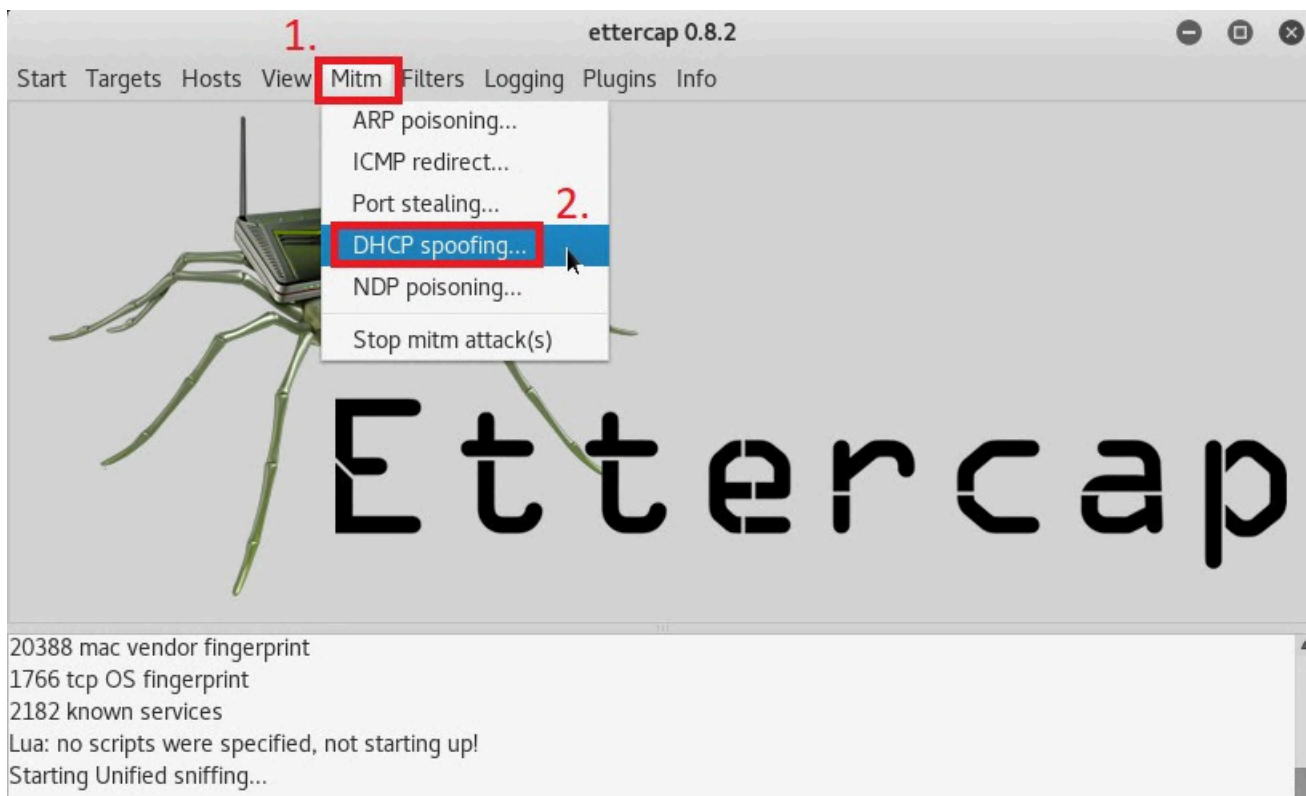
Now tool like this will open



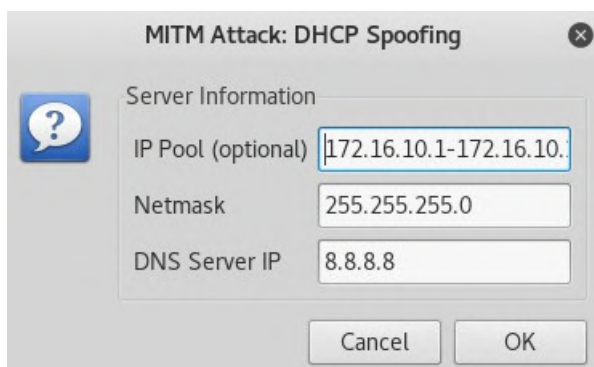
After opening to click on Sniff => unified sniffing => Network interface – eth0



Then, go to MITM > DHCP Spoofing



Now set the parameter and then click OK



Will create a Rouge DHCP server

Our existing network is 192.168.10.0/24but, Rogue DHCP Pool IP is 172.16.10.0/24

After, Rogue DHCP server creation pc willtake IP of that network

```
WRKST4> ip dhcp
DORA IP 172.16.10.1/24 GW 192.168.10.1

WRKST4> show ip

NAME       : WRKST4[1]
IP/MASK    : 172.16.10.1/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 1793, 1800/900/1575
MAC        : 00:50:79:66:68:04
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500
```

Now we have to secure this network from such attack with **DHCP snooping security**

DHCP SNOOPING SECURITY

In this security we are untrusting all the interfaces of switch except one which is directly connected to DHCP Server. So, we have to make that interface trust by applying command.

Switch(config)#ip dhcp snooping vlan 1(to set snooping parameter)

Switch(config)#ip dhcp snooping(to activate DHCP snooping)

Now we are trusting only one port which is connected to Actual DHCP server

Switch(config)#interface ethernet 0/3

Switch(config-if)#ip dhcp snooping trust

Switch(config-if)#exit

When we activate DHCP Snooping it will add on additional information to the packet- that information is VLAN ID and interface details. Now this additional information won't understand by actual DHCP server for that we have to put command to reply it back

DHCP_SERVER(config)#ip dhcp relay information trust-all

By this command router will understand the additional information

VERIFICATION

```
Switch#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
00:50:79:66:68:04  192.168.10.2   86370      dhcp-snooping  1     Ethernet1/0
Total number of bindings: 1
```

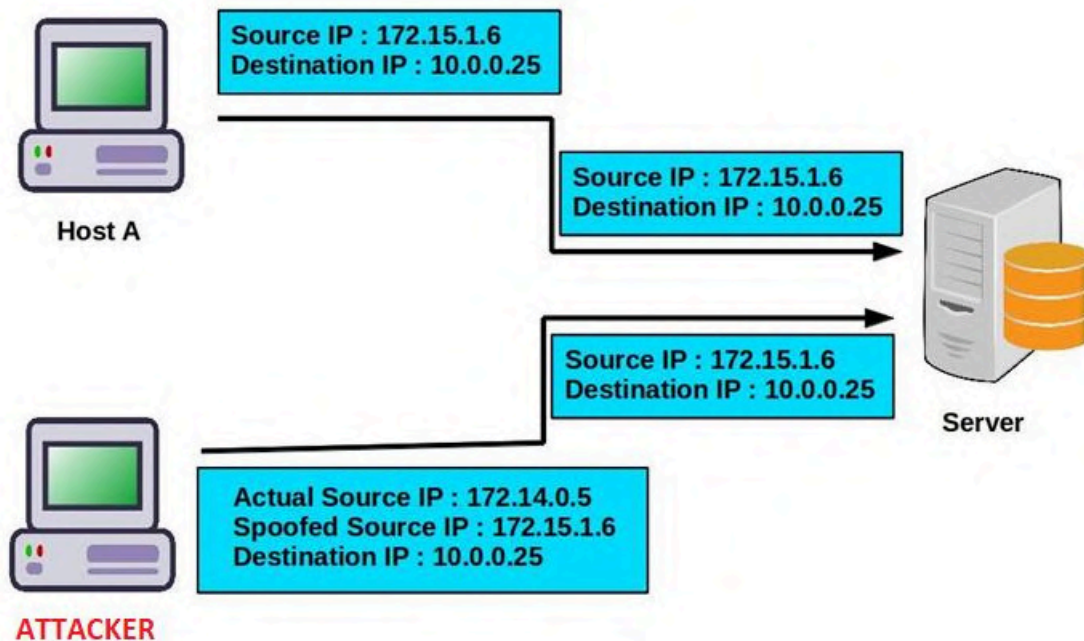
In this image you can see the **VLAN** & **INTERFACE** heading and it actually adds up that information to the data packet.

We implement this security on global mode to untrust all the ports. We make interface trust only for actual port which is connected to DHCP server and the trunk ports from where the traffic is ingresses.

6.IP SPPOFING

In this type of attack we spoof the IP of targeted pc with some malicious intent. if we able to spoof that IP we will represent ourselves as same as that pc

IP Address Spoofing



In this scenario attacker successfully spoofed the IP of Host A. by spoofing the IP attacker can bypass the ACL.

We change the IP by assigning static ip .then, only we can change the actual ip with spoofed ip.to stop all such attack we implement a security called **IP SOURCE GUARD**

IP SOURCE GUARD

IP Source Guard (IPSG) is a security feature that restricts IP traffic on untrusted Layer 2 ports by filtering traffic based on the DHCP snooping binding database or manually configured IP source bindings

To implement security

```
Switch(config-if)#interface ethernet 1/0
```

```
Switch(config-if)#ip verify source
```

```
Switch(config-if)#exit
```

VERIFICATION OF DHCP SNOOPING BINDING

```
Switch#show ip dhcp snooping binding
```

OR

Manual IP SOURCE BINDING

```
Switch(config)#ip source binding 00:50:00:00:02:00 vlan 1 192.168.10.100 interface ethernet 0/1
```

OR

This IP SOURCE GUARD we verify with the port-security

```
Switch(config-if)#interface ethernet 1/0
```

```
Switch(config-if)#ip verify source port-security
```

```
Switch(config-if)#exit
```

VERIFICATION

```
Switch# show ip verify source
```

DYNAMIC ARP INSPECTION (DAI)

Dynamic ARP inspection is a security feature we used reduce a risk platform regarding arp requests. Like ARP flooding, ARP poisoning, etc

Implementation

```
Switch(config)#ip arp inspection vlan 1
```

By applying this command switch untrust all the ports for ARP requests. we left all the ports as untrust for PC's . we most probably make port trust for the Servers and trunk ports which is only connected with the other switch.If port is in trust then it bypass arp inspection validation check.

How to do it

```
Switch(config)#interface ethernet 0/2
```

```
Switch(config-if)#ip arp inspection trust .....(to make interface trust for ARP inspection)
```

```
Switch(config-if)#ip arp inspection limit rate 5 .....(to limit the arp packet per sec)[OPTIONAL]
```

```
Switch(config-if)#exit
```

DYNAMIC ARP INSPECTION VERIFY THE ARP PACKET WITH THE DHCP SNOOPING BINDING TABLE

to statically verifies the ARP entries

```
Switch(config)#arp access-list ARP-LIST
```

```
Switch(config-arp-nacl)# permit ip host 192.168.10.253 mac host aabb.cc00.2000
```

```
Switch(config-arp-nacl)#exit
```

```
Switch(config)#ip arp inspection filter ARP-LIST vlan 1
```

VERIFICATION

```
Switch# show ip arp inspection
```

```
Switch# show ip arp inspection statistics
```

7.STP ATTACKS

Spanning tree protocol is also a attackable platform on a switch. so, what kind of attacks will perform on a switch. spanning tree root status stealing/spoofing , BPDU flooding ,etc.

Just to save our network from such types of attack we will implement some sort of security

BPDU GUARD

```
Switch(config)#interface ethernet 1/0
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpduguard enable
Switch(config-if)# exit
```

We enable BPDUGUARD with PORTFAST because, when enable PORTFAST switch won't send BPDU on that interface and when we enable BPDU GUARD switch won't receive any BPDU's on that interface

BPDU FILTER

```
Switch(config)#interface ethernet 1/0
Switch(config-if)#spanning-tree portfast
Switch(config-if)#spanning-tree bpdufilter enable
Switch(config-if)#exit
```

If we use BPDUFILTER only then what does BPDUFILTER do is it will listen the BPDU's but won't reply it back and PORTFAST won't send an update on that interface. and when we implement BPDUFILTER with PORTFAST it will create some mess around like if we receive a BPDU from an unexpected interface it will lose its status of PORTFAST and it will create a temporary looping.

So, in most cases we prefer BPDU GUARD with PORTFAST

ROOT GUARD

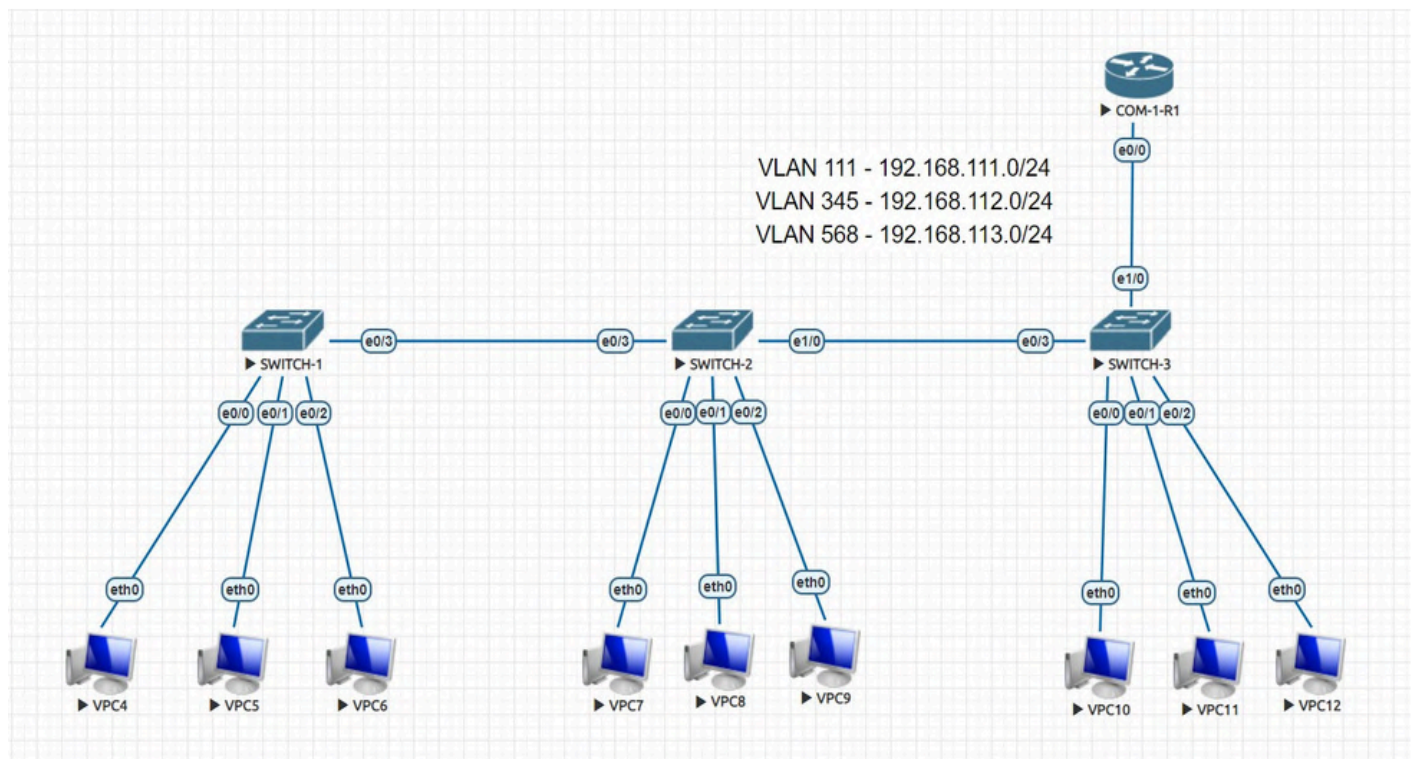
We do use root guard because we don't want that any lowest priority or lowest mac address will influence the switch STP selection process. what does root guard do is it will ignore the bpdu which consist the lowest priority and lowest mac

```
Switch(config)#interface ethernet 1/0
Switch(config-if)#spanning-tree guard root
Switch(config-if)#exit
```

VERIFICATION

```
Switch#show running-config | section interface ethernet 1/0
```

L2 SECURITY LAB



TASK – 16

Task 16.1 Host Name Should Be Same As Mention In First Look Name

Task 16.2 Create and Implement

VLAN - 111 => SALES_DEPT = 192.168.111.0/24

VLAN - 345 => MRKT_DEPT = 192.168.112.0/24

VLAN - 568 => IT_DEPT = 192.168.113.0/24

VTP DOMAIN - ccn.com

VTP PASSWORD - ccn@123

VTP mode - SWITCH-1=SERVER, SWITCH-2 =CLIENTS , SWITCH-3 = CLIENTS

Task 16.3 Secure Network from MAC BASED ATTACK

Task 16.4 Secure Network from DHCP SPOOFING ATTACK

Task 16.5 Secure Network from IP SPOOFING ATTACK

Task 16.6 Secure Network Form ARP ATTACKS

Task 16.7 secure network from BPDU FLOODING ATTACKS

Task 16.8 MAKE SWITCH-1 AS ROOT BRIDGE AND make sure no infirioe BPDU WILL AFFECT THE NETWORK

VERIFICATION

Switch-1

```
hostname SWITCH-1
```

```
ip arp inspection vlan 111,345,568
```

```
!  
!  
!
```

```
ip dhcp snooping vlan 111,345,568
```

```
spanning-tree vlan 111,345,568 priority 24576
```

```
!
```

```
interface Ethernet0/0
```

```
switchport access vlan 111
```

```
switchport mode access
```

```
switchport port-security violation restrict
```

```
switchport port-security mac-address sticky
```

```
switchport port-security mac-address sticky 0050.7966.6804
```

```
switchport port-security
```

```
spanning-tree portfast edge
```

```
spanning-tree bpduguard enable
```

```
spanning-tree guard root
```

```
ip verify source
```

```
!
```

```
interface Ethernet0/1
```

```
switchport access vlan 345
```

```
switchport mode access
```

```
switchport port-security violation restrict
```

```
switchport port-security mac-address sticky
```

```
switchport port-security mac-address sticky 0050.7966.6805
```

```
switchport port-security
```

```
spanning-tree portfast edge
```

```
spanning-tree bpduguard enable
```

```
spanning-tree guard root
```

```
ip verify source
```

```
interface Ethernet0/2
  switchport access vlan 568
  switchport mode access
  switchport port-security violation restrict
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 0050.7966.6806
  switchport port-security
  spanning-tree portfast edge
  spanning-tree bpduguard enable
  spanning-tree guard root
  ip verify source
!
interface Ethernet0/3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  spanning-tree guard root
  ip dhcp snooping trust
.
```

SWITCH-2

```
hostname SWITCH-2
```

```
ip arp inspection vlan 111,345,568
!
!
!
ip dhcp snooping vlan 111,345,568
```

```
interface Ethernet0/0
 switchport access vlan 111
 switchport mode access
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security
 spanning-tree portfast edge
 spanning-tree bpduguard enable
 ip verify source
!
interface Ethernet0/1
 switchport access vlan 345
 switchport mode access
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security
 spanning-tree portfast edge
 spanning-tree bpduguard enable
 ip verify source
,
```

```
interface Ethernet0/2
 switchport access vlan 568
 switchport mode access
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 switchport port-security
 spanning-tree portfast edge
 spanning-tree bpduguard enable
 ip verify source
!
interface Ethernet0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface Ethernet1/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 ip dhcp snooping trust
,
```

SWITCH-3

```
hostname SWITCH-3
```

```
ip arp inspection vlan 111,345,568
```

```
!  
!  
!
```

```
ip dhcp snooping vlan 111,345,568
```

```
interface Ethernet0/0  
  switchport access vlan 111  
  switchport mode access  
  switchport port-security violation restrict  
  switchport port-security mac-address sticky  
  switchport port-security mac-address sticky 0050.7966.680a  
  switchport port-security  
  spanning-tree portfast edge  
  spanning-tree bpduguard enable  
  ip verify source
```

```
!
```

```
interface Ethernet0/1  
  switchport access vlan 345  
  switchport mode access  
  switchport port-security violation restrict  
  switchport port-security mac-address sticky  
  switchport port-security  
  spanning-tree portfast edge  
  spanning-tree bpduguard enable  
  ip verify source
```

```
interface Ethernet0/2  
  switchport access vlan 568  
  switchport mode access  
  switchport port-security violation restrict  
  switchport port-security mac-address sticky  
  switchport port-security  
  spanning-tree portfast edge  
  spanning-tree bpduguard enable  
  ip verify source
```

```
!
```

```
interface Ethernet0/3  
  switchport trunk encapsulation dot1q  
  switchport mode trunk
```

```
!
```

```
interface Ethernet1/0  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  ip dhcp snooping trust
```

COM-1-R1

```
hostname COM-1-R1
```

```
ip dhcp relay information trust-all
```

```
ip dhcp pool CCN-SALES
network 192.168.111.0 255.255.255.0
default-router 192.168.111.254
dns-server 8.8.8.8
```

```
ip dhcp pool CCN-MRKT
network 192.168.112.0 255.255.255.0
dns-server 8.8.8.8
default-router 192.168.112.254
```

```
ip dhcp pool CCN-IT
network 192.168.113.0 255.255.255.0
default-router 192.168.113.254
dns-server 8.8.8.8
```

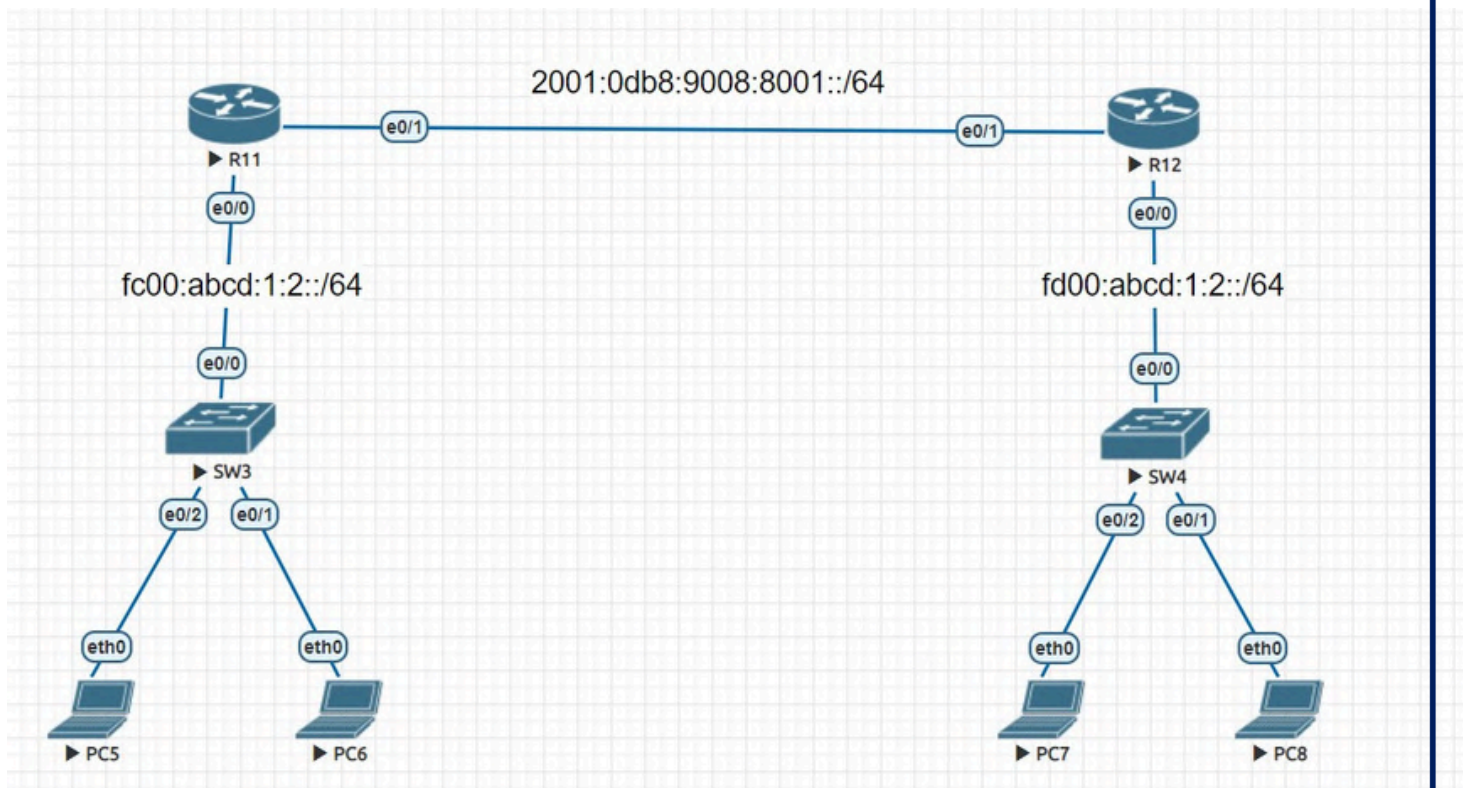
```
interface Ethernet0/0
no ip address
```

```
interface Ethernet0/0.111
encapsulation dot1Q 111
ip address 192.168.111.254 255.255.255.0
```

```
interface Ethernet0/0.345
encapsulation dot1Q 345
ip address 192.168.112.254 255.255.255.0
```

```
interface Ethernet0/0.568
encapsulation dot1Q 568
ip address 192.168.113.254 255.255.255.0
```

IPv6



TASK - 17

Task 17.1 Configure hostname as per the lab

Task 17.2 Configure the routers LAN interfaces with given IPv6 network and EUI-64. Now, fetch IP from it

Task 17.3 Configure the routers R11-eth 0/1, R12-eth 0/1

Task 17.4 Configure routers with the static ipv6 routing

Task 17.5 Configure routers with the OSPFv3

Task 17.6 Configure the router with EIGRP-ng

Task 17.7 Configure the routers with RIP-ng

After doing routing try to ping other network pc with each type of routing

VERIFICATION

R11

```
hostname R11
```

ROUTING

```
ipv6 unicast-routing
```

By applying this command router start IPv6 routing

```
!
ipv6 route FD00:ABCD:1:2::/64 2001:DB8:9008:8001::2
ipv6 router eigrp 90
  eigrp router-id 10.1.1.1
!
ipv6 router ospf 100
  router-id 10.1.1.1
!
ipv6 router rip IPV6
!
```

```
interface Ethernet0/0
  no ip address
  ipv6 address FC00:ABCD:1:2::/64 eui-64
  ipv6 enable
  ipv6 eigrp 90
  ipv6 rip IPV6 enable
  ipv6 ospf 100 area 1
!
interface Ethernet0/1
  no ip address
  ipv6 address 2001:DB8:9008:8001::1/64
  ipv6 enable
  ipv6 eigrp 90
  ipv6 rip IPV6 enable
  ipv6 ospf 100 area 0
!
```

```
VPCS> ip auto
GLOBAL SCOPE      : fc00:abcd:1:2:2050:79ff:fe66:6805/64
ROUTER LINK-LAYER : aa:bb:cc:00:10:00
```

R12

```
hostname R12
```

ROUTING

```
ipv6 unicast-routing
```

```
ipv6 route FC00:ABCD:1:2::/64 2001:DB8:9008:8001::1
ipv6 router eigrp 90
  eigrp router-id 10.1.1.2
!
ipv6 router ospf 100
  router-id 10.1.1.2
!
ipv6 router rip IPV6
```

```
interface Ethernet0/0
  no ip address
  ipv6 address FD00:ABCD:1:2::/64 eui-64
  ipv6 enable
  ipv6 eigrp 90
  ipv6 rip IPV6 enable
  ipv6 ospf 100 area 1
!
interface Ethernet0/1
  no ip address
  ipv6 address 2001:DB8:9008:8001::2/64
  ipv6 enable
  ipv6 eigrp 90
  ipv6 rip IPV6 enable
  ipv6 ospf 100 area 0
!
```

```
VPCS> ip auto
GLOBAL SCOPE      : fd00:abcd:1:2:2050:79ff:fe66:6807/64
ROUTER LINK-LAYER : aa:bb:cc:00:20:00
```

Ping form PC5 to PC7

```
VPCS> ping fd00:abcd:1:2:2050:79ff:fe66:6807

fd00:abcd:1:2:2050:79ff:fe66:6807 icmp6_seq=1 ttl=60 time=1.916 ms
fd00:abcd:1:2:2050:79ff:fe66:6807 icmp6_seq=2 ttl=60 time=0.990 ms
fd00:abcd:1:2:2050:79ff:fe66:6807 icmp6_seq=3 ttl=60 time=1.355 ms
fd00:abcd:1:2:2050:79ff:fe66:6807 icmp6_seq=4 ttl=60 time=1.090 ms
fd00:abcd:1:2:2050:79ff:fe66:6807 icmp6_seq=5 ttl=60 time=1.490 ms
```

How to check routing table

Router# show ipv6 route